United States Patent	12415860
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Lin; Shuoyen Jack et al.

Binding moiety for conditional activation of immunoglobulin molecules

Abstract

Disclosed herein are binding moieties that comprise non-CDR loops for masking the binding of a binding molecule to its target and CDRs for binding bulk serum proteins. Conditionally active target binding proteins that contain the binding moieties are also provided. Pharmaceutical compositions comprising the binding proteins disclosed herein and methods of using such formulations are further provided.

Inventors:	**Lin; Shuoyen Jack (San Bruno, CA), Austin; Richard J. (San Francisco, CA), Lemon; Bryan D. (Mountain View, CA), Kwant; Kathryn (San Bruno, CA), Rocha; Sony S. (San Francisco, CA), Wesche; Holger (San Francisco, CA)**
Applicant:	**Harpoon Therapeutics, Inc.** (South San Francisco, CA)

Family ID:	**1000008818649**
Assignee:	**Harpoon Therapeutics, Inc. (South San Francisco, CA)**
Appl. No.:	**17/055096**
Filed (or PCT Filed):	**May 14, 2019**
PCT No.:	**PCT/US2019/032307**
PCT Pub. No.:	**WO2019/222283**
PCT Pub. Date:	November 21, 2019

Prior Publication Data

Document Identifier	Publication Date
US 20210292421 A1	Sep. 23, 2021

Related U.S. Application Data

us-provisional-application US 62756453 20181106
us-provisional-application US 62756429 20181106
us-provisional-application US 62671344 20180514
us-provisional-application US 62671349 20180514

Publication Classification

Int. Cl.: **C07K16/28** (20060101); **C07K16/00** (20060101); **C07K16/30** (20060101); A61K39/00 (20060101)

U.S. Cl.:

CPC **C07K16/2863** (20130101); **C07K16/2809** (20130101); **C07K16/2878** (20130101); **C07K16/3069** (20130101); A61K2039/505 (20130101); C07K2317/24 (20130101); C07K2317/31 (20130101); C07K2317/33 (20130101); C07K2317/34 (20130101); C07K2317/565 (20130101); C07K2317/569 (20130101); C07K2317/60 (20130101); C07K2317/73 (20130101); C07K2317/92 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4816567	12/1988	Cabilly et al.	N/A	N/A
4943533	12/1989	Mendelsohn et al.	N/A	N/A
5061620	12/1990	Tsukamoto et al.	N/A	N/A
5199942	12/1992	Gillis	N/A	N/A
5225539	12/1992	Winter	N/A	N/A
5350674	12/1993	Boenisch et al.	N/A	N/A
5399346	12/1994	Anderson et al.	N/A	N/A
5530101	12/1995	Queen et al.	N/A	N/A
5565332	12/1995	Hoogenboom et al.	N/A	N/A
5580859	12/1995	Felgner et al.	N/A	N/A
5585089	12/1995	Queen et al.	N/A	N/A
5585362	12/1995	Wilson et al.	N/A	N/A
5589466	12/1995	Felgner et al.	N/A	N/A
5759808	12/1997	Casterman et al.	N/A	N/A
5766886	12/1997	Studnicka et al.	N/A	N/A
5773292	12/1997	Bander	N/A	N/A
5800988	12/1997	Casterman et al.	N/A	N/A
5840526	12/1997	Casterman et al.	N/A	N/A
5844093	12/1997	Kettleborough et al.	N/A	N/A

5858358	12/1998	June et al.	N/A	N/A
5859205	12/1998	Adair et al.	N/A	N/A
5874541	12/1998	Casterman et al.	N/A	N/A
5883223	12/1998	Gray	N/A	N/A
6005079	12/1998	Casterman et al.	N/A	N/A
6015695	12/1999	Casterman et al.	N/A	N/A
6107090	12/1999	Bander	N/A	N/A
6120766	12/1999	Hale et al.	N/A	N/A
6136311	12/1999	Bander	N/A	N/A
6326193	12/2000	Liu et al.	N/A	N/A
6331415	12/2000	Cabilly et al.	N/A	N/A
6352694	12/2001	June et al.	N/A	N/A
6407213	12/2001	Carter et al.	N/A	N/A
6534055	12/2002	June et al.	N/A	N/A
6548640	12/2002	Winter	N/A	N/A
6670453	12/2002	Frenken et al.	N/A	N/A
6692964	12/2003	June et al.	N/A	N/A
6759518	12/2003	Kontermann et al.	N/A	N/A
6767711	12/2003	Bander	N/A	N/A
6797514	12/2003	Berenson et al.	N/A	N/A
6867041	12/2004	Berenson et al.	N/A	N/A
6887466	12/2004	June et al.	N/A	N/A
6905680	12/2004	June et al.	N/A	N/A
6905681	12/2004	June et al.	N/A	N/A
6905874	12/2004	Berenson et al.	N/A	N/A
7060808	12/2005	Goldstein et al.	N/A	N/A
7067318	12/2005	June et al.	N/A	N/A
7144575	12/2005	June et al.	N/A	N/A
7163680	12/2006	Bander	N/A	N/A
7172869	12/2006	June et al.	N/A	N/A
7175843	12/2006	June et al.	N/A	N/A
7232566	12/2006	June et al.	N/A	N/A
7247301	12/2006	van de Winkel et al.	N/A	N/A
7262276	12/2006	Huang et al.	N/A	N/A
7595378	12/2008	van de Winkel et al.	N/A	N/A
7666414	12/2009	Bander	N/A	N/A
7723484	12/2009	Beidler et al.	N/A	N/A
7807162	12/2009	Silence	N/A	N/A
7850971	12/2009	Maddon et al.	N/A	N/A
7939072	12/2010	Yarden et al.	N/A	N/A
8114965	12/2011	Maddon et al.	N/A	N/A
8188223	12/2011	Beirnaert et al.	N/A	N/A
8236308	12/2011	Kischel et al.	N/A	N/A
8470330	12/2012	Schuelke et al.	N/A	N/A
8623356	12/2013	Christopherson et al.	N/A	N/A
8629244	12/2013	Kolkman et al.	N/A	N/A
8703135	12/2013	Beste et al.	N/A	N/A

8784821	12/2013	Kufer et al.	N/A	N/A
8846042	12/2013	Zhou	N/A	N/A
8907071	12/2013	Sullivan et al.	N/A	N/A
8937164	12/2014	Descamps et al.	N/A	N/A
8986972	12/2014	Stull et al.	N/A	N/A
9089615	12/2014	Stull et al.	N/A	N/A
9089616	12/2014	Stull et al.	N/A	N/A
9089617	12/2014	Stull et al.	N/A	N/A
9090683	12/2014	Stull et al.	N/A	N/A
9107961	12/2014	Stull et al.	N/A	N/A
9127071	12/2014	Yoshida et al.	N/A	N/A
9133271	12/2014	Stull et al.	N/A	N/A
9155803	12/2014	Stull et al.	N/A	N/A
9169316	12/2014	Baty et al.	N/A	N/A
9309327	12/2015	Humphreys et al.	N/A	N/A
9327022	12/2015	Zhang et al.	N/A	N/A
9334318	12/2015	Stull et al.	N/A	N/A
9340621	12/2015	Kufer et al.	N/A	N/A
9345787	12/2015	Hemminki et al.	N/A	N/A
9352051	12/2015	Stull et al.	N/A	N/A
9353182	12/2015	Stull et al.	N/A	N/A
9358304	12/2015	Stull et al.	N/A	N/A
9480757	12/2015	Stull et al.	N/A	N/A
9481724	12/2015	Ravetch et al.	N/A	N/A
9486537	12/2015	Stull et al.	N/A	N/A
9708412	12/2016	Baeuerle et al.	N/A	N/A
9764042	12/2016	Stull et al.	N/A	N/A
9770518	12/2016	Stull et al.	N/A	N/A
9775916	12/2016	Stull et al.	N/A	N/A
9855343	12/2017	Stull et al.	N/A	N/A
9861708	12/2017	Stull et al.	N/A	N/A
9867887	12/2017	Stull et al.	N/A	N/A
9878053	12/2017	Stull et al.	N/A	N/A
9920115	12/2017	Dubridge et al.	N/A	N/A
9931420	12/2017	Stull et al.	N/A	N/A
9931421	12/2017	Stull et al.	N/A	N/A
9937268	12/2017	Stull et al.	N/A	N/A
10066016	12/2017	Dubridge et al.	N/A	N/A
10100106	12/2017	Dubridge et al.	N/A	N/A
10137204	12/2017	Stull et al.	N/A	N/A
10428120	12/2018	Kontermann et al.	N/A	N/A
10543271	12/2019	Wesche et al.	N/A	N/A
10544221	12/2019	Dubridge et al.	N/A	N/A
10730954	12/2019	Wesche et al.	N/A	N/A
10815311	12/2019	Wesche et al.	N/A	N/A
10844134	12/2019	Baeuerle et al.	N/A	N/A
10849973	12/2019	Dubridge et al.	N/A	N/A
11111311	12/2020	Yoshida et al.	N/A	N/A
11180563	12/2020	Wesche et al.	N/A	N/A

11400157	12/2021	Igawa	N/A	C07K 14/00
2004/0101519	12/2003	June et al.	N/A	N/A
2005/0042664	12/2004	Wu et al.	N/A	N/A
2005/0048617	12/2004	Wu et al.	N/A	N/A
2005/0100543	12/2004	Hansen et al.	N/A	N/A
2005/0175606	12/2004	Huang et al.	N/A	N/A
2006/0034810	12/2005	Riley et al.	N/A	N/A
2006/0046971	12/2005	Stuhler et al.	N/A	N/A
2006/0121005	12/2005	Berenson et al.	N/A	N/A
2006/0228364	12/2005	Dennis et al.	N/A	N/A
2006/0252096	12/2005	Zha et al.	N/A	N/A
2007/0014794	12/2006	Carter et al.	N/A	N/A
2007/0178082	12/2006	Silence et al.	N/A	N/A
2007/0269422	12/2006	Beirnaert et al.	N/A	N/A
2008/0069772	12/2007	Stuhler et al.	N/A	N/A
2008/0260757	12/2007	Holt et al.	N/A	N/A
2009/0117108	12/2008	Wang et al.	N/A	N/A
2009/0136485	12/2008	Chu et al.	N/A	N/A
2009/0259026	12/2008	Tomlinson et al.	N/A	N/A
2010/0122358	12/2009	Brueggemann et al.	N/A	N/A
2010/0150918	12/2009	Kufer et al.	N/A	N/A
2010/0166734	12/2009	Dolk	N/A	N/A
2010/0189651	12/2009	Stagliano et al.	N/A	N/A
2010/0189727	12/2009	Rodeck et al.	N/A	N/A
2010/0266531	12/2009	Hsieh et al.	N/A	N/A
2010/0291112	12/2009	Kellner et al.	N/A	N/A
2010/0311119	12/2009	Hermans et al.	N/A	N/A
2011/0129458	12/2010	Dolk et al.	N/A	N/A
2011/0165621	12/2010	Dreier et al.	N/A	N/A
2011/0262439	12/2010	Kufer et al.	N/A	N/A
2011/0275787	12/2010	Kufer et al.	N/A	N/A
2011/0313135	12/2010	Vanhove et al.	N/A	N/A
2012/0039899	12/2011	Olsen et al.	N/A	N/A
2012/0231024	12/2011	Elsaesser-Beile et al.	N/A	N/A
2012/0328619	12/2011	Fey et al.	N/A	N/A
2013/0017200	12/2012	Scheer et al.	N/A	N/A
2013/0136744	12/2012	Bouche et al.	N/A	N/A
2013/0266568	12/2012	Brinkmann et al.	N/A	N/A
2013/0267686	12/2012	Brinkmann et al.	N/A	N/A
2013/0273055	12/2012	Borges et al.	N/A	N/A
2013/0315906	12/2012	Lowman et al.	N/A	N/A
2013/0330335	12/2012	Bremel et al.	N/A	N/A
2014/0004121	12/2013	Fanslow, III et al.	N/A	N/A
2014/0023664	12/2013	Lowman et al.	N/A	N/A
2014/0045195	12/2013	Daugherty et al.	N/A	N/A
2014/0073767	12/2013	Lee et al.	N/A	N/A
2014/0088295	12/2013	Smith et al.	N/A	N/A
2014/0205601	12/2013	Beirnaert et al.	N/A	N/A

2014/0242075	12/2013	Parren et al.	N/A	N/A
2014/0302037	12/2013	Borges et al.	N/A	N/A
2014/0308285	12/2013	Yan et al.	N/A	N/A
2014/0322218	12/2013	Xiao et al.	N/A	N/A
2015/0037334	12/2014	Kufer et al.	N/A	N/A
2015/0056206	12/2014	Zhou	N/A	N/A
2015/0064169	12/2014	Wang	536/23.53	C07K 16/36
2015/0079088	12/2014	Lowman et al.	N/A	N/A
2015/0079093	12/2014	Stuhler	N/A	N/A
2015/0093336	12/2014	Van Ginderachter et al.	N/A	N/A
2015/0174268	12/2014	Li et al.	N/A	N/A
2015/0175697	12/2014	Bonvini et al.	N/A	N/A
2015/0183875	12/2014	Cobbold et al.	N/A	N/A
2015/0232557	12/2014	Tan et al.	N/A	N/A
2015/0274836	12/2014	Ho et al.	N/A	N/A
2015/0274844	12/2014	Blankenship et al.	N/A	N/A
2015/0328332	12/2014	Stull et al.	N/A	N/A
2016/0017058	12/2015	Kim et al.	N/A	N/A
2016/0024174	12/2015	Odunsi et al.	N/A	N/A
2016/0032011	12/2015	Zhang et al.	N/A	N/A
2016/0032019	12/2015	Xiao et al.	N/A	N/A
2016/0039942	12/2015	Cobbold et al.	N/A	N/A
2016/0068605	12/2015	Nemeth et al.	N/A	N/A
2016/0115241	12/2015	Yan et al.	N/A	N/A
2016/0130331	12/2015	Stull et al.	N/A	N/A
2016/0215063	12/2015	Bernett et al.	N/A	N/A
2016/0251440	12/2015	Roobrouck et al.	N/A	N/A
2016/0257721	12/2015	Lieber et al.	N/A	N/A
2016/0319040	12/2015	Dreier et al.	N/A	N/A
2016/0340444	12/2015	Baeuerle et al.	N/A	N/A
2016/0355842	12/2015	Parks et al.	N/A	N/A
2017/0029502	12/2016	Raum et al.	N/A	N/A
2017/0037149	12/2016	Raum et al.	N/A	N/A
2017/0152316	12/2016	Cobbold et al.	N/A	N/A
2017/0204164	12/2016	Himmler et al.	N/A	N/A
2017/0275373	12/2016	Kufer et al.	N/A	N/A
2017/0298149	12/2016	Baeuerle et al.	N/A	N/A
2017/0334979	12/2016	Dubridge et al.	N/A	N/A
2017/0334997	12/2016	Dubridge et al.	N/A	N/A
2017/0369563	12/2016	Dubridge et al.	N/A	N/A
2018/0016323	12/2017	Brandenburg et al.	N/A	N/A
2018/0134789	12/2017	Baeuerle et al.	N/A	N/A
2018/0148508	12/2017	Wang et al.	N/A	N/A
2018/0318417	12/2017	Schuetz et al.	N/A	N/A
2018/0326060	12/2017	Wesche et al.	N/A	N/A
2018/0346601	12/2017	Dettling et al.	N/A	N/A
2019/0031749	12/2018	Dubridge et al.	N/A	N/A
2019/0046656	12/2018	Stull et al.	N/A	N/A

2019/0092862	12/2018	Cui et al.	N/A	N/A
2019/0112381	12/2018	Wesche et al.	N/A	N/A
2019/0135930	12/2018	Wesche et al.	N/A	N/A
2019/0225702	12/2018	Baeuerle et al.	N/A	N/A
2019/0247510	12/2018	Stull et al.	N/A	N/A
2020/0115461	12/2019	Evnin et al.	N/A	N/A
2020/0148771	12/2019	Paeuerle et al.	N/A	N/A
2020/0231672	12/2019	Dubridge et al.	N/A	N/A
2020/0270362	12/2019	Wesche et al.	N/A	N/A
2020/0289646	12/2019	Wesche et al.	N/A	N/A
2021/0047439	12/2020	Wesche et al.	N/A	N/A
2021/0095047	12/2020	Baeuerle et al.	N/A	N/A
2021/0100902	12/2020	Dubridge et al.	N/A	N/A
2021/0171649	12/2020	Wesche et al.	N/A	N/A
2021/0179735	12/2020	Baeuerle et al.	N/A	N/A
2021/0269530	12/2020	Lin et al.	N/A	N/A
2021/0284728	12/2020	Lin et al.	N/A	N/A
2021/0355219	12/2020	Lin et al.	N/A	N/A
2021/0380715	12/2020	Yoshida et al.	N/A	N/A
2022/0017626	12/2021	Wesche et al.	N/A	N/A
2022/0054544	12/2021	Lin et al.	N/A	N/A
2022/0098311	12/2021	Wesche et al.	N/A	N/A
2022/0112297	12/2021	Wesche et al.	N/A	N/A
2022/0267462	12/2021	Wesche et al.	N/A	N/A
2023/0257451	12/2022	Dubridge et al.	N/A	N/A
2024/0084009	12/2023	Dubridge et al.	N/A	N/A
2024/0084035	12/2023	Molloy et al.	N/A	N/A
2024/0100157	12/2023	Wesche et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
3040823	12/2017	CA	N/A
1563092	12/2004	CN	N/A
101557817	12/2008	CN	N/A
101646689	12/2009	CN	N/A
105968201	12/2015	CN	N/A
105968204	12/2015	CN	N/A
108137706	12/2017	CN	N/A
109593786	12/2018	CN	N/A
0239400	12/1986	EP	N/A
0519596	12/1991	EP	N/A
0592106	12/1993	EP	N/A
1378520	12/2003	EP	N/A
1736484	12/2005	EP	N/A
2336179	12/2010	EP	N/A
2817338	12/2013	EP	N/A
3038659	12/2015	EP	N/A
3093293	12/2015	EP	N/A
3093294	12/2015	EP	N/A

3095797	12/2015	EP	N/A
3107576	12/2015	EP	N/A
3261650	12/2017	EP	N/A
3337517	12/2017	EP	N/A
3458050	12/2018	EP	N/A
3556400	12/2018	EP	N/A
901228	12/1944	FR	N/A
2005501517	12/2004	JP	N/A
2016500655	12/2015	JP	N/A
2019052750	12/2018	JP	N/A
20180030477	12/2017	KR	N/A
WO-9109967	12/1990	WO	N/A
WO-9307105	12/1992	WO	N/A
WO-9404678	12/1993	WO	N/A
WO-9937681	12/1998	WO	N/A
WO-0043507	12/1999	WO	N/A
WO-0190190	12/2000	WO	N/A
WO-0196584	12/2000	WO	N/A
WO-02085945	12/2001	WO	N/A
WO-03025020	12/2002	WO	N/A
WO-03035694	12/2002	WO	N/A
WO-03064606	12/2002	WO	N/A
WO-2004003019	12/2003	WO	N/A
WO-2004041867	12/2003	WO	N/A
WO-2004042404	12/2003	WO	N/A
WO-2004049794	12/2003	WO	N/A
WO-2005040220	12/2004	WO	N/A
WO-2006020258	12/2005	WO	N/A
WO-2006122786	12/2005	WO	N/A
WO-2006122787	12/2005	WO	N/A
WO-2007024715	12/2006	WO	N/A
WO-2007042261	12/2006	WO	N/A
WO-2007062466	12/2006	WO	N/A
WO-2007115230	12/2006	WO	N/A
WO-2008028977	12/2007	WO	N/A
WO-2009025846	12/2008	WO	N/A
WO-2009030285	12/2008	WO	N/A
WO-2009035577	12/2008	WO	N/A
WO-2009147248	12/2008	WO	N/A
WO-2010003118	12/2009	WO	N/A
WO-2010037836	12/2009	WO	N/A
WO-2010037837	12/2009	WO	N/A
2010081173	12/2009	WO	N/A
WO-2011039368	12/2010	WO	N/A
WO-2011051327	12/2010	WO	N/A
WO-2011079283	12/2010	WO	N/A
WO-2011117423	12/2010	WO	N/A
WO-2011161260	12/2010	WO	N/A
WO-2012131053	12/2011	WO	N/A
WO-2012138475	12/2011	WO	N/A

WO-2012158818	12/2011	WO	N/A
WO-2012175400	12/2011	WO	N/A
WO-2013036130	12/2012	WO	N/A
WO-2013045707	12/2012	WO	N/A
WO-2013053725	12/2012	WO	N/A
WO-2013104804	12/2012	WO	N/A
WO-2013110531	12/2012	WO	N/A
WO-2013126712	12/2012	WO	N/A
WO-2013128027	12/2012	WO	N/A
WO-2013128194	12/2012	WO	N/A
WO-2014012085	12/2013	WO	N/A
WO-2014033304	12/2013	WO	N/A
WO-2014052064	12/2013	WO	N/A
WO-2014138306	12/2013	WO	N/A
WO-2014140358	12/2013	WO	N/A
WO-2014144689	12/2013	WO	N/A
WO-2014151910	12/2013	WO	N/A
WO-2015031693	12/2014	WO	N/A
WO-2015103072	12/2014	WO	N/A
WO-2015127407	12/2014	WO	N/A
WO-2015146437	12/2014	WO	N/A
WO-2015150447	12/2014	WO	N/A
WO-2015184207	12/2014	WO	N/A
WO-2016009029	12/2015	WO	N/A
WO-2016046778	12/2015	WO	N/A
WO-2016055551	12/2015	WO	N/A
WO-2016105450	12/2015	WO	N/A
WO-2016130819	12/2015	WO	N/A
WO-2016138038	12/2015	WO	N/A
WO-2016171999	12/2015	WO	N/A
2016187594	12/2015	WO	N/A
WO-2016179003	12/2015	WO	N/A
WO-2016182064	12/2015	WO	N/A
WO-2016187101	12/2015	WO	N/A
WO-2016210447	12/2015	WO	N/A
WO-2017021356	12/2016	WO	N/A
WO-2017025038	12/2016	WO	N/A
WO-2017025698	12/2016	WO	N/A
WO-2017027392	12/2016	WO	N/A
WO-2017031104	12/2016	WO	N/A
WO-2017041749	12/2016	WO	N/A
WO-2017079528	12/2016	WO	N/A
WO-2017136549	12/2016	WO	N/A
2017162587	12/2016	WO	N/A
WO 2017/156178	12/2016	WO	C07K 16/28
WO-2017156178	12/2016	WO	N/A
WO-2017157305	12/2016	WO	N/A
WO-2017161206	12/2016	WO	N/A
WO-2017201442	12/2016	WO	N/A
WO-2017201488	12/2016	WO	N/A

WO-2017201493	12/2016	WO	N/A
WO-2018017863	12/2017	WO	N/A
WO-2018026953	12/2017	WO	N/A
WO-2018067993	12/2017	WO	N/A
WO-2018071777	12/2017	WO	N/A
WO-2018098354	12/2017	WO	N/A
WO-2018098356	12/2017	WO	N/A
WO-2018119183	12/2017	WO	N/A
WO-2018136725	12/2017	WO	N/A
WO-2018160671	12/2017	WO	N/A
WO-2018160754	12/2017	WO	N/A
WO-2018165619	12/2017	WO	N/A
WO-2018204717	12/2017	WO	N/A
WO-2018209298	12/2017	WO	N/A
WO-2018209304	12/2017	WO	N/A
WO-2018232020	12/2017	WO	N/A
WO-2019075359	12/2018	WO	N/A
WO-2019075378	12/2018	WO	N/A
WO-2019222278	12/2018	WO	N/A
WO-2019222282	12/2018	WO	N/A
WO-2019222283	12/2018	WO	N/A
WO-2020060593	12/2019	WO	N/A
WO-2020061482	12/2019	WO	N/A
WO-2020061526	12/2019	WO	N/A
WO-2020069028	12/2019	WO	N/A
WO-2020232303	12/2019	WO	N/A
WO-2021097060	12/2020	WO	N/A
WO-2021231434	12/2020	WO	N/A

OTHER PUBLICATIONS

Agata et al. Expression of the PD-1 antigen on the surface of stimulated mouse T and B lymphocytes. Int. Immunol 8:765-75 (1996). cited by applicant
Al-Lazikani et al. Standard conformations for the canonical structures of immunoglobulins. J. Mol Biology 273(4):927-948 (1997). cited by applicant
Almagro et al. Humanization of antibodies. Front Biosci 13:1619-1633 (2008). cited by applicant
Altschul et al. Basic local alignment search tool. J Mol Biol 215(3):403-410 (1990). cited by applicant
Altschul, et al. Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. Nucleic Acids Res 25:3389-3402 (1977). cited by applicant
Argani et al. Mesothelin is overexpressed in the vast majority of ductal adenocarcinomas of the pancreas: identification of a new pancreatic cancer marker by serial analysis of gene expression (SAGE). Clin Cancer Res 7(12):3862-3868 (2001). cited by applicant
Austin et al. Cancer Research (Jul. 2018) vol. 78, No. 13, Supp. Supplement 1. Abstract No. 1781. Meeting Info: 2018 Annual Meeting of the American Association for Cancer Research, AACR 2018. Chicago, IL, United States. Apr. 14-Apr. 18, 2018). cited by applicant
Baca et al. Antibody humanization using monovalent phage display. J Biol Chem 272(16):10678-10684 (1997). cited by applicant
Baeuerle et al. Bispecific T-cell engaging antibodies for cancer therapy. Cancer Res 69:4941-4944 (2009). cited by applicant
Barrett et al. Treatment of advanced leukemia in mice with mRNA engineered T cells. Hum Gene

Ther 22:1575-1586 (2011). cited by applicant
Batzer et al. Enhanced evolutionary PCR using oligonucleotides with inosine at the 3′- terminus. Nucleic Acids Res. 19(18):5081 (1991). cited by applicant
Baum et al. Antitumor activities of PSMAxCD3 diabodies by redirected T-cell lysis of prostate cancer cells. Immunotherapy 5(1):27-38 (2013). cited by applicant
Bedouelle et al. Diversity and junction residues as hotspots of binding energy in an antibody neutralizing the dengue virus. Febs J 273(1):34-46 (2006). cited by applicant
Bendell et al. Abstract 5552: First-in-human phase I study of HPN424, a tri-specific half-life extended PSMA-targeting T-cell engager in patients with metastatic castration-resistant prostate cancer (mCRPC). J Clin Oncol 38(15):5552 (May 2020). cited by applicant
Bird et al. Single-chain antigen-binding proteins. Science 242(4877):423-426 (1988). cited by applicant
Blank et al. Interaction of PD-L1 on tumor cells with PD-1 on tumor-specific T cells as a mechanism of immune evasion: implications for tumor immunotherapy. Cancer Immunol Immunother 54:307-314 (2005). cited by applicant
Bortoletto et al. Optimizing anti-CD3 affinity for effective T cell targeting against tumor cells. Eur J Immunol 32:3102-3107 (2002). cited by applicant
Bracci et al. Cyclophosphamide enhances the antitumor efficacy of adoptively transferred immune cells through the induction of cytokine expression, B-cell and T-cell homeostatic proliferation, and specific tumor infiltration. Clin Cancer Res 13(2 Pt 1):644-653 (2007). cited by applicant
Brown et al. Tolerance of single, but not multiple, amino acid replacements in antibody VH CDR 2: a means of minimizing B cell wastage from somatic hypermutation? J Immunol 156(9):3285-3291 (1996). cited by applicant
Caldas et al. Design and synthesis of germline-based hemi-humanized single-chain Fv against the CD18 surface antigen. Protein Eng 13(5):353-360 (2000). cited by applicant
Caldas et al. Humanization of the anti-CD18 antibody 6.7: an unexpected effect of a framework residue in binding to antigen. Mol Immunol. 39(15):941-952 (2003). cited by applicant
Carter et al. Humanization of an anti-p185HER2 antibody for human cancer therapy. PNAS USA 89(10):4285-4289 (1992). cited by applicant
Carter et al. PD-1: PD-L inhibitory pathway affects both CD4(+) and CD8(+) T cells and is overcome by IL-2. Eur J Immunol 32:634-643 (2002). cited by applicant
Casset et al. A peptide mimetic of an anti-CD4 monoclonal antibody by rational design. Biochemical and Biophysical Research Communication 307:198-205 (2003). cited by applicant
Chang et al. Loop-sequence features and stability determinants in antibody variable domains by high-throughput experiments. Structure 22(1):9-21 (2014). cited by applicant
Chang et al. Molecular cloning of mesothelin, a differentiation antigen present on mesothelium, mesotheliomas, and ovarian cancers. PNAS USA 93:136-140 (1996). cited by applicant
Chatalic et al. A Novel 111 In-labeled Anti-PSMA Nanobody for Targeted SPECT/CT Imaging of Prostate Cancer. J Nucl Med 56(7):1094-1099 and Supplemental Data (2015). cited by applicant
Chen et al. Fusion protein linkers: Property, design and functionality. Advanced Drug Delivery Reviews 65:1357-1369 (2013). cited by applicant
Chen et al. Selection and analysis of an optimized anti-VEGF antibody: Crystal structure of an affinity-matured Fab in complex with antigen. J Mol Bio 293:865-881 (1999). cited by applicant
Chien et al. Significant structural and functional change of an antigen-binding site by a distant amino acid substitution: proposal of a structural mechanism. PNAS USA 86(14):5532-5536 (1989). cited by applicant
Cho et al. Targeting B Cell Maturation Antigen (BCMA) in Multiple Myeloma: Potential Uses of BCMA-Based Immunotherapy. Front Immunol 9:1821 (2018). cited by applicant
Choi et al. Engineering of Immunoglobulin Fc heterodimers using yeast surface-displayed combinatorial Fc library screening. PLOS One 10(12):e0145349 (2015). cited by applicant

Chothia et al. Canonical structures for the hypervariable regions of immunoglobulins. J Mol Biol 196(4):901-917 (1987). cited by applicant
Chothia, et al. Conformations of immunoglobulin hypervariable regions. Nature 342(6252):877-83 (1989). cited by applicant
Corso et al. Real-time detection of mesothelin in pancreatic cancer cell line supernatant using an acoustic wave immunosensor. Cancer Detect Prey 30:180-187 (2006). cited by applicant
Cougot et al. 'Cap-tabolism'. Trends in Biochem Sci 29:436-444 (2001). cited by applicant
Couto et al. Anti-BA46 monoclonal antibody Mc3: humanization using a novel positional consensus and in vivo and in vitro characterization. Cancer Res 55(8):1717-1722 (1995). cited by applicant
Couto et al. Designing human consensus antibodies with minimal positional templates. Cancer Res 55(23 Supp):5973s-5977s (1995). cited by applicant
Creaney et al. Detection of malignant mesothelioma in asbestos-exposed individuals: the potential role of soluble mesothelin-related protein. Hematol. Oncol. Clin. North Am. 19:1025-1040 (2005). cited by applicant
Cristaudo et al. Clinical significance of serum mesothelin in patients with mesothelioma and lung cancer. Clin. Cancer Res. 13:5076-5081 (2007). cited by applicant
Dao et al. Targeting the intracellular WT1 oncogene product with a therapeutic human antibody. Sci Transi Med 5(176):176ra33 (2013). cited by applicant
De Genst et al. Antibody repertoire development in camelids. Dev Comp Immunol 30(1-2):187-198 (2006). cited by applicant
De Pascalis et al. Grafting of "abbreviated" complementarity-determining regions containing specificity-determining residues essential for ligand contact to engineer a less immunogenic humanized monoclonal antibody. J Immunol. 169(6):3076-3084 (2002). cited by applicant
Dennis et al. Imaging Tumors with an Albumin-Binding Fab, a Novel Tumor-Targeting Agent. Cancer Res 67(1):254-61 (2007). cited by applicant
Document D28—Investigation of human CD38 variants binding to monoclonal antibodies. Submitted by Pfizer to the European Patent Register on Apr. 30, 2014 in connection with their opposition to the EP2155783 patent. (3 pages) (2014). cited by applicant
Document D78—CD3ε N-terminal peptide bound to the CDRs of SP24. Submitted by Janssen Biotech to the European Patent Register on Mar. 18, 2016 in connection with their opposition to the EP2155783 patent (1 page) (2016). cited by applicant
Document D79—Interactions between CD3εand SP34 CDR residues. CD3ε residues are in ellipses, SP34 CDR residues are in boxes. Submitted by Janssen Biotech to the European Patent Register on Mar. 18, 2016 in connection with their opposition to the EP2155783 patent (1 page) (2016). cited by applicant
Document D83—Alignment of variable domains from the prior art and the patent. Submitted by Janssen Biotech to the European Patent Register on Mar. 18, 2016 in connection with their opposition to the EP2155783 patent (1 page) (2016). cited by applicant
Dong et al. B7-H1 pathway and its role in the evasion of tumor immunity. J Mol Med 81:281-287 (2003). cited by applicant
Elango et al. Optimized transfection of mRNA transcribed from a d(A/T)100 tail-containing vector. Biochim Biophys Res Commun 330:958-966 (2005). cited by applicant
Foote et al. Antibody Framework Residues Affecting the Conformation of the Hypervariable Loops. J. Mol. Biol. 224(2):487-99 (1992). cited by applicant
Frankel et al. Targeting T cells to tumor cells using bispecific antibodies. Curr Opin Chem Biol 17(3):385-392 (2013). cited by applicant
Freeman et al. Engagement of the PD-1 immunoinhibitory receptor by a novel B7 family member leads to negative regulation of lymphocyte activation. J Exp Med 192:1027-1034 (2000). cited by applicant

Garland et al. The use of Teflon cell culture bags to expand functionally active CD8+ cytotoxic T lymphocytes. J Immunol Meth 227(1-2):53-63 (1999). cited by applicant
Giusti et al. Somatic diversification of S107 from an antiphosphocholine to an anti-DNA autoantibody is due to a single base change in its heavy chain variable region. PNAS USA 84(9):2926-30 (1987). cited by applicant
Glaser et al. Novel antibody hinge regions for efficient production of CH2 domain-deleted antibodies. J. Biol. Chem. 280:41494-503 (2005). cited by applicant
Goldman et al. Enhancing Stability of Camelid and Shark Single Domain Antibodies: An Overview. Front. Immunol. 8:865 (2017). cited by applicant
Goodman et al. The Pharmaceutical Basis of Therapeutics. 6th ed. pp. 21-25 (1980). cited by applicant
Goswami et al. Developments and Challenges for mAb-Based Therapeutics. Antibodies 2:452-500 (2013). cited by applicant
Gross et al. Endowing T cells with antibody specificity using chimeric T cell receptors. Faseb J. 6(15):3370-3378 (1992). cited by applicant
Grupp et al. Chimeric antigen receptor-modified T cells for acute lymphoid leukemia. NEJM 368:1509-1518 (2013). cited by applicant
Gubbels et al. Mesothelin-MUC16 binding is a high affinity, N-glycan dependent interaction that facilitates peritoneal metastasis of ovarian tumors. Mol Cancer 5:50 (2006). cited by applicant
Gussow et al. Chapter 5: Humanization of Monoclonal Antibodies. Methods in Enzymology 203:99-121 (1991). cited by applicant
Haanen et al. Selective expansion of cross-reactive CD8(+) memory T cells by viral variants. J Exp Med 190(9):1319-1328 (1999). cited by applicant
Halaby et al. The immunoglobulin fold family: sequence analysis and 3D structure comparisons. Prot Eng 12(7):563-571 (1999). cited by applicant
Han et al. Masked Chimeric Antigen Receptor for Tumor-Specific Activation. Molecular Therapy 25(1):274-284 (2017). cited by applicant
Harding et al. The immunogenicity of humanized and fully human antibodies: residual immunogenicity resides in the CDR regions. MAbs 2(3):256-265 (2010). cited by applicant
Harmsen et al. Properties, production, and applications of camelid single-domain antibody fragments. Appl. Microbiol. Biotechnol. 77:13-22 (2007). cited by applicant
Hassan et al. Detection and quantitation of serum mesothelin, a tumor marker for patients with mesothelioma and ovarian cancer. Clin Cancer Res 12:447-453 (2006). cited by applicant
Hassan et al. Mesothelin: a new target for immunotherapy. Clin Cancer Res 10:3937-3942 (2004). cited by applicant
Hassan et al. Mesothelin targeted cancer immunotherapy. Eur J Cancer 44:46-53 (2008). cited by applicant
Hassan et al. Phase I study of SS1P, a recombinant anti-mesothelin immunotoxin given as a bolus I.V. infusion to patients with mesothelin-expressing mesothelioma, ovarian, and pancreatic cancers. Clin Cancer Res 13(17):5144-5149 (2007). cited by applicant
Hassan et al. Preclinical evaluation of MORAb-009, a chimeric antibody targeting tumor-associated mesothelin. Cancer Immun. 7:20 (2007). cited by applicant
Hellstrom et al. Mesothelin variant 1 is released from tumor cells as a diagnostic marker. Cancer Epidemiol Biomarkers Prey 15:1014-1020 (2006). cited by applicant
Hipp et al. A novel BCMA/CD3 bispecific T-cell engager for the treatment of multiple myeloma induces selective lysis in vitro and in vivo. Leukemia 31(8):1743-1751 (2017). cited by applicant
Ho et al. A novel high-affinity human monoclonal antibody to mesothelin. Int J Cancer 128:2020-2030 (2011). cited by applicant
Ho et al. Mesothelin expression in human lung cancer. Clin Cancer Res 13:1571-1575 (2007). cited by applicant

Ho et al. Mesothelin is shed from tumor cells. Cancer Epidemiol Biomarkers Prey 15:1751 (2006). cited by applicant
Holliger, et al. “Diabodies”: small bivalent and bispecific antibody fragments. Proc Natl Acad Sci USA. Jul. 15, 1993; 90(14): 6444-6448. doi: 10.1073/pnas.90.14.6444. cited by applicant
Holm et al. Functional mapping and single chain construction of the anti-cytokeratin 8 monoclonal antibody TS1. Mol Immunol 44(6):1075-1084 (2007). cited by applicant
Holt et al. Anti-serum albumin domain antibodies for extending the half-lives of short-lived drugs. Protein Eng Des Sel 21(5):283-288 (2008). cited by applicant
Hopp et al. The effects of affinity and valency of an albumin-binding domain (ABD) on the half-life of a single-chain diabody-ABD fusion protein. Protein Eng. Des. Sel. 23(11):827-34 (2010). cited by applicant
Huck et al. Sequence of a human immunoglobulin gamma 3 heavy chain constant region gene: comparison with the other human C gamma genes. Nucl. Acids Res. 14:1779-89 (1986). cited by applicant
Huston et al. Protein engineering of antibody binding sites: recovery of specific activity in an anti-digoxin single-chain Fv analogue produced in *Escherichia coli*. PNAS USA 85(16):5879-5883 (1988). cited by applicant
Hutchinson et al. Mutagenesis at a specific position in a DNA sequence. J Biol Chem 253:6551-6560 (1978). cited by applicant
Izumoto et al. Phase II clinical trial of Wilms tumor 1 peptide vaccination for patients with recurrent glioblastoma multiforme. J Neurosurg 108:963-971 (2008). cited by applicant
Janssen letter—Submission under Rule 116 EPC. Submitted by Janssen Biotech to the European Patent Register on Mar. 18, 2016 in connection with their opposition to the EP2155783 patent (6 pages) (2016). cited by applicant
Jones et al. Replacing the complementarity-determining regions in a human antibody with those from a mouse. Nature 321:522-525 (1986). cited by applicant
Kabat et al. Identical V region amino acid sequences and segments of sequences in antibodies of different specificities. Relative contributions of VH and VL genes, minigenes, and complementarity-determining regions to binding of antibody-combining sites. J Immunol 147:1709-1719 (1991). cited by applicant
Kabat et al. Sequences of proteins of immunological interest. NIH Publ. No. 91-3242 1:647-669 (1991). cited by applicant
Kalos et al. T cells with chimeric antigen receptors have potent antitumor effects and can establish memory in patients with advanced leukemia. Sci Transl Med 3(95):95ra73 (2011). cited by applicant
Kojima et al. Molecular cloning and expression of megakaryocyte potentiating factor cDNA. J Biol Chem 270:21984-21990 (1995). cited by applicant
Konishi et al. B7-H1 expression on non-small cell lung cancer cells and its relationship with tumor-infiltrating lymphocytes and their PD-1 expression. Clin Cancer Res 10:5094-5100 (2004). cited by applicant
Škrlec et al. Non-immunoglobulin scaffolds: a focus on their targets. Trends in Biotechnol 33:408-418 (2015). cited by applicant
Laabi et al. The BCMA gene, preferentially expressed during B lymphoid maturation, is bidirectionally transcribed. Nucleic Acids Res 22(7):1147-1154 (1994). cited by applicant
Latchman et al. PD-L2 is a second ligand for PD-1 and inhibits T cell activation. Nat Immunol 2:261-268 (2001). cited by applicant
Le Gall et al. Immunosuppressive properties of anti-CD3 single-chain Fv and diabody. J Immunol Methods 285(1):111-127 (2004). cited by applicant
Li et al. Development of novel tetravalent anti-CD20 antibodies with potent antitumor activity. Cancer Res 68:2400-2408 (2008). cited by applicant

Liu et al. A New Format of Single Chain Tri-specific Antibody with Diminished Molecular Size Efficiently Induces Ovarian Tumor Cell Killing. Biotechnology Letters 27(22):1821-1827 (2005). cited by applicant
Liu et al. MGD011, a CD19 x CD3 Dual Affinity Re-Targeting Bi-specific Molecule Incorporating Extended Circulating Half-life for the Treatment of B-cell Malignancies. Clin Cancer Res 23(6):1506-1518 (epub 2016) (2017). cited by applicant
Lowman et al. Monovalent phage display: A method for selecting variant proteins from random libraries. Methods 3:205-216 (1991). cited by applicant
Lu et al. In vitro and in vivo antitumor effect of a trivalent bispecific antibody targeting ErbB2 and CD16. Cancer Biol Ther. 7(11):1744-1750 (2008). . cited by applicant
Lutterbuese et al. T cell-engaging BiTE antibodies specific for EGFR potently eliminate KRAS- and BRAF-mutated colorectal cancer cells. PNAS 107:12605-12610 (2007). cited by applicant
Maccallum et al. Antibody-antigen interactions: contact analysis and binding site topography. J Mol Biol. 262(5):732-745 (1996). cited by applicant
Mariuzza et al. The structural basis of antigen-antibody recognition. Annu Rev Biophys Biophys Chem 16:139-159 (1987). cited by applicant
Milone et al. Chimeric receptors containing CD137 signal transduction domains mediate enhanced survival of T cells and increased antileukemic efficacy in vivo. Mol Ther 17(8):1453-1464 (2009). cited by applicant
Mirsky et al. Antibody-Specific Model of Amino Acid Substitution for Immunological Inferences from Alignments of Antibody Sequences. Mol. Biol. Evol. 32(3):806-819 (2014). cited by applicant
Müller et al. Improved Pharmacokinetics of Recombinant Bispecific Antibody Molecules by Fusion to Human Serum Albumin. J. Biol. Chem. 282(17):12650-60 (2007). cited by applicant
Morea et al. Antibody modeling: implications for engineering and design. Methods 20(3):267-279 (2000). cited by applicant
Moschella et al. Unraveling cancer chemoimmunotherapy mechanisms by gene and protein expression profiling of responses to cyclophosphamide. Cancer Res 71(10):3528-3539 (2011). cited by applicant
Muller et al. Improving the pharmacokinetic properties of biologics by fusion to an anti-HSA shark VNAR domain. MAbs 4(6):673-685 (2012). cited by applicant
Mumtaz et al. Design of liposomes for circumventing the reticuloendothelial cells. Glycobiology 5:505-10 (1991). cited by applicant
Muul et al. Persistence and expression of the adenosine deaminase gene for 12 years and immune reaction to gene transfer components: long-term results of the first clinical gene therapy trial. Blood 101(7):2563-2569 (2003). cited by applicant
Muyldermans. Nanobodies: natural single-domain antibodies. Annu Rev Biochem 82:775-797 (2013). cited by applicant
Nacheva et al. Preventing nondesired RNA-primed RNA extension catalyzed by T7 RNA polymerase. Eur J Biochem 270:1458-1465 (2003). cited by applicant
Nazarian et al. Characterization of bispecific T-cell Engager (BiTE) antibodies with a high-capacity T-cell dependent cellular cytotoxicity (TDCC) assay. J Biomol Screen 20:519-527 (2015). cited by applicant
Needleman et al. A general method applicable to the search for similarities in the amino acid sequence of two proteins. J. Mol. Biol. 48:443-453 (1970). cited by applicant
Nelson et al. Antibody fragments Hope and Hype. mAbs 2(1):77-83 (2010). cited by applicant
Nicholson et al. Construction and characterisation of a functional CD19 specific single chain Fv fragment for immunotherapy of B lineage leukaemia and lymphoma. Mol Immun 34(16-17):1157-1165 (1997). cited by applicant
Nishikawa et al. Nonviral vectors in the new millennium: delivery barriers in gene transfer. Human Gene Therapy. 12:861-870 (2001). cited by applicant

Nunez-Prado et al. The coming of age of engineered multivalent antibodies. Drug Discovery Today 20(5):588-594 (2015). cited by applicant
Ohiro et al. A homogeneous and noncompetitive immunoassay based on the enhanced fluorescence resonance energy transfer by leucine zipper interaction. Anal Chem 74(22):5786-5792 (2002). cited by applicant
Ohtsuka et al. An alternative approach to deoxyoligonucleotides as hybridization probes by insertion of Deoxyinosine at Ambiguous Codon Positions. J Biol Chem 260(5):2605-2608 (Mar. 10, 1985). cited by applicant
O'Keefe et al. Chapter 18: Prostate specific membrane antigen. In: Chung L.W.K., Isaacs W.B., Simons J.W. (eds) Prostate Cancer. Contemporary Cancer Research. Humana Press, Totowa, NJ (pp. 307-326) (2001). cited by applicant
Ordonez. Application of mesothelin immunostaining in tumor diagnosis. Am J Surg Pathol 27:1418-1428 (2003). cited by applicant
Padlan. Anatomy of the Antibody Molecule. Mol Immunol 31(3):169-217 (1994). cited by applicant
Padlan, et al., A possible procedure for reducing the immunogenicity of antibody variable domains while preserving their ligand-binding properties. Mol Immunol 28(4-5):489-498 (1991). cited by applicant
Padlan et al. Structure of an antibody-antigen complex: Crystal structure of the HyHEL-10 Fab-lysozyme complex. PNAS USA 86:5938-5942 (1989). cited by applicant
Pawluczkowycz et al. Binding of submaximal C1q promotes complement-dependent cytotoxicity (CDC) of B cells opsonized with anti-CD20 mAbs ofatumumab (OFA) or rituximab (RTX): considerably higher levels of CDC are induced by OFA than by RTX. J Immunol 183:749-758 (2009). cited by applicant
PCT/US2016/033644 International Search Report and Written Opinion dated Sep. 6, 2016. cited by applicant
PCT/US2017/033665 International Search Report and Written Opinion dated Oct. 18, 2017. cited by applicant
PCT/US2017/033673 International Search Report and Written Opinion dated Oct. 18, 2017. cited by applicant
PCT/US2017/056530 International Search Report and Written Opinion dated Jan. 23, 2018. cited by applicant
PCT/US2017/063121 International Search Report and Written Opinion dated Mar. 26, 2018. cited by applicant
PCT/US2017/063126 International Search Report and Written Opinion dated Apr. 5, 2018. cited by applicant
PCT/US2018/014396 International Search Report and Written Opinion dated Jun. 14, 2018. cited by applicant
PCT/US2018/020185 International Search Report and Written Opinion dated Jun. 15, 2018. cited by applicant
PCT/US2018/020307 International Search Report and Written Opinion dated Aug. 24, 2018. cited by applicant
PCT/US2018/030983 International Search Report and Written Opinion dated Sep. 25, 2018. cited by applicant
PCT/US2018/032418 International Search Report and Written Opinion dated Sep. 24, 2018. cited by applicant
PCT/US2018/032427 International Search Report and Written Opinion dated Sep. 13, 2018. cited by applicant
PCT/US2018/055659 International Search Report and Written Opinion dated Feb. 21, 2019. cited by applicant

PCT/US2018/055682 International Search Report and Written Opinion dated Mar. 1, 2019. cited by applicant
PCT/US2019/032224 International Search Report and Written Opinion dated Aug. 28, 2019. cited by applicant
PCT/US2019/032302 International Search Report and Written Opinion dated Aug. 22, 2019. cited by applicant
PCT/US2019/032306 International Search Report and Written Opinion dated Aug. 22, 2019. cited by applicant
PCT/US2019/032307 International Search Report and Written Opinion dated Aug. 22, 2019. cited by applicant
PCT/US2019/052206 International Search Report and Written Opinion dated Feb. 14, 2020. cited by applicant
PCT/US2019/052270 International Search Report and Written Opinion dated Mar. 5, 2020. cited by applicant
PCT/US2019/053017 International Search Report and Written Opinion dated Jan. 31, 2020. cited by applicant
PCT/US/2020/032985 International Search Report and Written Opinion dated Oct. 15, 2020. cited by applicant
Pearson et al. Improved Tools for Biological Sequence Comparison. PNAS USA 85:2444-48 (1988). cited by applicant
Pedersen et al. Comparison of surface accessible residues in human and murine immunoglobulin Fv domains. Implication for humanization of murine antibodies. J Mol Biol 235(3):959-973 (1994). cited by applicant
Pfizer letter—Opposition to European Patent EP2155783 (Application 08735001.3). Submitted by Pfizer to the European Patent Register on Apr. 30, 2014 in connection with their opposition to the EP2155783 patent. (pp. 1-23 and Appendix 1 on pp. 24-26) (2014). cited by applicant
Porter et al. Chimeric antigen receptor T cells persist and induce sustained remissions in relapsed refractory chronic lymphocytic leukemia. Sci Trans Med 7(303):303ra319 (2015). cited by applicant
Porter et al. Chimeric antigen receptor-modified T cells in chronic lymphoid leukemia. NEJM 365:725-733 (2011). cited by applicant
Presta. Antibody Engineering. Curr Op Struct Biol 2:593-596 (1992). cited by applicant
Presta et al. Humanization of an antibody directed against IgE. J Immunol 151:2623-2632 (1993). cited by applicant
Ramadoss et al. An Anti-B Cell Maturation Antigen Bispecific Antibody for Multiple Myeloma. J. Ann. Chem. Soc. 137(16):5288-91 (2015). cited by applicant
Riechmann et al. Reshaping human antibodies for therapy. Nature, 332.6162:323-7 (1988). cited by applicant
Riechmann et al. Single domain antibodies: comparison of camel VH and camelised human VH domains. J Immunol Methods 231(1-2):25-38 (1999). cited by applicant
Roguska et al. A comparison of two murine monoclonal antibodies humanized by CDR-grafting and variable domain resurfacing. Protein Eng 9(10):895-904 (1996). cited by applicant
Roguska et al. Humanization of murine monoclonal antibodies through variable domain resurfacing. PNAS 91:969-973 (1994). cited by applicant
Rosenberg et al. Use of tumor-infiltrating lymphocytes and interleukin-2 in the immunotherapy of patients with metastatic melanoma. A preliminary report. NEJM 319:1676 (1988). cited by applicant
Rosok et al. A Combinatorial Library Strategy for the Rapid Humanization of Anticarcinoma BR96 Fab. J Biol Chem 271:22611-22618 (1996). cited by applicant
Rossolini et al. Use of deoxyinosine-containing primers vs degenerate primers for polymerase

chain reaction based on ambiguous sequence information. Mol Cell Probes 8(2):91-98 (1994). cited by applicant
Rozan et al. Single-domain antibody-based and linker-free bispecific antibodies targeting FcyRIII induce potent antitumor activity without recruiting regulatory T cells. Mol Cancer Ther 12(8):1481-1491 (2013). cited by applicant
Rudikoff et al. Single amino acid substitution altering antigen-binding Specificity. PNAS USA 79:1979-1983 (1982). cited by applicant
Rump et al. Binding of ovarian cancer antigen CA125/MUC16 to mesothelin mediates cell adhesion. J Biol Chem 279:9190-9198 (2004). cited by applicant
Running Deer et al. High-level expression of proteins in mammalian cells using transcription regulatory sequences from the Chinese hamster EF-1alpha gene. Biotechnol Prog. 20:880-889 (2004). cited by applicant
Sadelain et al. Targeting tumours with genetically enhanced T lymphocytes. Nat Rev Cancer 3(1):35-45 (2003). cited by applicant
Sadelain et al. The basic principles of chimeric antigen receptor design. Cancer Discov. 3(4):388-98 (2013). cited by applicant
Saerens et al. Identification of a universal VHH framework to graft non-canonical antigen-binding loops of camel single-domain antibodies. J. Mol. Biol. 352(3):597-607 (2005). cited by applicant
Sandhu. A rapid procedure for the humanization of monoclonal antibodies. Gene 150(2):409-410 (1994). cited by applicant
Sastry et al. Targeting hepatitis B virus-infected cells with a T-cell receptor-like antibody. J Virol 85(5):1935-1942 (2011). cited by applicant
Schenborn et al. A novel transcription property of SP6 and T7 RNA polymerases: dependence on template structure. Nuc Acids Res 13:6223-6236 (1985). cited by applicant
Scheraga. Predicting three-dimensional structures of oligopeptides. Rev Computational Chem 3:73-142 (1992). cited by applicant
Schmidt et al. Cloning and Characterization of Canine Prostate-Specific Membrane Antigen. The Prostate 73:642-650 (2013). cited by applicant
Schmittgen et al. Expression of prostate specific membrane antigen and three alternatively spliced variants of PSMA in prostate cancer patients. Int J Cancer 107:323-329 (2003). cited by applicant
Sergeeva et al. An anti-PR1/HLA-A2 T-cell receptor-like antibody mediates complement-dependent cytotoxicity against acute myeloid leukemia progenitor cells. Blood 117(16):4262-4272 (2011). cited by applicant
Sims et al. A humanized CD18 antibody can block function without cell destruction. J Immunol. 151:2296-2308 (1993). cited by applicant
Smirnova et al. Identification of new splice variants of the genes BAFF and BCMA. Mol. Immunol. 45 (4):1179-83 (2008). cited by applicant
Smith et al. Comparison of Biosequences. Advances in Applied Mathematics. 2:482-489 (1981). cited by applicant
Song et al. CD27 costimulation augments the survival and antitumor activity of redirected human T cells in vivo. Blood 119(3):696-706 (2012). cited by applicant
Spiess et al. Alternative molecular formats and therapeutic applications for bispecific antibodies. Mol. Immunol. 67(2 Pt A):95-106 (2015). cited by applicant
Stepinski et al. Synthesis and properties of mRNAs containing the novel ‘anti-reverse’ cap analogs 7-methyl(3′0-methyl)GpppG and 7-methyl(e'-deoxy)GpppG. RNA 7:1486-1495 (2001). cited by applicant
Sternjak et al. Cancer Research, (Jul. 2017) vol. 77, No. 13, Supp. Supplement 1. Abstract No. 3630. Meeting Info: American Association for Cancer Research Annual Meeting 2017. Washington, DC, United States. Apr. 1, 2017-Apr. 5, 2017. cited by applicant
Stork et al. A novel tri-functional antibody fusion protein with improved pharmacokinetic

properties generated by fusing a bispecific single-chain diabody with an albumin-binding domain from streptococcal protein G. Protein Eng. Des. Sel. 20(11):569-76 (2007). cited by applicant
Strop. Veracity of microbial transglutaminase. Bioconjugate Chem. 25(5):855-862 (2014). cited by applicant
Studnicka et al. Human-engineered monoclonal antibodies retain full specific binding activity by preserving non-CDR complementarity-modulating residues. Pro Eng 7(6):805-814 (1994). cited by applicant
Su et al. PSMA specific single chain antibody-mediated targeted knockdown of Notch1 inhibits human prostate cancer cell proliferation and tumor growth. Cancer Lett. 338 (2): 282-291 (2013). cited by applicant
Tan et al. Influence of the hinge region on complement activation, C1q binding, and segmental flexibility in chimeric human immunoglobulins. PNAS USA 87:162-166 (1990). cited by applicant
Tan et al. “Superhumanized” antibodies: reduction of immunogenic potential by complementarity-determining region grafting with human germline sequences: application to an anti-CD28. J Immunol 169:1119-1125 (2002). cited by applicant
Tang et al. A human single-domain antibody elicits potent antitumor activity by targeting an epitope in mesothelin close to the cancer cell surface. Mol. Cancer Thera 12(4):416-426 (2013). cited by applicant
Tassev et al. Retargeting NK92 cells using an HLA-A2-restricted, EBNA3C-specific chimeric antigen receptor. Cancer Gene Ther 19(2):84-100 (2012). cited by applicant
Ten Berg et al. Selective expansion of a peripheral blood CD8+ memory T cell subset expressing both granzyme B and L-selectin during primary viral infection in renal allograft recipients. Transplant Proc 30(8):3975-3977 (1998). cited by applicant
Thomas et al. Mesothelin-specific CD8(+) T cell responses provide evidence of in vivo cross-priming by antigen-presenting cells in vaccinated pancreatic cancer patients. J Exp Med 200:297-306 (2004). cited by applicant
Tijink et al. Improved tumor targeting of anti-epidermal growth factor receptor nanobodies through albumin binding: taking advantage of modular Nanobody technology. Mol. Cancer Ther. 7(8):2288-97 (2008). cited by applicant
Tiller et al. Facile Affinity Maturation of Antibody Variable Domains Using Natural Diversity Mutagenesis. Front. Immunol. 8:986 (2017). cited by applicant
Tutt et al. Trispecific F(ab')3 derivatives that use cooperative signaling via the TCR/CD3 complex and CD2 to activate and redirect resting cytotoxic T cells. J Immunol. 147(1):60-69 (Jul. 1, 1991). cited by applicant
Ui-Tei et al. Sensitive assay of RNA interference in *Drosophila* and Chinese hamster cultured cells using firefly luciferase gene as target. Febs Letters 479: 79-82 (2000). cited by applicant
U.S. Appl. No. 15/160,984 Office Action dated Feb. 24, 2017. cited by applicant
U.S. Appl. No. 15/160,984 Office Action dated Sep. 22, 2016. cited by applicant
U.S. Appl. No. 15/600,264 Office Action dated Apr. 25, 2019. cited by applicant
U.S. Appl. No. 15/600,264 Office Action dated Apr. 26, 2018. cited by applicant
U.S. Appl. No. 15/600,264 Office Action dated Nov. 27, 2018. cited by applicant
U.S. Appl. No. 15/600,264 Office Action dated Oct. 3, 2017. cited by applicant
U.S. Appl. No. 15/600,582 Office Action dated Nov. 16, 2017. cited by applicant
U.S. Appl. No. 15/630,259 Office Action dated Dec. 30, 2019. cited by applicant
U.S. Appl. No. 15/630,259 Office Action dated Sep. 30, 2020. cited by applicant
U.S. Appl. No. 15/704,620 Office Action dated Oct. 26, 2017. cited by applicant
U.S. Appl. No. 15/821,498 Office Action dated Apr. 21, 2020. cited by applicant
U.S. Appl. No. 15/821,498 Office Action dated May 3, 2019. cited by applicant
U.S. Appl. No. 15/821,498 Office Action dated Oct. 26, 2018. cited by applicant
U.S. Appl. No. 15/821,530 Office Action dated Apr. 22, 2020. cited by applicant

U.S. Appl. No. 15/821,530 Office Action dated Apr. 3, 2019. cited by applicant
U.S. Appl. No. 15/821,530 Office Action dated Sep. 25, 2018. cited by applicant
U.S. Appl. No. 15/977,968 Office Action dated Feb. 21, 2019. cited by applicant
U.S. Appl. No. 15/977,988 Office Action dated Aug. 20, 2019. cited by applicant
U.S. Appl. No. 15/977,988 Office Action dated Mar. 26, 2019. cited by applicant
U.S. Appl. No. 15/977,988 Pre-Interview First Office Action dated Jan. 25, 2019. cited by applicant
U.S. Appl. No. 16/159,545 Office Action dated Aug. 6, 2019. cited by applicant
U.S. Appl. No. 16/159,545 Office Action dated Dec. 2, 2019. cited by applicant
U.S. Appl. No. 16/159,554 Office Action dated Jun. 7, 2019. cited by applicant
U.S. Appl. No. 16/159,554 Office Action dated Oct. 1, 2019. cited by applicant
U.S. Appl. No. 16/159,554 Office Action dated Oct. 5, 2020. cited by applicant
U.S. Appl. No. 16/583,070 Office Action dated Mar. 3, 2020. cited by applicant
Vajdos et al. Comprehensive functional maps of the antigen-binding site of an anti-ErbB2 antibody obtained with shotgun scanning mutagenesis. J Mol Biol 320:415-428 (2002). cited by applicant
Van Den Beuchken et al. Building novel binding ligands to B7.1 and B7.2 based on human antibody single variable light chain domains. J Mol biol 310:591-601 (2001). cited by applicant
Van Der Linden et al. Induction of immune responses and molecular cloning of the heavy chain antibody repertoire of Lama glama. J Immunol Methods 240:185-195 (2000) . cited by applicant
Vaughan et al. Human antibodies by design. Nature Biotech 16:535-539 (1998). cited by applicant
Verhoeyen et al. Reshaping human antibodies: Grafting an antilysozyme activity. Science 239:1534-1536 (1988). cited by applicant
Verma et al. TCR mimic monoclonal antibody targets a specific peptide/HLA class I complex and significantly impedes tumor growth in vivo using breast cancer models. J Immunol 184(4):2156-2165 (2010). cited by applicant
Vincke et al. General strategy to humanize a camelid single-domain antibody and identification of a universal humanized nanobody scaffold. J. Biol. Chem. 284(5):3273-3284 (2009). cited by applicant
Wang et al. A New Recombinant Single Chain Trispecific Antibody Recruits T Lymphocytes to Kill CEA (Carcinoma Embryonic Antigen) Positive Tumor Cells In Vitro Efficiently. Journal of Biochemistry 135(4):555-565 (2004). cited by applicant
Willemsen et al. A phage display selected fab fragment with MHC class I-restricted specificity for MAGE-A1 allows for retargeting of primary human T lymphocytes. Gene Ther 8(21):1601-1608 (2001). cited by applicant
Winkler et al. Changing the antigen binding specificity by single point mutations of an anti-p24 (HIV-1) antibody. J Immunol. 165(8):4505-4514 (2000). cited by applicant
Wu et al. Humanization of a murine monoclonal antibody by simultaneous optimization of framework and CDR residues. J.Mol.Biol. 294:151-162 (1999). cited by applicant
Yan et al. Engineering upper hinge improves stability and effector function of a human IgG1. J. Biol. Chem. 287:5891 (2012). cited by applicant
Yee et al. Adoptive T cell therapy using antigen-specific CD8+ T cell clones for the treatment of patients with metastatic melanoma: in vivo persistence, migration, and antitumor effect of transferred T cells. PNAS USA 99(25):16168-16173 (2002). cited by applicant
Yoshinaga et al. Ig L-chain shuffling for affinity maturation of phage library-derived human anti-human MCP-1 antibody blocking its chemotactic activity. J Biochem 143(5):593-601 (2008). cited by applicant
Yu et al. Rationalization and design of the complementarity determining region sequences in an antibody-antigen recognition interface. PLoS One 7(3):e33340 (2012). cited by applicant
Zabetakis et al. Contributions of the complementarity determining regions to the thermal stability of a single-domain antibody. PLoS One 8(10):e77678 (2013). cited by applicant
Zare et al. Production of nanobodies against prostate-specific membrane antigen (PSMA)

recognizing LnCaP cells. Int. J. Biol. Markers 29(2):e169-e179 (2014). cited by applicant
Zhang et al. New High Affinity Monoclonal Antibodies Recognize Non-Overlapping Epitopes on Mesothelin for Monitoring and Treating Mesothelioma. Sci Rep 5:9928 (2015). cited by applicant
Zhu et al. Combody: one-domain antibody multimer with improved avidity. Immunology and Cell Biology 88(6):667-675 (2010). cited by applicant
Chen et al. Preparation and characterization of dexamethasone acetate-loaded solid lipid nanoparticles. Chinese J Pharm 39(4):261-264 (2008) (English abstract). cited by applicant
Dondelinger et al., Understanding the Significance and Implications of Antibody Numbering and Antigen-Binding Surface/Residue Definition. Frontiers in Immunology 9(2278):1-15 (2018). cited by applicant
GenBank AHA34196.1, immunoglobulin variable heavy regionJIY-F10 RTA-F10, partial [Vicugna pacos] (Nov. 18, 2013). cited by applicant
Hu et al. Over-expression of human Notch ligand Delta-like 3 promotes proliferation of human gastric cancer cells in vitro. Nan Fang Yi Ke Da Xue Xue Bao 38(1):14-19 (2018) (English Abstract). cited by applicant
U.S. Appl. No. 17/165,760 Office Action dated Jan. 19, 2024. cited by applicant
U.S. Appl. No. 17/276,796 Office Action dated Feb. 1, 2024. cited by applicant
Zhao et al. Novel Antibody Therapeutics Targeting Mesothelin In Solid Tumors. Clin Cancer Drugs 3(2):76-86 (2016). cited by applicant
Jiang et al. Protritac: A protease cleavable T cell Engager Platform. Scientific Reports 6(Suppl 1):115 Available at https://calidibio.com/wp-content/uploads/2019/10/609-Abstract-_ SITC-2018.pdf (2018). cited by applicant
Lin et al. ProTriTAC: A Protease-Activatable T Cell Engager Platform that Links Half-Life Extension to Functional Masking Society for Immunotherapy of Cancer (SITC) Annual Meeting. Nov. 2018, Available at https://www.harpoontx.com/file.cfm/43/docs/SITC_2018_ProTriTAC_Poster.pdf. cited by applicant
Balzar et al. Epidermal growth factor-like repeats mediate lateral and reciprocal interactions of Ep-CAM molecules in homophilic adhesions. Mol Cell Biol. 21(7):2570-80 (2001). cited by applicant
Balzar et al. The biology of the 17-1A antigen (Ep-CAM). J. Mol. Med. 77:699-712 (1999). cited by applicant
Brauchle et al. Characterization of a Novel FLT3 Bite Molecule for the Treatment of Acute Myeloid Leukemia. Mol Cancer Ther 19:1875-88 (2020). cited by applicant
Chaubal et al. Ep-CAM—a marker for the detection of disseminated tumor cells in patients suffering from SCCHN. Anticancer Res 19:2237-2242 (1999). cited by applicant
Eyvazi et al. Antibody Based EpCAM Targeted Therapy of Cancer, Review and Update. Curr Cancer Drug Targets. 18(9):857-868 (2018). cited by applicant
Gastl et al. Ep-CAM overexpression in breast cancer as a predictor of survival. Lancet. 356:1981-1982 (2000). cited by applicant
Goettlinger et al. The epithelial cell surface antigen 17-1A, a target for antibody-mediated tumor therapy: its biochemical nature, tissue distribution and recognition by different monoclonal antibodies. Int J Cancer. 38:47-53 (1986). cited by applicant
Julian et al. Efficient affinity maturation of antibody variable domains requires co-selection of compensatory mutations to maintain thermodynamic stability. Sci Rep 7:45259 (2017). cited by applicant
Kim et al. Strategies and 1-17 Advancement in Antibody-Drug Conjugate Optimization for Targeted Cancer Therapeutics. Biomol Ther (Seoul) 23(6):493-509 (2015). cited by applicant
Koprowski et al. Colorectal carcinoma antigens detected by hybridoma antibodies. Somatic Cell Genet. 5:957-971 (1979). cited by applicant
Krzywinska et al. CD45 Isoform Profile Identifies Natural Killer (NK) Subsets with Differential

Activity. PLoS One 11(4):e0150434 (2016). cited by applicant
Leibl et al. Ovarian granulosa cell tumors frequently express EGFR (Her-1), Her-3, and Her-4: An immunohistochemical study. Gynecol Oncol 101(1):18-23 (2006). cited by applicant
Litvinov et al. Epithelial cell adhesion molecule (Ep-CAM) modulates cell-cell interactions mediated by classic cadherins. J Cell Biol. 139:1337-1348 (1997). cited by applicant
Litvinov et al. Expression of Ep-CAM in cervical squamous epithelia correlates with an increased proliferation and the disappearance of markers for terminal differentiation. Am. J. Pathol. 148:865-75 (1996). cited by applicant
Lucchi et al. The Masking Game: Design of Activatable Antibodies and Mimetics for Selective Thera-peutics Cell Control. ACS Cent Sci 7(5):724-738 (2021). cited by applicant
Mason et al. CD79a: a novel marker for B-cell neoplasms in routinely processed tissue samples. Blood 86(4):1453-1459 (1995). cited by applicant
Osta et al. EpCAM is overexpressed in breast cancer and is a potential target for breast cancer gene therapy. Cancer Res 64:5818-24 (2004). cited by applicant
PCT/US2020/060184 International Search Report and Written Opinion dated Mar. 4, 2021. cited by applicant
Piyathilake et al. The expression of Ep-CAM (17-1A) in squamous cell cancers of the lung. Hum Pathol. 31:482-487 (2000). cited by applicant
Poczatek et al. Ep-Cam levels in prostatic adenocarcinoma and prostatic intraepithelial neoplasia. J Urol. 162:1462-1464 (1999). cited by applicant
Quak et al. Production of a monoclonal antibody (K 931) to a squamous cell carcinoma associated antigen identified as the 17-1A antigen. Hybridoma 9:377-387 (1990). cited by applicant
Sandler et al. Nondermatologic adverse events associated with anti-EGFR therapy. Oncology (Williston Park) 20(5 Suppl 2):35-40 (2006). cited by applicant
Sheng et al. Novel Transgenic Mouse Model for Studying Human Serum Albumin as a Biomarker of Carcinogenic Exposure. Chem. Res. Toxicol. 29(5):797-809 (2016). cited by applicant
Simon et al. Epithelial glycoprotein is a member of a family of epithelial cell surface antigens homologous to nidogen, a matrix adhesion protein. PNAS USA 87:2755-2759 (1990). cited by applicant
Stehle et al. Albumin-based drug carriers: comparison between serum albumins of different species on pharmacokinetics and tumor uptake of the conjugate. Anticancer Drugs. 10(8):785-90 (1999). cited by applicant
Stirewalt et al. The role of FLT3 in haematopoietic malignancies. Nat Rev Cancer 3:650-665 (2003). cited by applicant
Thomas. Cetuximab: adverse event profile and recommendations for toxicity management. Clin J Oncol Nurs. 9(3):332-8 (2005). cited by applicant
Trail et al. Antibody drug 1-17 conjugates for treatment of breast cancer: Novel targets and diverse approaches in ADC design. Pharmacol Ther 181:126-142 (2018). cited by applicant
Trebak et al. Oligomeric state of the colon carcinoma-associated glycoprotein GA733-2 (Ep-CAM/EGP40) and its role in GA733-mediated homotypic cell-cell adhesion. J Biol Chem. 276:2299-2309 (2001). cited by applicant
U.S. Appl. No. 16/159,554 Office Action dated Mar. 16, 2021. cited by applicant
Lin, S.. Jack et al., ProTriTAC: A Protease-Activatable T Cell Engager Platform that Links Half-Life Extension to Functional Masking Society for Immunotherapy of Cancer, (SITC) Annual Meeting, Abstract P608, 2 pages, 2018. cited by applicant
Lucchi, Roberta et al., The Masking Game: Design of Activatable Antibodies and Mimetics for Selective Therapeutics and Cell Control, ACS Cent. Sci., 7, 724-738, 2021. cited by applicant

Primary Examiner: Boesen; Agnieszka

Background/Summary

CROSS-REFERENCE (1) This application is the U.S. National Stage Application of International Application No. PCT/US2019/032307, filed May 14, 2019, and claims the benefit of U.S. Provisional Application Nos. 62/671,344, filed May 14, 2018; 62/671,349, filed May 14, 2018; 62/756,429, filed Nov. 6, 2018; and 62/756,453 filed Nov. 6, 2018, all of which are incorporated by reference herein in their entirety.

INCORPORATION BY REFERENCE
(1) All publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference, and as if set forth in their entireties.
SEQUENCE LISTING
(2) The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Dec. 6, 2023, is named 47517-729_831_SL.txt and is 1,222,504 bytes in size.
BACKGROUND OF THE INVENTION
(3) T cell engagers transiently tether T cells to tumor cells and mediate T cell-directed tumor killing. T cell engagers, such as blinatumomab (BLINCYTO®), have demonstrated clinical activity in several hematological malignancies. Adoption of T cell engagers in solid tumors is limited by the scarcity of tumor antigens with sufficient differential expression between tumor and normal tissue. T cell engagers that are preferentially active in the tumor microenvironment may enable the safe targeting of more solid tumor antigens.
(4) There is a need to extend the half-life of a therapeutic, diagnostic, or imaging molecule in circulation and also improve its ability to reach its target within an intended location (e.g., a tumor cell) without non-specific binding.
SUMMARY OF THE INVENTION
(5) One embodiment provides a binding moiety comprising a non-CDR loop and a cleavable linker, wherein the moiety is capable of masking the binding of binding molecule to its target, wherein the binding molecule comprises an immunoglobulin molecule or a non-immunoglobulin molecule. In some embodiments, the moiety is a natural peptide, a synthetic peptide, an engineered scaffold, or an engineered bulk serum protein. In some embodiments, the engineered scaffold comprises a sdAb, a scFv, a Fab, a VHH, a fibronectin type III domain, immunoglobulin-like scaffold, DARPin, cystine knot peptide, lipocalin, three-helix bundle scaffold, protein G-related albumin-binding module, or a DNA or RNA aptamer scaffold. In some embodiments, the moiety is capable of binding to a bulk serum protein. In some embodiments, the non-CDR loop is from a variable domain, a constant domain, a C1-set domain, a C2-set domain, an I-domain, or any combinations thereof. In some embodiments, the moiety further comprises complementarity determining regions (CDRs). In some embodiments, the moiety is capable of binding to the bulk serum protein. In some embodiments, the bulk serum protein is a half-life extending protein. In some embodiments, the bulk serum protein is albumin, transferrin, IgG1, IgG2, IgG4, IgG3, IgA monomer, Factor XIII, Fibrinogen, IgE, or pentameric IgM. In some embodiments, the bulk serum protein is albumin, transferrin, Factor XIII, or Fibrinogen. In some embodiments, the CDRs within the binding moiety provide binding site specific for the bulk serum protein. In some embodiments, the binding moiety is capable of masking the binding of the target antigen binding domain (such as an immunoglobulin

molecule) or a non-immunoglobulin binding molecule to its target via specific intermolecular interactions between the binding moiety and the target antigen binding domain or the non-immunoglobulin binding moiety. In some embodiments, the non-CDR loop within the binding moiety provides a binding site specific for binding of the binding moiety to the target antigen binding domain (such as an immunoglobulin molecule) or the non-immunoglobulin binding molecule.

(6) In some embodiments, the binding moiety comprises a binding site specific for an immunoglobulin light chain. In some embodiments, the immunoglobulin light chain is an Igκ free light chain. In some embodiments, the CDRs provide the binding site specific for the bulk serum protein or the immunoglobulin light chain. In some embodiments, the immunoglobulin molecule is a target antigen binding domain. In some embodiments, the moiety is bound to the target antigen binding domain. In some embodiments, the moiety is covalently linked to the target antigen binding domain. In some embodiments, the moiety is capable of masking the binding of the target antigen binding domain to its target via specific intermolecular interactions between the binding moiety and the target antigen binding domain. In some embodiments, the non-CDR loop provides a binding site specific for binding of the moiety to the target antigen binding domain. In some embodiments, upon cleavage of the cleavable linker, the binding moiety is separated from the target antigen binding domain and the target antigen binding domain binds to its target. In some embodiments, the target antigen domain binds to a tumor antigen. In some embodiments, the tumor antigen comprises EpCAM, EGFR, HER-2, HER-3, c-Met, FoIR, PSMA, CD38, BCMA, and CEA. 5T4, AFP, B7-H3, CDH-6, CAIX, CD117, CD123, CD138, CD166, CD19, CD20, CD205, CD22, CD30, CD33, CD352, CD37, CD44, CD52, CD56, CD70, CD71, CD74, CD79b, DLL3, EphA2, FAP, FGFR2, FGFR3, GPC3, gpA33, FLT-3, gpNMB, HPV-16 E6, HPV-16 E7, ITGA2, ITGA3, SLC39A6, MAGE, mesothelin, Muc1, Muc16, NaPi2b, Nectin-4, CDH-3, CDH-17, EPHB2, ITGAV, ITGB6, NY-ESO-1, PRLR, PSCA, PTK7, ROR1, SLC44A4, SLITRK5, SLITRK6, STEAP1, TIM1, Trop2, or WT1. In some embodiments, the target antigen domain binds to an immune checkpoint protein. In some embodiments, the immune checkpoint protein is CD27, CD137, 2B4, TIGIT, CD155, ICOS, HVEM, CD40L, LIGHT, OX40, DNAM-1, PD-L1, PD1, PD-L2, CTLA-4, CD8, CD40, CEACAM1, CD48, CD70, A2AR, CD39, CD73, B7-H3, B7-H4, BTLA, IDO1, IDO2, TDO, KIR, LAG-3, TIM-3, or VISTA. In some embodiments, the target antigen binding domain binds to a T-cell. In some embodiments, the target antigen binding domain binds to CD3. In some embodiments, the cleavable linker comprises a cleavage site. In some embodiments, the cleavage site is recognized by a protease. In some embodiments, the protease cleavage site is recognized by a serine protease, a cysteine protease, an aspartate protease, a threonine protease, a glutamic acid protease, a metalloproteinase, a gelatinase, or a asparagine peptide lyase. In some embodiments, the protease cleavage site is recognized by a Cathepsin B, a Cathepsin C, a Cathepsin D, a Cathepsin E, a Cathepsin K, a Cathepsin L, a kallikrein, a hK1, a hK10, a hK15, a plasmin, a collagenase, a Type IV collagenase, a stromelysin, a Factor Xa, a chymotrypsin-like protease, a trypsin-like protease, a elastase-like protease, a subtilisin-like protease, an actinidain, a bromelain, a calpain, a caspase, a caspase-3, a Mir1-CP, a papain, a HIV-1 protease, a HSV protease, a CMV protease, a chymosin, a renin, a pepsin, a matriptase, a legumain, a plasmepsin, a nepenthesin, a metalloexopeptidase, a metalloendopeptidase, a matrix metalloprotease (MMP), a MMP1, a MMP2, a MMP3, a MMP7, a MMP8, a MMP9, a MMP10, a MMP11, a MMP12, a MMP13, a MMP14, an ADAM9, an ADAM10, an ADAM12, an urokinase plasminogen activator (uPA), an enterokinase, a prostate-specific target (PSA, hK3), an interleukin-1β converting enzyme, a thrombin, a FAP (FAP-α), a dipeptidyl peptidase, a type II transmembrane serine protease (TTSP), a neutrophil elastase, a cathepsin G, a proteinase 3, a neutrophil serine protease 4, a mast cell chymase, and a mast cell tryptase.

(7) One embodiment provides a conditionally active binding protein comprising a binding moiety (M) which comprises a non-CDR loop, a cleavable linker (L), a first target antigen binding domain

(T1), and a second target antigen binding domain (T2), wherein the first target antigen binding domain (T1) comprises an immunoglobulin molecule, wherein the non-CDR loop is capable of binding to the first target antigen binding domain, and wherein the binding moiety is capable of masking the binding of the first target antigen binding domain to its target. In some embodiments, the binding moiety is capable of binding to a half-life extending protein. In some embodiments, the binding moiety is a natural peptide, a synthetic peptide, an engineered scaffold, or an engineered serum bulk protein. In some embodiments, the engineered scaffold comprises a sdAb, a scFv, a Fab, a VHH, a fibronectin type III domain, immunoglobulin-like scaffold, DARPin, cystine knot peptide, lipocalin, three-helix bundle scaffold, protein G-related albumin-binding module, or a DNA or RNA aptamer scaffold. In some embodiments, the non-CDR loop is from a variable domain, a constant domain, a C1-set domain, a C2-set domain, an I-domain, or any combinations thereof. In some embodiments, the binding moiety further comprises complementarity determining regions (CDRs). In some embodiments, the binding moiety comprises a binding site specific for a bulk serum protein. In some embodiments, the bulk serum protein is albumin, transferrin, IgG1, IgG2, IgG4, IgG3, IgA monomer, Factor XIII, Fibrinogen, IgE, or pentameric IgM. In some embodiments, the binding moiety further comprises a binding site specific for an immunoglobulin light chain. In some embodiments, the immunoglobulin light chain is an Igκ free light chain. T In some embodiments, the CDRs provide the binding site specific for the bulk serum protein or the immunoglobulin light chain, or any combinations thereof. In some embodiments, the binding moiety is capable of masking the binding of the first target antigen binding domain to its target via specific intermolecular interactions between the binding moiety and the first target antigen binding domain. In some embodiments, the non-CDR loop provides a binding site specific for binding of the binding moiety to the first target antigen binding domain. In some embodiments, the first or the second target antigen binding domain binds to a tumor antigen). In some embodiments, the tumor antigen comprises at least one of. EpCAM (exemplary protein sequence comprises UniProtkB ID No. P16422), EGFR (exemplary protein sequence comprises UniProtkB ID No. P00533), HER-2(exemplary protein sequence comprises UniProtkB ID No. P04626), HER-3(exemplary protein sequence comprises UniProtkB ID No. P21860), c-Met (exemplary protein sequence comprises UniProtkB ID No. P08581), FoIR (exemplary protein sequence comprises UniProtkB ID No. P15238), PSMA (exemplary protein sequence comprises UniProtkB ID No. Q04609), CD38 (exemplary protein sequence comprises UniProtkB ID No. P28907), BCMA (exemplary protein sequence comprises UniProtkB ID No. Q02223), and CEA (exemplary protein sequence comprises UniProtkB ID No. P06731, 5T4 (exemplary protein sequence comprises UniProtkB ID No. Q13641), AFP (exemplary protein sequence comprises UniProtkB ID No. P02771), B7-H3 (exemplary protein sequence comprises UniProtkB ID No. Q5ZPR3), CDH-6 (exemplary protein sequence comprises UniProtkB ID No. P97326), CAIX (exemplary protein sequence comprises UniProtkB ID No. Q16790), CD117 (exemplary protein sequence comprises UniProtkB ID No. P10721), CD123 (exemplary protein sequence comprises UniProtkB ID No. P26951), CD138 (exemplary protein sequence comprises UniProtkB ID No. P18827), CD166 (exemplary protein sequence comprises UniProtkB ID No. Q13740), CD19 (exemplary protein sequence comprises UniProtkB ID No. P15931), CD20 (exemplary protein sequence comprises UniProtkB ID No. P11836), CD205 (exemplary protein sequence comprises UniProtkB ID No. 060449), CD22 (exemplary protein sequence comprises UniProtkB ID No. P20273), CD30 (exemplary protein sequence comprises UniProtkB ID No. P28908), CD33 (exemplary protein sequence comprises UniProtkB ID No. P20138), CD352 (exemplary protein sequence comprises UniProtkB ID No. Q96DU3), CD37 (exemplary protein sequence comprises UniProtkB ID No. P11049), CD44 (exemplary protein sequence comprises UniProtkB ID No. P16070), CD52 (exemplary protein sequence comprises UniProtkB ID No. P31358), CD56 (exemplary protein sequence comprises UniProtkB ID No. P13591), CD70 (exemplary protein sequence comprises UniProtkB ID No. P32970), CD71 (exemplary protein sequence comprises UniProtkB ID No. P02786), CD74 (exemplary protein sequence comprises UniProtkB ID No. P04233), CD79b (exemplary protein sequence comprises UniProtkB ID No. P40259), DLL3 (exemplary protein sequence comprises UniProtkB ID No. Q9NYJ7), EphA2 (exemplary protein sequence comprises UniProtkB ID No. P29317), FAP (exemplary protein sequence comprises UniProtkB ID No. Q12884), FGFR2 (exemplary protein sequence comprises UniProtkB ID No. P21802), FGFR3 (exemplary protein sequence comprises UniProtkB ID No. P22607), GPC3 (exemplary protein sequence comprises UniProtkB ID No. P51654), gpA33 (exemplary protein sequence comprises UniProtkB ID No. Q99795), FLT-3 (exemplary protein sequence comprises UniProtkB ID No. P36888), gpNMB (exemplary protein sequence comprises UniProtkB ID No. Q14956), HPV-16 E6 (exemplary protein sequence comprises UniProtkB ID No. P03126), HPV-16 E7 (exemplary protein sequence comprises UniProtkB ID No. P03129), ITGA2 (exemplary protein sequence comprises UniProtkB ID No. P17301), ITGA3 (exemplary protein sequence comprises UniProtkB ID No. P26006), SLC39A6 (exemplary protein sequence comprises UniProtkB ID No. Q13433), MAGE (exemplary protein sequence comprises UniProtkB ID No. Q9HC15), mesothelin (exemplary protein sequence comprises UniProtkB ID No. Q13421), Muc1 (exemplary protein sequence comprises UniProtkB ID No. P15941), Muc16 (exemplary protein sequence comprises UniProtkB ID No. Q8WX17), NaPi2b (exemplary protein sequence comprises UniProtkB ID No. 095436), Nectin-4 (exemplary protein sequence comprises UniProtkB ID No. Q96918), CDH-3 (exemplary protein sequence comprises UniProtkB ID No. Q8WX17), CDH-17 (exemplary protein sequence comprises UniProtkB ID No. E5RJT3), EPHB2 (exemplary protein sequence comprises UniProtkB ID No. P29323), ITGAV (exemplary protein sequence comprises UniProtkB ID No. P06756), ITGB6 (exemplary protein sequence comprises UniProtkB ID No. P18564), NY-ESO-1 (exemplary protein sequence comprises UniProtkB ID No. P78358), PRLR (exemplary protein sequence comprises UniProtkB ID No. P16471), PSCA (exemplary protein sequence comprises UniProtkB ID No. 043653), PTK7 (exemplary protein sequence comprises UniProtkB ID No. Q13308), ROR1 (exemplary protein sequence comprises UniProtkB ID No. Q01973), SLC44A4 (exemplary protein sequence comprises UniProtkB ID No. Q53GD3), SLITRK5 (exemplary protein sequence comprises UniProtkB ID No. Q8IW52), SLITRK6 (exemplary protein sequence comprises UniProtkB ID No. Q9HY7), STEAP1 (exemplary protein sequence comprises UniProtkB ID No. Q9UHE8), TIM1 (exemplary protein sequence comprises UniProtkB ID No. Q96D42), Trop2 (exemplary protein sequence comprises UniProtkB ID No. P09758), or WT1 (exemplary protein sequence comprises UniProtkB ID No. P19544), or any combinations thereof. In some embodiments, the first or the second target antigen binding domain binds to an immune checkpoint protein. In some embodiments, the immune checkpoint protein is at least one of: CD27 (exemplary protein sequence comprises UniProtkB ID No. P26842), CD137 (exemplary protein sequence comprises UniProtkB ID No. Q07011), 2B4 (exemplary protein sequence comprises UniProtkB ID No. Q9bZW8), TIGIT (exemplary protein sequence comprises UniProtkB ID No. Q495A1), CD155 (exemplary protein sequence comprises UniProtkB ID No. P15151), ICOS (exemplary protein sequence comprises UniProtkB ID No. Q9Y6W8), HVEM (exemplary protein sequence comprises UniProtkB ID No. 043557), CD40L (exemplary protein sequence comprises UniProtkB ID No. P29965), LIGHT (exemplary protein sequence comprises UniProtkB ID No. 043557), OX40 (exemplary protein sequence comprises UniProtkB ID No.), DNAM-1 (exemplary protein sequence comprises UniProtkB ID No. Q15762), PD-L1 (exemplary protein sequence comprises UniProtkB ID No. Q9ZQ7), PD1 (exemplary protein sequence comprises UniProtkB ID No. Q15116), PD-L2 (exemplary protein sequence comprises UniProtkB ID No. Q9BQ51), CTLA-4 (exemplary protein sequence comprises UniProtkB ID No. P16410), CD8 (exemplary protein sequence comprises UniProtkB ID No. P10966, P01732), CD40 (exemplary protein sequence comprises UniProtkB ID No. P25942), CEACAM1 (exemplary protein sequence comprises UniProtkB ID No. P13688), CD48 (exemplary protein sequence comprises UniProtkB ID No. P09326), CD70 (exemplary protein sequence comprises UniProtkB ID No. P32970), AA2AR (exemplary protein sequence comprises UniProtkB ID No. P29274), CD39 (exemplary protein sequence comprises UniProtkB ID No. P49961), CD73 (exemplary protein sequence comprises UniProtkB ID No. P21589), B7-H3 (exemplary protein sequence comprises UniProtkB ID No. Q5ZPR3), B7-H4 (exemplary protein sequence comprises UniProtkB ID No. Q7Z7D3), BTLA (exemplary protein sequence comprises UniProtkB ID No. Q76A9), IDO1 (exemplary protein sequence comprises UniProtkB ID No. P14902), IDO2 (exemplary protein sequence comprises UniProtkB ID No. Q6ZQW0), TDO (exemplary protein sequence comprises UniProtkB ID No. P48755), KIR (exemplary protein sequence comprises UniProtkB ID No. Q99706), LAG-3 (exemplary protein sequence comprises UniProtkB ID No. P18627), TIM-3 (also known as HAVCR2, exemplary protein sequence comprises UniProtkB ID No. Q8TDQ0), or VISTA (exemplary protein sequence comprises UniProtkB ID No. Q9D659). In some embodiments, the first or the second target antigen binding domain binds to an immune cell.

(8) In some embodiments, the first or the second target antigen binding domain binds to a T-cell. In some embodiments, the first or the second target antigen binding domain binds to CD3. In some embodiments, the binding moiety (M), the cleavable linker (L), the first target antigen binding domain (T1), and the second target antigen binding domain (T2) are in one of the following configurations: M:L:T1:T2, and T2:T1:L:M. In some embodiments, the binding moiety comprises an albumin binding domain (anti-Alb), the first target antigen binding domain (T1) comprises a CD3 binding domain (e.g., an anti-CD3 scFV), and a ProTriTAC molecule has the following orientation: anti-Alb: anti-CD3: T2. In some embodiments, the binding moiety comprises an albumin binding domain (anti-Alb), the second target antigen binding domain (T2) comprises a CD3 binding domain (e.g., an anti-CD3 scFV), and a ProTriTAC molecule has the following orientation: anti-Alb: T1: anti-CD3. The T1 domain, in certain examples, is a tumor antigen binding domain, such as, but not limited to, an anti-EGFR domain, an anti-MSLN domain, an anti-BCMA domain, an anti-EpCAM domain, an anti-PSMA domain, or an anti-DLL3 domain.

(9) In some embodiments, the cleavable linker comprises a cleavage site. In some embodiments, the cleavage site is recognized by a protease. In some embodiments, the protease cleavage site is recognized by a serine protease, a cysteine protease, an aspartate protease, a threonine protease, a glutamic acid protease, a metalloproteinase, a gelatinase, or an asparagine peptide lyase. In some embodiments, the protease cleavage site is recognized by a Cathepsin B, a Cathepsin C, a Cathepsin D, a Cathepsin E, a Cathepsin K, a Cathepsin L, a kallikrein, a hK1, a hK10, a hK15, a plasmin, a collagenase, a Type IV collagenase, a stromelysin, a Factor Xa, a chymotrypsin-like protease, a trypsin-like protease, a elastase-like protease, a subtilisin-like protease, an actinidain, a bromelain, a calpain, a caspase, a caspase-3, a Mir1-CP, a papain, a HIV-1 protease, a HSV protease, a CMV protease, a chymosin, a renin, a pepsin, a matriptase, a legumain, a plasmepsin, a nepenthesin, a metalloexopeptidase, a metalloendopeptidase, a matrix metalloprotease (MMP), a MMP1, a MMP2, a MMP3, a MMP7, a MMP8, a MMP9, a MMP10, a MMP11, a MMP12, a MMP13, a MMP14, an ADAM9, an ADAM10, an ADAM12, an urokinase plasminogen activator (uPA), an enterokinase, a prostate-specific target (PSA, hK3), an interleukin-1β converting enzyme, a thrombin, a FAP (FAP-α), a dipeptidyl peptidase, a type II transmembrane serine protease (TTSP), a neutrophil elastase, a cathepsin G, a proteinase 3, a neutrophil serine protease 4, a mast cell chymase, and a mast cell tryptase. In some embodiments, the conditionally active protein further comprises a half-life extension domain bound to the binding moiety, wherein the half-life extension domain provides the binding protein with a safety switch, and wherein upon cleavage of the linker the binding protein is activated by separation of the binding moiety and the half-life extension domain from the first target antigen binding domain, and the binding protein is thereby separated from the safety switch. In some embodiments, the cleavage of the linker is in a tumor microenvironment.

(10) One embodiment provides a conditionally active binding protein, comprising a binding moiety linked to a target antigen binding domain via a non-CDR loop within the binding moiety, wherein

the binding moiety is further linked to a half-life extension domain and comprises a cleavable linker, wherein the target antigen binding domain comprises an immunoglobulin molecule, wherein the binding protein has an extended half-life prior to its activation by cleavage of the linker, and wherein upon activation the binding moiety and the half-life extension domain are separated from the target antigen binding domain, and wherein the binding protein, in its activated state, does not have an extended half-life. In some embodiments, the cleavage of the linker is in a tumor microenvironment.
(11) In some embodiments, the non-CDR loop comprises a CC′ loop of at least one of: a camelid VHH domain, a human VH domain, a humanized VH domain, or a single domain antibody. In some embodiments, the binding moiety comprises a binding site specific for a CD3e domain, and wherein the binding site for the CD3e domain comprises at least one of the following motifs: QDGNE (SEQ ID NO: 921), QDGNEE (SEQ ID NO: 801), DGNE (SEQ ID NO: 922), and DGNEE (SEQ ID NO: 923).

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The novel features of the invention are set forth with particularity in the appended claims. A better understanding of the features and advantages of the present invention will be obtained by reference to the following detailed description that sets forth illustrative embodiments, in which the principles of the invention are utilized, and the accompanying drawings of which.
(2) FIG. **1** illustrates a variable domain of an exemplary immunoglobulin domain, comprising complementarity determining regions (CDR1, CDR2, and CDR3), and non-CDR loops connecting the beta strand (AB, CC′, C″ D, EF, and DE).
(3) FIGS. **2**A-**2**B provide exemplary arrangements of various domains of a conditionally active binding protein of this disclosure. FIG. **2**A: Version 1. FIG. **2**B: Version 2.
(4) FIG. **3** shows an exemplary conditionally active target binding protein of this disclosure.
(5) FIGS. **4**A-**4**C show activation and possible mode of action of trispecific molecules (ProTriTAC). FIG. **4**A shows ProTriTAC molecules in circulation, in tumor environment, and in circulation. FIG. **4**B shows an exemplary sequence (SEQ ID NO: 929) for the protease cleavable site in a linker tethered to an anti-albumin binding moiety and FIG. **4**C shows an SDS-PAGE gel showing the ProTriTAC in its activatable (prodrug) and activated (active drug) states.
(6) FIGS. **5**A-B illustrate a process for making and purifying molecules described herein.
(7) FIG. **5**A shows a schematic flowchart for manufacturing a ProTriTAC molecule and FIG. **5**C shows an SDS-PAGE gel showing three purified ProTriTAC molecules.
(8) FIGS. **6**A-B show analytical size exclusion chromatograms on a ProTriTAC molecule exposed to different stress conditions in graph form in FIG. **6**A with the corresponding data in FIG. **6**B. FIG. **6**B provides the data for FIG. **6**A.
(9) FIG. **7** shows protease-dependent, anti-tumor activity of exemplary ProTriTAC molecules in HCT116 Colorectal Tumor Xenograft Model in NSG Mice.
(10) FIGS. **8**A-**8**D show various designs of exemplary ProTriTAC and control molecules.
(11) FIG. **8**A: Control #1; FIG. **8**B: Control #2; FIG. **8**C: ProTriTAC; and FIG. **8**D: Activated ProTriTAC.
(12) FIG. **9** shows pharmacokinetic profiles for exemplary ProTriTAC and control molecules.
(13) FIG. **10** shows conversion and half-life of exemplary ProTriTAC molecule.
(14) FIG. **11** shows plasma clearance of an exemplary ProTriTAC molecule and its converted active drug format.
(15) FIG. **12** shows CD3 binding potential of an exemplary ProTriTAC molecule, its converted active drug format, and a control non-cleavable ProTriTAC molecule.

(16) FIG. **13** shows human primary T cell binding potential of an exemplary ProTriTAC molecule, its converted active drug format, and a control non-cleavable ProTriTAC molecule.
(17) FIG. **14** shows T cell killing potential of an exemplary ProTriTAC molecule, its converted active drug format, and a control non-cleavable ProTriTAC molecule.
(18) FIG. **15** shows the schematic structure of an exemplary trispecific molecule containing a binding moiety as described herein (also referred to herein as ProTriTAC or activatable ProTriTAC).
(19) FIGS. **16**A-**16**E shows exemplary schematic structures for Prodrug molecules combining functional masking and half-life extension. FIG. **16**A shows a ProDrug molecule comprising an anti-albumin moiety which includes a masking moiety, and a cleavable linker connecting the anti-albumin moiety and the drug. FIG. **16**B shows a ProDrug molecule comprising an anti-albumin moiety comprising two peptide motifs linked by linker, one of which includes a masking moiety, and a cleavable linker connecting the albumin binding moiety to a drug. FIG. **16**C shows a ProDrug molecule comprising a modified albumin (containing a masking moiety) linked to a drug by a cleavable linker. FIG. **16**D shows a ProDrug molecule comprising a modified albumin (containing a masking moiety and a protease cleavage site) linked to a drug. FIG. **16**E shows an activated ProDrug. In each schematic structure (FIGS. **16**A-**16**D) the drug molecule is functionally masked by the anti-albumin moiety or the modified albumin from binding its target or from being activated at an undesired site or from binding at non-target sites and thereby creating a drug sink.
(20) FIGS. **17**A-**17**E show exemplary schematic structures for ProTriTAC molecules combining functional masking and half-life extension. FIG. **17**A shows a ProTriTAC molecule comprising an anti-albumin moiety which includes a masking moiety, and a cleavable linker connecting the anti-albumin moiety and a T cell engager molecule. FIG. **17**B shows a ProTriTAC molecule comprising an anti-albumin moiety comprising two peptide motifs linked by linker, one of which includes a masking moiety, and a cleavable linker connecting the albumin binding moiety to a T cell engager molecule. FIG. **17**C shows a ProTriTAC molecule comprising a modified albumin (containing a masking moiety) linked to a T cell engager by a cleavable linker. FIG. **17**D shows a ProTriTAC molecule comprising a modified albumin (containing a masking moiety and a protease cleavage site) linked to a T cell engager. FIG. **17**E shows an activated ProTriTAC. In each schematic structure (FIGS. **17**A-**17**D) a target binding interface within the ProTriTAC molecule is functionally masked by the anti-albumin moiety or the modified albumin from binding its target or from being activated at an undesired site or from binding at non-target sites and thereby creating a sink.
(21) FIG. **18** shows anti-tumor activity of exemplary ProTriTAC molecules and TriTAC molecules of this disclosure.
(22) FIG. **19** shows pharmacokinetic profile of exemplary ProTriTAC molecules and TriTAC molecules of this disclosure.
(23) FIGS. **20**A-**20**F show admix xenograft individual tumor volumes following administering exemplary ProTriTAC molecules or TriTAC molecules of this disclosure. FIG. **20**A illustrates results from a GFP TriTAC. FIG. **20**B. illustrates results from EGFR ProTriTAC (NCLV). FIG. **20**C illustrates results from EGFR ProTriTAC (L001). FIG. **20**D illustrates results from EGFR ProTriTAC (L041). FIG. **20**E illustrates results from EGFR ProTriTAC (L040). FIG. **20**F illustrates results from EGFR ProTriTAC (L045).
(24) FIGS. **21**A-**21**C shows cytokine levels (IFN-gamma (FIG. **21**A), IL-6 (FIG. **21**B), and IL-10; FIG. **21**C) following administering exemplary ProTriTAC molecules or TriTAC molecules of this disclosure.
(25) FIGS. **22**A-**22**E show body weight percent change in mice, following administering exemplary ProTriTAC molecules and TriTAC molecules of this disclosure. FIG. **22**A illustrates results of 30 μg/kg; FIG. **22**B illustrates results of 100 μg/kg; FIG. **22**C illustrates results of 300 μg/kg; FIG. **22**D illustrates results of 1000 μg/kg; and FIG. **22**E provides fold protection at the

various concentrations.
(26) FIGS. **23**A-**23**C show body weight percent change in mice, following administering varying concentrations of exemplary ProTriTAC molecules of this disclosure, containing non-cleavable or cleavable linkers. FIG. **23**A illustrates results of 300 μg/kg; FIG. **23**B illustrates results of 1000 μg/kg; FIG. **23**C provides fold protection at the various concentrations.
(27) FIGS. **24**A-**24**C show serum concentrations of aspartate aminotransferase (AST), in mice, following administering varying concentrations of a ProTriTAC molecule containing a non-cleavable linker (ProTriTAC (NCLV)) (FIG. **24**C), a TriTAC molecule (FIG. **24**A), or a ProtriTAC molecule (FIG. **24**B) containing a cleavable linker.
(28) FIGS. **25**A-**25**C show serum concentrations of alanine aminotransferase (ALT), in mice, following administering varying concentrations of a ProTriTAC molecule containing a non-cleavable linker (ProTriTAC (NCLV)) (FIG. **25**C), a TriTAC molecule (FIG. **25**A), or a ProtriTAC molecule (FIG. **25**B) containing a cleavable linker.
(29) FIGS. **26**A-**26**B show serum concentrations of ALT (right panel; FIG. **26**B) or AST (left panel; FIB. 26A), in cynomolgus monkeys, following administering varying concentrations of an EGFR ProTriTAC molecule, or an EGFR ProTriTAC (NCLV) molecule.
(30) FIGS. **27**A-**27**D show tumor volume in mice following administration of a GFP TriTAC molecule, an EGFR TriTAC molecule, or an EGFR ProTriTAC molecule, in varying concentrations. FIG. **27**A shows GFP TriTAC (at 300 μg/kg) and an EGFR TriTAC (at 10 μg/kg). FIG. **27**B shows the EGFR TriTAC (at 30 μg/kg and at 100 μg/kg). FIG. **27**C shows the EGFR TriTAC (at 300 μg/kg) and an EGFR ProTriTAC (at 30 μg/kg and 100 μg/kg). FIG. **27**D shows the EGFR ProTriTAC (at 300 μg/kg and 1000 μg/kg)
(31) FIG. **28** shows serum concentrations of ALT and AST, in mice, following administration of varying concentrations of a GFP TriTAC, an EGFR TriTAC, and an EGFR ProTriTAC molecule.
(32) FIG. **29** shows grafting of a CD36 epitope into the CC′ loop of a binding moiety of this disclosure. HuCD3e: SEQ ID NO: 902; CC10: SEQ ID NO: 260; CC12: SEQ ID NO: 259; and CC16: SEQ ID NO: 261. WT disclosed as SEQ ID NO: 795.
(33) FIG. **30** shows separation of a binding moiety of this disclosure, from a ProTriTAC molecule that contained the binding moiety, upon tumor associated protease activation by matriptase.
(34) FIG. **31** shows CD3 binding of ProTriTAC molecules, with or without activation, containing an exemplary binding moiety of this disclosure.
(35) FIG. **32** shows cell killing potential of a ProTriTAC molecule, with or without activation, containing an exemplary binding moiety of this disclosure.
(36) FIG. **33** shows the soft library mutagenesis approach carried out to explore the non-CDR loops within an exemplary binding moiety of this disclosure.
(37) FIG. **34** illustrates the results of the soft library mutagenesis approach carried out to explore the non-CDR loops within an exemplary binding moiety of this disclosure, after panning against HSA (human serum albumin or also referred to herein as albumin). Figure discloses “Naïve Library” sequences as SEQ ID NOS 930-1018, respectively, in order of appearance and discloses “After panning with HSA” sequences as SEQ ID NOS 1019-1140, respectively, in order of appearance.
(38) FIG. **35** illustrates that a binding moiety of this disclosure is able to expand the therapeutic window of a molecule containing it (e.g., a ProTriTAC molecule) by both steric masking and specific masking.
(39) FIG. **36** provides results from a representative T cell dependency cellular cytotoxicity assay with NCI-H508 cells using exemplary fusion proteins of this disclosure containing an anti-EpCAM domain as described herein and an anti-CD3 domain.
(40) FIG. **37** provides results from a representative T cell dependency cellular cytotoxicity assay using exemplary fusion proteins of this disclosure containing an anti-EpCAM domain as described herein and an anti-CD3 domain.

(41) FIG. **38** provides results from a representative T cell dependency cellular cytotoxicity assay using exemplary fusion proteins of this disclosure containing an anti-EpCAM domain as described herein and an anti-CD3 domain.
(42) FIG. **39** provides results from a representative T cell dependency cellular cytotoxicity assay using exemplary fusion proteins of this disclosure containing an anti-EpCAM domain as described herein and an anti-CD3 domain.
(43) FIG. **40** provides results from a representative T cell dependency cellular cytotoxicity assay using exemplary fusion proteins of this disclosure containing a humanized anti-EpCAM domain as described herein and an anti-CD3 domain.
(44) FIGS. **41**A-**41**C show body weight percent change in mice, following administering exemplary EpCAM ProTriTAC molecules and EpCAM TriTAC molecules of this disclosure.

DETAILED DESCRIPTION OF THE INVENTION

(45) While preferred embodiments of the present invention have been shown and described herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. Numerous variations, changes, and substitutions will now occur to those skilled in the art without departing from the invention. It should be understood that various alternatives to the embodiments of the invention described herein may be employed in practicing the invention. It is intended that the following claims define the scope of the invention and that methods and structures within the scope of these claims and their equivalents be covered thereby.
(46) Provided herein in certain embodiments are ProTriTAC molecules (also referred to herein as protrisecific molecules) that are T cell engager prodrugs designed to be conditionally active in a tumor microenvironment. In some cases, this enables targeting of a wider selection of tumor antigens (e.g., solid tumor antigens). The ProTriTAC molecules, in some examples, combine the desirable attributes of several prodrug approaches, including, but not limited to: combination of steric and specific masking, wherein the steric masking is, in some cases, is through albumin that is recognized by an anti-albumin domain in a ProTriTAC molecule, and the specific masking, in some cases, is through specific intermolecular interactions between an anti-albumin domain (in some examples) and a target antigen binding domain of the ProTriTAC molecule (such as, an anti-CD3 scFv domain, in some examples); additional safety imparted by half-life differential of prodrug versus an active drug, derived by activation of the conditionally activated ProTriTAC molecule; ability to plug-and-play with different tumor target binders.
(47) Certain Definitions
(48) The terminology used herein is for the purpose of describing particular cases only and is not intended to be limiting. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Furthermore, to the extent that the terms “including”, “includes”, “having”, “has”, “with”, or variants thereof are used in either the detailed description and/or the claims, such terms are intended to be inclusive in a manner similar to the term “comprising.”
(49) The term “about” or “approximately” means within an acceptable error range for the particular value as determined by one of ordinary skill in the art, which will depend in part on how the value is measured or determined, e.g., the limitations of the measurement system. For example, “about” can mean within 1 or more than 1 standard deviation, per the practice in the given value. Where particular values are described in the application and claims, unless otherwise stated the term “about” should be assumed to mean an acceptable error range for the particular value.
(50) The terms “individual,” “patient,” or “subject” are used interchangeably. None of the terms require or are limited to situation characterized by the supervision (e.g. constant or intermittent) of a health care worker (e.g. a doctor, a registered nurse, a nurse practitioner, a physician's assistant, an orderly, or a hospice worker).
(51) A “single chain Fv” or “scFv”, as used herein, refers to a binding protein in which the variable domains of the heavy chain and of the light chain of a traditional two chain antibody are joined to

form one chain. Typically, a linker peptide is inserted between the two chains to allow for proper folding and creation of an active binding site.
(52) A "cleavage site for a protease," or "protease cleavage site", as meant herein, is an amino acid sequence that can be cleaved by a protease, such as, for example, a matrix metalloproteinase or a furin. Examples of such sites include Gly-Pro-Leu-Gly-Ile-Ala-Gly-Gln (SEQ ID NO: 924) or Ala-Val-Arg-Trp-Leu-Leu-Thr-Ala (SEQ ID NO: 925), which can be cleaved by metalloproteinases, and Arg-Arg-Arg-Arg-Arg-Arg (SEQ ID NO: 926), which is cleaved by a furin. In therapeutic applications, the protease cleavage site can be cleaved by a protease that is produced by target cells, for example cancer cells or infected cells, or pathogens.
(53) As used herein, "elimination half-time" is used in its ordinary sense, as is described in *Goodman and Gillman's The Pharmaceutical Basis of Therapeutics* 21-25 (Alfred Goodman Gilman, Louis S. Goodman, and Alfred Gilman, eds., 6th ed. 1980). Briefly, the term is meant to encompass a quantitative measure of the time course of drug elimination. The elimination of most drugs is exponential (i.e., follows first-order kinetics), since drug concentrations usually do not approach those required for saturation of the elimination process. The rate of an exponential process may be expressed by its rate constant, k, which expresses the fractional change per unit of time, or by its half-time, $t_{1/2}$ the time required for 50% completion of the process. The units of these two constants are $time^{-1}$ and time, respectively. A first-order rate constant and the half-time of the reaction are simply related ($k \times t_{1/2} = 0.693$) and may be interchanged accordingly. Since first-order elimination kinetics dictates that a constant fraction of drug is lost per unit time, a plot of the log of drug concentration versus time is linear at all times following the initial distribution phase (i.e. after drug absorption and distribution are complete). The half-time for drug elimination can be accurately determined from such a graph.
(54) A "therapeutic agent," as used herein, includes a "binding molecule."
(55) The term "binding molecule," as used herein is any molecule, or portion or fragment thereof, that can bind to a target molecule, cell, complex and/or tissue, and which includes proteins, nucleic acids, carbohydrates, lipids, low molecular weight compounds, and fragments thereof, each having the ability to bind to one or more of a soluble protein, a cell surface protein, a cell surface receptor protein, an intracellular protein, a carbohydrate, a nucleic acid, a hormone, or a low molecular weight compound (small molecule drug), or a fragment thereof. The binding molecule, in some instances, is a protein belonging to the immunoglobulin superfamily, or a non-immunoglobulin binding molecule. The "binding molecule" does do not include a cytokine.
(56) The term "proteins belonging to immunoglobulin superfamily," or "immunoglobulin molecules," as used herein, include proteins that comprise an immunoglobulin fold, such as antibodies and target antigen binding fragments thereof, antigen receptors, antigen presenting molecules, receptors on natural killer cells, antigen receptor accessory molecules, receptors on leukocytes, IgSF cellular adhesion molecules, growth factor receptors, and receptor tyrosine kinases/phosphatases.
(57) The term "antibodies" include antibodies or immunoglobulins of any isotype, fragments of antibodies that retain specific binding to antigen, including, but not limited to, Fab, Fv, scFv, and Fd fragments, chimeric antibodies, humanized antibodies, single-chain antibodies (scAb), single domain antibodies (dAb), single domain heavy chain antibodies, a single domain light chain antibodies, bi-specific antibodies, multi-specific antibodies, and fusion proteins comprising an antigen-binding (also referred to herein as antigen binding) portion of an antibody and a non-antibody protein. The antibodies, in some examples, are detectably labeled, e.g., with a radioisotope, an enzyme that generates a detectable product, a fluorescent protein, and the like. The antibodies, in some cases, are further conjugated to other moieties, such as members of specific binding pairs, e.g., biotin (member of biotin-avidin specific binding pair), and the like. The antibodies, in some cases, are bound to a solid support, including, but not limited to, polystyrene plates or beads, and the like. Also encompassed by the term are Fab′, Fv, F(ab′)2, and or other

antigen binding fragments that retain specific binding to antigen, and monoclonal antibodies. As used herein, a monoclonal antibody is an antibody produced by a group of identical cells, all of which were produced from a single cell by repetitive cellular replication. That is, the clone of cells only produces a single antibody species. While a monoclonal antibody can be produced using hybridoma production technology, other production methods known to those skilled in the art can also be used (e.g., antibodies derived from antibody phage display libraries). An antibody, in some instances, is monovalent or bivalent. An antibody, in some instances, is an Ig monomer, which is a “Y-shaped” molecule that consists of four polypeptide chains: two heavy chains and two light chains connected by disulfide bonds.

(58) The term “non-immunoglobulin binding molecules,” as used herein, include, but is not limited to examples such as a growth factor, a hormone, a signaling protein, an inflammatory mediator, ligand, a receptor, or a fragment thereof, a native hormone or a variant thereof being able to bind to its natural receptor; a nucleic acid or polynucleotide sequence being able to bind to complementary sequence or a soluble cell surface or intracellular nucleic acid/polynucleotide binding proteins, a carbohydrate binding moiety being able to bind to other carbohydrate binding moieties, cell surface or intracellular proteins, a low molecular weight compound (drug) that binds to a soluble or cell surface or intracellular target protein. The non-immunoglobulin binding molecules, in some cases, include coagulation factors, plasma proteins, fusion proteins, and imaging agents. The non-immunoglobulin binding molecules do not include a cytokine.

(59) A “cytokine,” as meant herein, refers to intercellular signaling molecules, and active fragments and portions thereof, which are involved in the regulation of mammalian somatic cells. A number of families of cytokines, for example, interleukins, interferons, and transforming growth factors are included.

(60) As used herein, “non-CDR loops” within immunoglobulin (Ig) molecules are regions of a polypeptide other than the complementarity determining regions (CDRs) of an antibody. These regions may be derived from an antibody or an antibody fragment. These regions may also be synthetically or artificially derived, such as through mutagenesis or polypeptide synthesis.

(61) In an Ig, Ig-like, or beta-sandwich scaffold that has 9 beta-strands (e.g., a VH, a VL, a camelid VHH, a sdAb), the non-CDR loops can refer to the AB, CC′, C″D, EF loops or loops connecting beta-strands proximal to the C-terminus. In an Ig, Ig-like, or beta-sandwich scaffold that has 7 beta-strands (e.g., a CH, a CL, an adnectin, a Fn-III), the non-CDR loops can refer to the AB, CD, and EF loops or loops connecting beta-strands proximal the C-terminus. In other Ig-like or beta-sandwich scaffolds, the non-CDR loops are the loops connecting beta-strands proximal to the C-terminus or topologically equivalent residues using the framework established in the Halaby 1999 publication (Prot Eng Des Sel 12:563-571).

(62) In a non-beta-sandwich scaffold (e.g., a DARPin, an affimer, an affibody), the “non-CDR loops” refer to an area that is (1) amenable for sequence randomization to allow engineered specificities to a second antigen, and (2) distal to the primary specificity determining region(s) typically used on the scaffold to allow simultaneous engagement of the scaffold to both antigens without steric interference. For this purpose, the primary specificity determining region(s) can be defined using the framework established in the Skrlec 2015 publication (Trends in Biotechnol, 33:408-418). An excerpt of the framework is listed below.

(63) TABLE-US-00001 Scaffold Primary specificity determining region(s) Affibody 13 residues in two helices Affimer 12-36 residues Anticalin Four loops (up to 24 aa) Avimer 11 residues Scaffold Primary specificity determining region(s) Centyrin 13 residues DARPin 7 residues in each n-repeat, or 8 residues in each n-repeat Fynomer 6 residues in the RT- and n-Src-loop Kunitz domain 1-2 loops

(64) “Target antigen binding domain”, as used herein, refers to a region which targets a specific antigen. A target antigen binding domain comprises, for example an sdAb, an scFv, a variable heavy chain antibody (VHH), a variable heavy (VH) or a variable light domain (VL), a full length

antibody, or any other peptide that has a binding affinity towards a specific antigen. The target antigen binding domain does do not include a cytokine.

(65) "TriTAC," as used herein refers to a trispecific binding protein that is not conditionally activated.

(66) Binding Moiety, Cleavable Linker and Conditionally Active Binding Proteins

(67) This disclosure provides, in some embodiments, binding moieties that are capable of masking the interaction of binding molecules with their targets. In some embodiments, a binding moiety of this disclosure comprises a masking moiety and a cleavable linker, such as a protease cleavable linker. In some embodiments, a binding moiety of this disclosure comprises a masking moiety (e.g., a modified non-CDR loop sequence) and a non-cleavable linker. As illustrated in FIG. **35**, the binding moiety is capable of synergistically expanding a therapeutic window of a molecule that comprises the moiety, by both steric masking and specific masking. In some examples, the binding molecule is a protein belonging to an immunoglobulin superfamily, such as a target antigen binding domain comprising an immunoglobulin fold. In some embodiments, the binding molecule is a non-immunoglobulin protein. In some embodiments, the binding moiety combines both steric masking (for example, via binding to a bulky serum albumin) and specific masking (for example, via non-CDR loops binding to the CDRs of an anti-CD3 scFv domain). In some cases, modifying the non-CDR loops within the binding moiety does not affect albumin binding. The protease cleavable linker, in some cases, enables activation of a prodrug molecule comprising the binding moiety (such as a ProTriTAC molecule comprising a binding moiety as described herein, a CD3 binding domain, and an albumin binding domain), in a single proteolytic event, thereby allowing more efficient conversion of the prodrug molecule in tumor microenvironment. Further, tumor-associated proteolytic activation, in some cases, reveals active T cell engager (such as a ProTriTAC molecule comprising a binding moiety as described herein, a CD3 binding domain, and an albumin binding domain) with minimal off-tumor activity after activation. The present disclosure, in some embodiments, provides a half-life extended T cell engager format (ProTriTAC) comprising a binding moiety as described herein, which in some cases represents a new and improved approach to engineer conditionally active T cell engagers.

(68) FIG. **4**B provides a schematic for an exemplary ProTriTAC molecule comprising an exemplary binding moiety as described herein (the αalbumin sdAb) and a gel showing the ProTriTAC before and after activation by cleaving of the protease cleavable linker, and FIG. **4**A shows a possible mode of action of the same. FIG. **15** shows the schematic structure of an exemplary trispecific molecule containing a binding moiety as described herein (also referred to herein as ProTriTAC or activatable ProTriTAC), with engineered non-CDR loops. The exemplary trispecific molecule contains an anti-albumin domain comprising a cleavable linker (such a linker comprising a protease cleavable site, also referred to herein as a substrate linker) and a masking domain; an anti-CD3 binding domain; and an anti-target domain (specific for a tumor antigen) which in some cases is a non-immunoglobulin molecule. In some cases, non-CDR loops in the anti-albumin domain is capable of binding and masking the anti-target domain. In some cases, non-CDR loops in the anti-albumin domain is capable of binding and masking the anti-CD3 domain. The binding moiety, in some embodiments, comprises a CDR loop specific for binding albumin.

(69) Provided herein, in a first embodiment, is a binding moiety that masks the binding of a target antigen binding domain, and is capable of binding to a bulk-serum protein, such as a half-life extending protein. The binding moiety of the first embodiment, in certain instances, further comprises a cleavable linker attached to it. The cleavable linker, for example, comprises a protease cleavage site or a pH dependent cleavage site. The cleavable linker, in certain instances, is cleaved only in a tumor micro-environment. Thus, in some examples, the binding moiety of the first embodiment, bound to the half-life extending protein, connected to the cleavable linker, and further bound to the target antigen binding domain, maintains the target antigen binding domain in an inert state in circulation until the cleavable linker is cleaved off in a tumor microenvironment. The half-

life of the target antigen binding domain, such as an antibody or an antigen binding fragment thereof, is thus extended in systemic circulation by using the binding moiety of the first embodiment which acts as a safety switch that keeps the target antigen binding moiety in an inert state until it reaches the tumor microenvironment where it is conditionally activated by cleavage of the linker and is able to bind its target antigen.

(70) In a second embodiment is provided a binding moiety that masks the interaction between a non-immunoglobulin binding molecule and its target. The binding moiety of the second embodiment, in certain instances, is capable of binding to a bulk serum protein. In some instances, the binding moiety of the second embodiment further comprises a cleavable linker attached to it. The cleavable linker, for example, comprises a protease cleavage site or a pH dependent cleavage site. The cleavable linker, in certain instances, is cleaved only in a tumor micro-environment. The non-immunoglobulin binding molecule is, in some cases, maintained in an inert state by the binding moiety of the second embodiment and activated by cleavage of the linker, for example in a target environment. In some instances, the cleavable linker is cleaved off in a tumor microenvironment and in such cases the tumor microenvironment is the target environment. The half-life of the non-immunoglobulin binding molecule is thus extended in systemic circulation by using the binding moiety of the second embodiment which acts as a safety switch that keeps the non-immunoglobulin binding molecule in an inert state until it reaches the target environment where it is conditionally activated by cleavage of the linker. In some examples of the second embodiment where the non-immunoglobulin binding molecule is an imaging agent, said agent is activated in a target environment upon cleavage of the cleavable linker. The target environment, in such cases, is a tissue or a cell or any biological environment that is to be imaged using the imaging agent.

(71) The safety switch described above provides several advantages; some examples including (i) expanding the therapeutic window of the immunoglobulin molecule, such as a target antigen binding domain, a non-immunoglobulin binding molecule; (ii) reducing target-mediated drug disposition by maintaining the immunoglobulin molecule, such as a target antigen binding domain, the non-immunoglobulin binding molecule, in an inert state when a conditionally active protein comprising a binding moiety according to the first or second embodiments is in systemic circulation; (iii) reducing the concentration of undesirable activated proteins in systemic circulation, thereby minimizing the spread of chemistry, manufacturing, and controls related impurities, e.g., pre-activated drug product, endogenous viruses, host-cell proteins, DNA, leachables, anti-foam, antibiotics, toxins, solvents, heavy metals; (iv) reducing the concentration of undesirable activated proteins in systemic circulation, thereby minimizing the spread of product related impurities, aggregates, breakdown products, product variants due to: oxidation, deamidation, denaturation, loss of C-term Lys in MAbs; (v) preventing aberrant activation of the immunoglobulin molecule, such as a target antigen binding domain, or the non-immunoglobulin binding molecule in circulation; (vi) reducing the toxicities associated with the leakage of activated species from diseased tissue or other pathophysiological conditions, e.g., tumors, autoimmune diseases, inflammations, viral infections, tissue remodeling events (such as myocardial infarction, skin wound healing), or external injury (such as X-ray, CT scan, UV exposure); and (vii) reducing non-specific binding of the immunoglobulin molecule, such as a target antigen binding domain, or the non-immunoglobulin binding molecule. Furthermore, post-activation, or in other words post breaking of the safety switch, the immunoglobulin molecule, such as a target antigen binding domain, the non-immunoglobulin binding molecule is separated from the safety switch which provided extended half-life, and thus is cleared from circulation.

(72) In addition, the binding moieties of the first, second, and the third embodiments, in some cases, are used to generate a “biobetter” version of a biologic. Generally, preparing a biobetter form of a molecule, e.g., an antibody or an antigen binding fragment thereof, involves taking the originator molecule and making specific alterations in it to improve its parameters and thereby

make it a more efficacious, less frequently dosed, better targeted, and/or a better tolerated drug. Thus, a target antigen binding domain masked by the binding moiety of the first embodiment which is bound to a half-life extending protein, and conditionally activated in a tumor microenvironment by cleavage of the cleavable linker, gives the target antigen binding domain a significantly longer serum half-life and reduces the likelihood of its undesirable activation in circulation, thereby producing a “biobetter” version of the target antigen binding domain. Similarly, the binding moieties of the second embodiment are, in some cases, utilized to generate biobetter versions of the non-immunoglobulin binding molecules. Accordingly, in various embodiments, biobetter versions of immunoglobulin molecules, non-immunoglobulin binding molecules are provided, wherein the biobetter function is attributed to a binding moiety, respectively, according to the first or second embodiments.

(73) The binding moieties described herein comprise at least one non-CDR loop. In some embodiments, a non-CDR loop provides a binding site for binding of the binding moiety of the first embodiment to a target antigen binding domain. In some examples of the first embodiment, a non-CDR loop provides a binding site for binding of the binding moiety of the first embodiment to an immunoglobulin molecules, such as a target antigen binding domain. In some examples of the second embodiment, a non-CDR loop provides a binding site for binding of the binding moiety of the second embodiment to a non-immunoglobulin binding molecule. In some cases, the binding moiety of the first embodiment masks binding of the target binding domain to the target antigen, e.g. via steric occlusion, via specific intermolecular interactions, or a combination of both. The binding moieties of the second embodiment also, in some cases, mask binding of a non-immunoglobulin binding molecule to their targets, via steric occlusion, via specific intermolecular interactions, or a combination of both.

(74) In some embodiments, the binding moieties described herein further comprise complementarity determining regions (CDRs). In some instances, the binding moieties are domains derived from an immunoglobulin molecule (Ig molecule). The Ig may be of any class or subclass (IgG1, IgG2, IgG3, IgG4, IgA, IgE, IgM etc). A polypeptide chain of an Ig molecule folds into a series of parallel beta strands linked by loops. In the variable region, three of the loops constitute the “complementarity determining regions” (CDRs) which determine the antigen binding specificity of the molecule. An IgG molecule comprises at least two heavy (H) chains and two light (L) chains inter-connected by disulfide bonds, or an antigen binding fragment thereof. Each heavy chain is comprised of a heavy chain variable region (abbreviated herein as VH) and a heavy chain constant region. The heavy chain constant region is comprised of three domains, CH1, CH2 and CH3. Each light chain is comprised of a light chain variable region (abbreviated herein as VL) and a light chain constant region. The light chain constant region is comprised of one domain, CL. The VH and VL regions can be further subdivided into regions of hypervariability, termed complementarity determining regions (CDRs) with are hypervariable in sequence and/or involved in antigen recognition and/or usually form structurally defined loops, interspersed with regions that are more conserved, termed framework regions (FR). Each VH and VL is composed of three CDRs and four FRs, arranged from amino-terminus to carboxy-terminus in the following order: FR1, CDR1, FR2, CDR2, FR3, CDR3, FR4. In some embodiments of this disclosure, at least some or all of the amino acid sequences of FR1, FR2, FR3, and FR4 are part of the “non-CDR loop” of the binding moieties described herein. As shown in FIG. **1**, a variable domain of an immunoglobulin molecule has several beta strands that are arranged in two sheets. The variable domains of both light and heavy immunoglobulin chains contain three hypervariable loops, or complementarity-determining regions (CDRs). The three CDRs of a V domain (CDR1, CDR2, CDR3) cluster at one end of the beta barrel. The CDRs are the loops that connect beta strands B-C, C′-C″, and F-G of the immunoglobulin fold, whereas the bottom loops that connect beta strands AB, CC′, C″-D and E-F of the immunoglobulin fold, and the top loop that connects the D-E strands of the immunoglobulin fold are the non-CDR loops. In some embodiments of this disclosure, at least some amino acid

residues of a constant domain, CH1, CH2, or CH3, are part of the “non-CDR loop” of the binding moieties described herein. Non-CDR loops comprise, in some embodiments, one or more of AB, CD, EF, and DE loops of a C1-set domain of an Ig or an Ig-like molecule; AB, CC′, EF, FG, BC, and EC′ loops of a C2-set domain of an Ig or an Ig-like molecule; DE, BD, GF, A(A1A2)B, and EF loops of I(Intermediate)-set domain of an Ig or Ig-like molecule.

(75) Within the variable domain, the CDRs are believed to be responsible for antigen recognition and binding, while the FR residues are considered a scaffold for the CDRs. However, in certain cases, some of the FR residues play an important role in antigen recognition and binding. Framework region residues that affect Ag binding are divided into two categories. The first are FR residues that contact the antigen, thus are part of the binding-site, and some of these residues are close in sequence to the CDRs. Other residues are those that are far from the CDRs in sequence, but are in close proximity to it in the 3-D structure of the molecule, e.g., a loop in heavy chain.

(76) In some embodiments, the non-CDR loop is modified to generate an antigen binding site specific for a bulk serum protein, such as albumin. It is contemplated that various techniques can be used for modifying the non-CDR loop, e.g., site-directed mutagenesis, random mutagenesis, insertion of at least one amino acid that is foreign to the non-CDR loop amino acid sequence, amino acid substitution. An antigen peptide is inserted into a non-CDR loop, in some examples. In some examples, an antigenic peptide is substituted for the non-CDR loop. The modification, to generate an antigen binding site, is in some cases in only one non-CDR loop. In other instances, more than one non-CDR loop are modified. For instance, the modification is in any one of the non-CDR loops shown in FIG. **1**, i.e., AB, CC′, C″ D, EF, and D-E. In some cases, the modification is in the DE loop. In other cases the modifications are in all four of AB, CC′, C″-D, E-F loops. In certain examples, the binding moieties described herein are bound to the immunoglobulin molecules, e.g., a target antigen binding domain, the non-immunoglobulin binding molecules via their AB, CC′, C″ D, or EF loop and are bound to a bulk-serum protein, such as albumin, via their B-C, C′-C″, or F-G loop. In certain examples, the binding moiety of the first embodiment is bound to the target antigen binding domain via its AB, CC′, C″ D, and EF loop and is bound to a bulk-serum protein, such as albumin, via its BC, C′C″, and FG loop.

(77) In certain examples, the binding moiety of the first embodiment is bound to a target antigen binding domain via one or more of AB, CC′, C″ D, and E-F loop and is bound to a bulk-serum protein, such as albumin, via one or more of BC, C′C″, and FG loop. In certain examples, the binding moiety of the first embodiment is bound to a bulk serum protein, such as albumin, via its AB, CC′, C″ D, or EF loop and is bound to a target antigen binding domain via its BC, C′C″, or FG loop. In certain examples, the binding moiety of the first embodiment is bound to a bulk serum protein, such as albumin, via its AB, CC′, C″ D, and EF loop and is bound to a target antigen binding domain via its BC, C′C″, and FG loop. In certain examples, the binding moiety of the first embodiment is bound to a bulk serum protein, such as albumin, via one or more of AB, CC′, C″ D, and E-F loop and is bound to the target antigen binding protein, via one or more of BC, C′C″, and FG loop.

(78) In certain examples, the binding moiety of the second embodiment is bound to a non-immunoglobulin molecule via one or more of AB, CC′, C″ D, and E-F loop and is bound to a bulk-serum protein, such as albumin, via one or more of BC, C′C″, and FG loop. In certain examples, the binding moiety of the second embodiment is bound to a bulk serum protein, such as albumin, via its AB, CC′, C″ D, or EF loop and is bound to a non-immunoglobulin molecule via its BC, C′C″, or FG loop. In certain examples, the binding moiety of the second embodiment is bound to a bulk serum protein, such as albumin, via its AB, CC′, C″ D, and EF loop and is bound to a non-immunoglobulin molecule via its BC, C′C″, and FG loop. In certain examples, the binding moiety of the second embodiment is bound to a bulk serum protein, such as albumin, via one or more of AB, CC′, C″ D, and E-F loop and is bound to a non-immunoglobulin molecule, via one or more of BC, C′C″, and FG loop.

(79) The binding moieties are any kinds of polypeptides. For example, in certain instances the binding moieties are natural peptides, synthetic peptides, or fibronectin scaffolds, or engineered bulk serum proteins. The bulk serum protein comprises, for example, albumin, fibrinogen, or a globulin. In some embodiments, the binding moieties are engineered scaffolds. Engineered scaffolds comprise, for example, sdAb, a scFv, a Fab, a VHH, a fibronectin type III domain, immunoglobulin-like scaffold (as suggested in Halaby et al., 1999. Prot Eng 12(7):563-571), DARPin, cystine knot peptide, lipocalin, three-helix bundle scaffold, protein G-related albumin-binding module, or a DNA or RNA aptamer scaffold.
(80) In some cases, the binding moiety of the first embodiment binds to at least one target antigen binding domain. In further embodiments, the non-CDR loops within the binding moiety of the first embodiment provide a binding site for the at least one target antigen binding domain. The target antigen binding domain, in some cases, binds to target antigens expressed on the surface of a diseased cell or tissue, for example a tumor or a cancer cell. Target antigens include but are not limited to EpCAM, EGFR, HER-2, HER-3, c-Met, FoIR, PSMA, CD38, BCMA, and CEA. 5T4, AFP, B7-H3, CDH-6, CAIX, CD117, CD123, CD138, CD166, CD19, CD20, CD205, CD22, CD30, CD33, CD352, CD37, CD44, CD52, CD56, CD70, CD71, CD74, CD79b, DLL3, EphA2, FAP, FGFR2, FGFR3, GPC3, gpA33, FLT-3, gpNMB, HPV-16 E6, HPV-16 E7, ITGA2, ITGA3, SLC39A6, MAGE, mesothelin, Muc1, Muc16, NaPi2b, Nectin-4, CDH-3, CDH-17, EPHB2, ITGAV, ITGB6, NY-ESO-1, PRLR, PSCA, PTK7, ROR1, SLC44A4, SLITRK5, SLITRK6, STEAP1, TIM1, Trop2, or WT1.
(81) In some cases, the binding moiety of the first embodiment is bound to a first target antigen binding domain via its non-CDR loops and the first target antigen binding domain is further connected to a second target antigen binding domain. Examples of first and second target antigen binding domains include, but are not limited to, a T cell engager, a bispecific T cell engager, a dual-affinity re-targeting antibody, a variable heavy domain (VH), a variable light domain (VL), a scFv comprising a VH and a VL domain, a soluble TCR fragment comprising a Valpha and Vbeta domain, a single domain antibody (sdAb), or a variable domain of camelid derived nanobody (VHH), a non-Ig binding domain, i.e., antibody mimetic, such as anticalins, affilins, affibody molecules, affimers, affitins, alphabodies, avimers, DARPins, fynomers, kunitz domain peptides, and monobodies, a ligand or peptide. In some examples, the first or the second target antigen binding domain is a VHH domain. In some examples, the first or the second target antigen binding domain is a sdAb. In some instances, the first target antigen binding domain is specific for a tumor antigen, such as EGFR, and the second target antigen binding domain is specific for CD3. The binding of the first target antigen binding domain to its target, e.g., a tumor antigen such as EGFR, is masked by the binding moiety of the first embodiment, via its non-CDR loops. One exemplary conditionally active protein, comprising a binding moiety according to the first embodiment, is shown in FIG. **3**.
(82) In some cases, the non-CDR loops within the binding moiety of the second embodiment provide a binding site for a non-immunoglobulin binding molecule.
(83) In some cases, the binding moieties comprise a binding site for a bulk serum protein. In some embodiments, the CDRs within the binding moieties provide a binding site for the bulk serum protein. The bulk serum protein is, in some examples, a globulin, albumin, transferrin, IgG1, IgG2, IgG4, IgG3, IgA monomer, Factor XIII, Fibrinogen, IgE, or pentameric IgM. In some embodiments, the binding moieties comprise a binding site for an immunoglobulin light chain. In some embodiments, the CDRs provide a binding site for the immunoglobulin light chain. The immunoglobulin light chain is, in some examples, an IgK free light chain or an Ig, free light chain.
(84) In some examples, the binding moieties comprise any type of binding domain, including but not limited to, domains from a monoclonal antibody, a polyclonal antibody, a recombinant antibody, a human antibody, a humanized antibody. In some embodiments, the binding moiety is a single chain variable fragment (scFv), a soluble TCR fragment, a single-domain antibody such as a

heavy chain variable domain (VH), a light chain variable domain (VL) and a variable domain (VHH) of camelid derived nanobody. In other embodiments, the binding moieties are non-Ig binding domains, i.e., antibody mimetic, such as anticalins, affilins, affibody molecules, affimers, affitins, alphabodies, avimers, DARPins, fynomers, kunitz domain peptides, and monobodies.
(85) TABLE-US-00002 TABLE 1 Exemplary Sequences for Masking Sequences within the Binding Moieties of this Disclosure are Provided In SEQ ID Nos. 50, 259-301 And 795. MaskID Sequence Alt. Names SEQ ID No. MASK001 APGKG WT SEQ ID No. 795 MASK002 GGQDGNEEMGGG CC12 SEQ ID No. 259 MASK004 GGQDGNEEGG CC10 SEQ ID No. 260 MASK006 GGGGQDGNEEMGGGGG CC16 SEQ ID No. 261 MASK007 APFGSEM SEQ ID No. 262 MASK008 AWNGPYE SEQ ID No. 263 MASK009 AQDNGDTKTG SEQ ID No. 264 MASK010 AEAKETQG SEQ ID No. 265 MASK011 ATRREQVEG SEQ ID No. 266 MASK012 AQAPSQP SEQ ID No. 267 MASK013 ATRSRTRNDG SEQ ID No. 268 MASK014 ADVDAPDGLG SEQ ID No. 269 MASK015 AADISDPGG SEQ ID No. 270 MASK016 ALSVDPSG SEQ ID No. 271 MASK017 ARLSVDPG SEQ ID No. 272 MASK018 AVEAADRG SEQ ID No. 273 MASK020 GG**P**DGNEEMGGG CC12-Q1P SEQ ID No. 274 MASK021 GG**F**DGNEEMGGG CC12-Q1F SEQ ID No. 275 MASK022 GG**G**DGNEEMGGG CC12-Q1G SEQ ID No. 276 MASK023 GGEMDGEGQNGG CC12- SEQ ID No. 277 scramble MASK024 GGGGGPDGNEEPGG SEQ ID No. 278 MASK025 GGGGSLDGNEEPGG SEQ ID No. 279 MASK026 GGGGALDGNEEPGG SEQ ID No. 280 MASK027 GGGGGLDGNEEPGG SEQ ID No. 281 MASK028 GGGALDGNEEPGG SEQ ID No. 282 MASK029 GGGGGPDGNEEPGGG SEQ ID No. 283 MASK030 GGSGALDGNEEPGG SEQ ID No. 284 MASK031 GGSGSLDGNEEPGG SEQ ID No. 285 MASK038 GGSGGPDGNEEPGG SEQ ID No. 286 MASK039 GGVRDGPDGNEEPGG SEQ ID No. 287 MASK040 GGSGGPDGNEEPGGGG SEQ ID No. 288 MASK041 GGGRGPDGNEEPGG SEQ ID No. 289 MASK042 GGSGGLDGNEEPGG SEQ ID No. 290 MASK043 GGGVGPDGNEEPGG SEQ ID No. 291 MASK044 GGGEGPDGNEEPGG SEQ ID No. 292 MASK046 GGGVALDGNEEPGG SEQ ID No. 293 MASK047 GGGRALDGNEEPGG SEQ ID No. 294 MASK048 GGYAGLDGNEEPGG SEQ ID No. 295 MASK049 GGAGGPDGNEEPGG SEQ ID No. 296 MASK051 GGRGGPDGNEEPGG SEQ ID No. 297 MASK052 GGGGPDGNEEPGGGG SEQ ID No. 298 MASK053 GGGEALDGNEEPGG SEQ ID No. 299 MASK054 GGDASLDGNEEPGG SEQ ID No. 300 MASK055 GGRDAPDGNEEGG SEQ ID No. 301
(86) It is contemplated herein that in some embodiments of this disclosure the binding moieties described herein comprise at least one cleavable linker. In one aspect, the cleavable linker comprises a polypeptide having a sequence recognized and cleaved in a sequence-specific manner. The cleavage, in certain examples, is enzymatic, based on pH sensitivity of the cleavable linker, or by chemical degradation. The conditionally active binding proteins contemplated herein, in some cases, comprise a protease cleavable linker recognized in a sequence-specific manner by a matrix metalloprotease (MMP), for example a MMP9. In some cases, the protease cleavable linker is recognized by a MMP9 comprises a polypeptide having an amino acid sequence PR(S/T)(L/I)(S/T) (SEQ ID NO: 3). In some cases, the protease cleavable linker recognized by a MMP9 comprises a polypeptide having an amino acid sequence LEATA (SEQ ID NO: 4). In some cases, the protease cleavable linker is recognized in a sequence-specific manner by a MMP11. In some cases, the protease cleavable linker recognized by a MMP11 comprises a polypeptide having an amino acid sequence GGAANLVRGG (SEQ ID NO: 5). In some cases, the protease cleavable linker is recognized by a protease disclosed in Table 3. In some cases, the protease cleavable linker is recognized by a protease disclosed in Table 3 comprises a polypeptide having an amino acid

sequence selected from a sequence disclosed in Table 3 (SEQ ID NOS: 1-42, 53, and 58-62). In some cases, the cleavable linker has an amino acid sequence as set forth in SEQ ID No. 59. In some cases, the cleavable linker is recognized by MMP9, matriptase, Urokinase plasminogen activator (uPA) and has an amino acid sequence as set forth in SEQ ID No. 59.
(87) In some embodiments of this disclosure the binding moieties described herein comprise at least one non-cleavable linker. The non-cleavable linker comprises, in some examples, a sequence as set forth in SEQ ID No. 51, SEQ ID No. 302, SEQ ID No. 303, SEQ ID No. 304, or SEQ ID No. 305.
(88) TABLE-US-00003 TABLE 2 Exemplary Non-cleavable Linker Sequences Non-cleavable Linker ID Sequence SEQ ID No. L002 SGGGGSGGVV SEQ ID No. 302 L016 SGGGGSGGGGSGGGGS SEQ ID No. 303 L017 SGGGGSGGGGSGGGGGS SEQ ID No. 304 L046 SGGGGSGGGS SEQ ID No. 305
(89) Proteases are proteins that cleave proteins, in some cases, in a sequence-specific manner. Proteases include but are not limited to serine proteases, cysteine proteases, aspartate proteases, threonine proteases, glutamic acid proteases, metalloproteases, asparagine peptide lyases, serum proteases, cathepsins, Cathepsin B, Cathepsin C, Cathepsin D, Cathepsin E, Cathepsin K, Cathepsin L, kallikreins, hK1, hK10, hK15, plasmin, collagenase, Type IV collagenase, stromelysin, Factor Xa, chymotrypsin-like protease, trypsin-like protease, elastase-like protease, subtilisin-like protease, actinidain, bromelain, calpain, caspases, caspase-3, Mir1-CP, papain, HIV-1 protease, HSV protease, CMV protease, chymosin, renin, pepsin, matriptase, legumain, plasmepsin, nepenthesin, metalloexopeptidases, metal loendopeptidases, matrix metalloproteases (MMP), MMP1, MMP2, MMP3, MMP7, MMP8, MMP9, MMP13, MMP11, MMP14, urokinase plasminogen activator (uPA), enterokinase, prostate-specific antigen (PSA, hK3), interleukin-1β converting enzyme, thrombin, FAP (FAP-α), dipeptidyl peptidase, type II transmembrane serine proteases (TTSP), neutrophil serine protease, cathepsin G, proteinase 3, neutrophil serine protease 4, mast cell chymase, and mast cell tryptases.
(90) TABLE-US-00004 TABLE 3 Exemplary Proteases and Protease Recognition Sequences SEQ Cleavage ID Protease Domain Sequence NO: MMP7 KRALGLPG 1 MMP7 (DE).sub.8RPLALWRS(DR).sub.8 2 MMP9 PR(S/T)(L/I)(S/T) 3 MMP9 LEATA 4 MMP11 GGAANLVRGG 5 MMP14 SGRIGFLRTA 6 MMP PLGLAG 7 MMP PLGLAX 8 MMP PLGC(me)AG 9 MMP ESPAYYTA 10 MMP RLQLKL 11 MMP RLQLKAC 12 MMP2, MMP9, MMP14 EP(Cit)G(Hof)YL 13 Urokinase plasminogen SGRSA 14 activator (uPA) Urokinase plasminogen DAFK 15 activator (uPA) Urokinase plasminogen GGGRR 16 activator (uPA) Lysosomal Enzyme GFLG 17 Lysosomal Enzyme ALAL 18 Lysosomal Enzyme FK 19 Cathepsin B NLL 20 Cathepsin D PIC(Et)FF 21 Cathepsin K GGPRGLPG 22 Prostate Specific Antigen HSSKLQ 23 Prostate Specific Antigen HSSKLQL 24 Prostate Specific Antigen HSSKLQEDA 25 Herpes Simplex Virus LVLASSSFGY 26 Protease HIV Protease GVSQNYPIVG 27 CMV Protease GVVQASCRLA 28 Thrombin F(Pip)RS 29 Thrombin DPRSFL 30 Thrombin PPRSFL 31 Caspase-3 DEVD 32 Caspase-3 DEVDP 33 Caspase-3 KGSGDVEG 34 Interleukin 1β converting GWEHDG 35 enzyme Enterokinase EDDDDKA 36 FAP KQEQNPGST 37 Kallikrein 2 GKAFRR 38 Plasmin DAFK 39 Plasmin DVLK 40 Plasmin DAFK 41 TOP ALLLALL 42 MMP9 + matriptase KPLGLQARVV 58 MMP9 + matriptase + uPA PQASTGRSGG 59 MMP9 + matriptase + uPA PQGSTGRAAG 60 Matriptase + uPA PPASSGRAGG 61 MMP9 + matriptase PIPVQGRAH 62 MMP9 + matriptase PQGSTARSAG 909
(91) Proteases are known to be secreted by some diseased cells and tissues, for example tumor or cancer cells, creating a microenvironment that is rich in proteases or a protease-rich microenvironment. In some case, the blood of a subject is rich in proteases. In some cases, cells surrounding the tumor secrete proteases into the tumor microenvironment. Cells surrounding the tumor secreting proteases include but are not limited to the tumor stromal cells, myofibroblasts,

blood cells, mast cells, B cells, NK cells, regulatory T cells, macrophages, cytotoxic T lymphocytes, dendritic cells, mesenchymal stem cells, polymorphonuclear cells, and other cells. In some cases, proteases are present in the blood of a subject, for example proteases that target amino acid sequences found in microbial peptides. This feature allows for targeted therapeutics such as antigen binding proteins to have additional specificity because T cells will not be bound by the antigen binding protein except in the protease rich microenvironment of the targeted cells or tissue.
(92) The binding moiety comprising the cleavable linker thus masks the binding of a first or a second target antigen binding domain to their respective targets. In some embodiments, the binding moiety is bound to a first target antigen binding domain, which is further bound to a second target antigen binding domain, in the following order: binding moiety (M): cleavable linker (L): first target antigen binding domain (T1): second antigen binding domain (T2). In other examples, the domains are organized in any one of the following orders: M:L:T2:T1; T2:T1:L:M, T1:T2:L:M. The binding moiety is further bound to a half-life extending protein, such as albumin or any other of its targets as described below. In some instances, the binding moiety is albumin or comprises a binding site for albumin. In some instances the binding moiety comprises a binding site for IgE. In some embodiments, the binding moiety comprises a binding site for Igκ free light chain.
(93) TABLE-US-00005 TABLE 4 Exemplary sequences for the binding moieties comprising a cleavable linker are provided in SEQ ID Nos. 796-800. Sequence SEQ ID No. 796
EVOLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQGGGGGLDGNE
EPGGLEWVSSISGSGRDTLYADSVKGRFTISRDNAKTTLYLQMNSLRPE
DTAVYYCTIGGSLSVSSQGTLVTVSSGGGGKPLGLQARVVGGGGT SEQ ID No. 797
EVOLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQGGGGGLDGNE
EPGGLEWVSSISGSGRDTLYADSVKGRFTISRDNAKTTLYLQMNSLRPE
DTAVYYCTIGGSLSVSSQGTLVTVSSGGGGPQASTGRSGGGGGGT SEQ ID No. 798
EVOLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQGGGGGLDGNE
EPGGLEWVSSISGSGRDTLYADSVKGRFTISRDNAKTTLYLQMNSLRPE
DTAVYYCTIGGSLSVSSQGTLVTVSSGGGGPQGSTGRAAGGGGGT SEQ ID No. 799
EVOLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQGGGGGLDGNE
EPGGLEWVSSISGSGRDTLYADSVKGRFTISRDNAKTTLYLQMNSLRPE
DTAVYYCTIGGSLSVSSQGTLVTVSSGGGGPPASSGRAGGGGGT SEQ ID No. 800
EVOLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQGGGGGLDGNE
EPGGLEWVSSISGSGRDTLYADSVKGRFTISRDNAKTTLYLQMNSLRPE
DTAVYYCTIGGSLSVSSQGTLVTVSSGGGGPIPVQGRAHGGGGT
Targets of Conditionally Active Binding Proteins
(94) The conditionally active binding proteins described herein are activated by cleavage of the at least one cleavable linker attached to the binding moieties within said conditionally active proteins. It is contemplated that in some cases the activated binding protein binds to a target antigen involved in and/or associated with a disease, disorder or condition. In particular, target antigens associated with a proliferative disease, a tumorous disease, an inflammatory disease, an immunological disorder, an autoimmune disease, an infectious disease, a viral disease, an allergic reaction, a parasitic reaction, a graft-versus-host disease or a host-versus-graft disease are contemplated to be the target for the activated binding proteins disclosed herein.
(95) In some embodiments, the target antigen is a tumor antigen expressed on a tumor cell. Tumor antigens are well known in the art and include, for example, EpCAM, EGFR, HER-2, HER-3, c-Met, FoIR, PSMA, CD38, BCMA, and CEA. 5T4, AFP, B7-H3, CDH-6, CAIX, CD117, CD123, CD138, CD166, CD19, CD20, CD205, CD22, CD30, CD33, CD352, CD37, CD44, CD52, CD56, CD70, CD71, CD74, CD79b, DLL3, EphA2, FAP, FGFR2, FGFR3, GPC3, gpA33, FLT-3, gpNMB, HPV-16 E6, HPV-16 E7, ITGA2, ITGA3, SLC39A6, MAGE, mesothelin, Muc1, Muc16, NaPi2b, Nectin-4, CDH-3, CDH-17, EPHB2, ITGAV, ITGB6, NY-ESO-1, PRLR, PSCA, PTK7,

ROR1, SLC44A4, SLITRK5, SLITRK6, STEAP1, TIM1, Trop2, or WT1.
(96) In some embodiments, the target antigen is an immune checkpoint protein. Examples of immune checkpoint proteins include but are not limited to CD27, CD137, 2B4, TIGIT, CD155, ICOS, HVEM, CD40L, LIGHT, OX40, DNAM-1, PD-L1, PD1, PD-L2, CTLA-4, CD8, CD40, CEACAM1, CD48, CD70, A2AR, CD39, CD73, B7-H3, B7-H4, BTLA, IDO1, IDO2, TDO, KIR, LAG-3, TIM-3, or VISTA.
(97) In some embodiments, a target antigen is a cell surface molecule such as a protein, lipid or polysaccharide. In some embodiments, a target antigen is a on a tumor cell, virally infected cell, bacterially infected cell, damaged red blood cell, arterial plaque cell, inflamed or fibrotic tissue cell.
(98) In some embodiments, the target antigen comprises an immune response modulator that is not a cytokine. Examples of immune response modulator include but are not limited to B7-1 (CD80), B7-2 (CD86), CD3, or GITR.
(99) In some embodiments, the first target antigen binding domain or the second target antigen binding domain comprises an anti-EGFR domain, an anti-EpCAM domain, an anti-DLL3 domain, an anti-MSLN domain, an anti-PSMA domain, an anti-BDMA domain, or any combinations thereof. In some embodiments, the first target antigen binding domain or the second target antigen binding domain comprises an anti-EGFR sdAb, an anti-EpCAM sdAb, an anti-DLL3 sdAb, an anti-MSLN sdAb, an anti-PSMA sdAb, an anti-BDMA sdAb, or any combinations thereof.
(100) In some embodiments, an anti-EGFR domain of this disclosure comprises an amino acid selected from the group consisting of SEQ ID Nos. 55, and 737-785. In some embodiments, an anti-PSMA domain of this disclosure comprises an amino acid selected from the group consisting of SEQ ID Nos. 57-73. In some embodiments, an anti-BCMA domain of this disclosure comprises an amino acid selected from the group consisting of SEQ ID Nos. 91-214. In some embodiments, an anti-MSLN domain of this disclosure comprises an amino acid selected from the group consisting of SEQ ID Nos. 215-258. In some embodiments, an anti-DLL3 domain of this disclosure comprises an amino acid selected from the group consisting of SEQ ID Nos. 306-736. In some embodiments, an anti-EpCAM domain of this disclosure comprises an amino acid selected from the group consisting of SEQ ID Nos. 804-841.
(101) In some embodiments, the first target antigen binding domain or the second target antigen binding domain comprises an anti-CD3 domain. In some embodiments, the anti-CD3 domain comprises an anti-CD3 scFV. In some embodiments, the anti-CD3 scFv comprises an amino acid sequence selected from the group consisting of: SEQ ID Nos. 74-90, and 794.
(102) Binding Protein Variants
(103) As used herein, the term “binding protein variants” refers to variants and derivatives of the conditionally active target-binding proteins described herein, containing a binding moiety as described above, comprising non-CDR loops that bind to an immunoglobulin binding molecule, such as a first or a second target antigen binding domain, a non-immunoglobulin binding molecule. In certain embodiments, amino acid sequence variants of the conditionally active target-binding proteins described herein are contemplated. For example, in certain embodiments amino acid sequence variants of the conditionally active target-binding proteins described herein are contemplated to improve the binding affinity and/or other biological properties of the binding proteins. Exemplary method for preparing amino acid variants include, but are not limited to, introducing appropriate modifications into the nucleotide sequence encoding the antibody, or by peptide synthesis. Such modifications include, for example, deletions from, and/or insertions into and/or substitutions of residues within the amino acid sequences of the antibody.
(104) Any combination of deletion, insertion, and substitution can be made to the various domains to arrive at the final construct, provided that the final construct possesses the desired characteristics, e.g., antigen-binding. In certain embodiments, binding protein variants having one or more amino acid substitutions are provided. Sites of interest for substitution mutagenesis include

the CDRs and framework regions. Amino acid substitutions may be introduced into the variable domains of a conditionally active protein of interest and the products screened for a desired activity, e.g., retained/improved antigen binding, decreased immunogenicity, or improved antibody-dependent cell mediated cytotoxicity (ADCC) or complement dependent cytotoxicity (CDC). Both conservative and non-conservative amino acid substitutions are contemplated for preparing the antibody variants.

(105) In another example of a substitution to create a variant conditionally active antibody, one or more hypervariable region residues of a parent antibody are substituted. In general, variants are then selected based on improvements in desired properties compared to a parent antibody, for example, increased affinity, reduced affinity, reduced immunogenicity, increased pH dependence of binding. For example, an affinity matured variant antibody can be generated, e.g., using phage display-based affinity maturation techniques such as those described herein and known in the field.

(106) In another example, substitutions are made in hypervariable regions (HVR) of a parent conditionally active antibody to generate variants and variants are then selected based on binding affinity, i.e., by affinity maturation. In some embodiments of affinity maturation, diversity is introduced into the variable genes chosen for maturation by any of a variety of methods (e.g., error-prone PCR, chain shuffling, or oligonucleotide-directed mutagenesis). A secondary library is then created. The library is then screened to identify any antibody variants with the desired affinity. Another method to introduce diversity involves HVR-directed approaches, in which several HVR residues (e.g., 4-6 residues at a time) are randomized. HVR residues involved in antigen binding may be specifically identified, e.g., using alanine scanning mutagenesis or modeling. Substitutions can be in one, two, three, four, or more sites within a parent antibody sequence.

(107) In some embodiments, a conditionally active binding protein, as described herein comprises a VL domain, or a VH domain, or both, with amino acid sequences corresponding to the amino acid sequence of a naturally occurring VL or VH domain, respectively, but that has been “humanized”, i.e., by replacing one or more amino acid residues in the amino acid sequence of said naturally occurring VL or VH domains (and in particular in the framework sequences) by one or more of the amino acid residues that occur at the corresponding position(s) in a VL or VH domain from a conventional 4-chain antibody from a human being (e.g., as indicated above). This can be performed in a manner known in the field, which will be clear to the skilled person, for example on the basis of the further description herein. Again, it should be noted that such humanized conditionally active target-binding antibodies of the disclosure are obtained in any suitable manner known per se and thus are not strictly limited to polypeptides that have been obtained using a polypeptide that comprises a naturally occurring VL and/or VH domain as a starting material. In some additional embodiments, an conditionally active target-binding antibody, as described herein, comprises a VL and a VH domain with amino acid sequences corresponding to the amino acid sequence of a naturally occurring VL or VH domain, respectively, but that has been “camelized”, i.e., by replacing one or more amino acid residues in the amino acid sequence of a naturally occurring VL or VH domain from a conventional 4-chain antibody by one or more of the amino acid residues that occur at the corresponding position(s) in a VL or a VH domain of a heavy chain antibody. Such “camelizing” substitutions are preferably inserted at amino acid positions that form and/or are present at the VH-VL interface, and/or at the so-called Camelidae hallmark residues (see for example WO 94/04678 and Davies and Riechmann (1994 and 1996)). Preferably, the VH sequence that is used as a starting material or starting point for generating or designing the camelized single domain is preferably a VH sequence from a mammal, more preferably the VH sequence of a human being, such as a VH3 sequence. However, it should be noted that such camelized conditionally active antibodies of the disclosure, in certain embodiments, are obtained in any suitable manner known in the field and thus are not strictly limited to polypeptides that have been obtained using a polypeptide that comprises a naturally occurring VL and/or VH domain as a starting material. For example, both “humanization” and “camelization” is performed by providing

a nucleotide sequence that encodes a naturally occurring VL and/or VH domain, respectively, and then changing, one or more codons in said nucleotide sequence in such a way that the new nucleotide sequence encodes a “humanized” or “camelized” conditionally active antibody, respectively. This nucleic acid can then be expressed, so as to provide the desired target-antigen binding capability. Alternatively, in other embodiments, a “humanized” or “camelized” conditionally active antibody is synthesized de novo using known peptide synthesis technique from the amino acid sequence of a naturally occurring antibody comprising a VL and/or VH domain. In some embodiments, a “humanized” or “camelized” conditionally active antibody is synthesized de novo using known peptide synthesis technique from the amino acid sequence or nucleotide sequence of a naturally occurring antibody comprising a VL and/or VH domain, respectively, a nucleotide sequence encoding the desired humanized or camelized conditionally active domain antibody of the disclosure, respectively, is designed and then synthesized de novo using known techniques for nucleic acid synthesis, after which the nucleic acid thus obtained is expressed in using known expression techniques, so as to provide the desired conditionally active antibody of the disclosure.

(108) Other suitable methods and techniques for obtaining the conditionally active binding protein of the disclosure and/or nucleic acids encoding the same, starting from naturally occurring sequences for VL or VH domains for example comprises combining one or more parts of one or more naturally occurring VL or VH sequences (such as one or more framework (FR) sequences and/or complementarity determining region (CDR) sequences), and/or one or more synthetic or semi-synthetic sequences, and/or a naturally occurring sequence for a CH2 domain, and a naturally occurring sequence for a CH3 domain comprising amino acid substitutions that favor formation of heterodimer over homodimer, in a suitable manner, so as to provide a conditionally active binding protein of the disclosure or a nucleotide sequence or nucleic acid encoding the same.

(109) Affinity Maturation

(110) In designing conditionally active binding proteins for therapeutic applications, it is desirable to create proteins that, for example, modulate a functional activity of a target, and/or improved binding proteins such as binding proteins with higher specificity and/or affinity and/or and binding proteins that are more bioavailable, or stable or soluble in particular cellular or tissue environments.

(111) The conditionally active binding proteins described in the present disclosure exhibit improved the binding affinities towards the target, for example a tumor antigen expressed on a cell surface. In some embodiments, the conditionally active binding protein of the present disclosure is affinity matured to increase its binding affinity to the target, using any known technique for affinity-maturation (e.g., mutagenesis, chain shuffling, CDR amino acid substitution). Amino acid substitutions may be conservative or semi-conservative. For example, the amino acids glycine, alanine, valine, leucine and isoleucine can often be substituted for one another (amino acids having aliphatic side chains). Of these possible substitutions, typically glycine and alanine are used to substitute for one another since they have relatively short side chains and valine, leucine and isoleucine are used to substitute for one another since they have larger aliphatic side chains which are hydrophobic. Other amino acids which may often be substituted for one another include but are not limited to: phenylalanine, tyrosine and tryptophan (amino acids having aromatic side chains); lysine, arginine and histidine (amino acids having basic side chains); aspartate and glutamate (amino acids having acidic side chains); asparagine and glutamine (amino acids having amide side chains); and cysteine and methionine (amino acids having sulphur-containing side chains). In some embodiments, the conditionally active target-binding proteins are isolated by screening combinatorial libraries, for example, by generating phage display libraries and screening such libraries for antibodies possessing the desired binding characteristics towards a target antigen, such as a tumor antigen expressed on a cell surface.

(112) Conditionally Active Binding Protein Modifications

(113) The conditionally active binding proteins described herein encompass derivatives or analogs

in which (i) an amino acid is substituted with an amino acid residue that is not one encoded by the genetic code, (ii) the mature polypeptide is fused with another compound such as polyethylene glycol, or (iii) additional amino acids are fused to the protein, such as a leader or secretory sequence or a sequence to block an immunogenic domain and/or for purification of the protein.
(114) Typical modifications include, but are not limited to, acetylation, acylation, ADP-ribosylation, amidation, covalent attachment of flavin, covalent attachment of a heme moiety, covalent attachment of a nucleotide or nucleotide derivative, covalent attachment of a lipid or lipid derivative, covalent attachment of phosphatidylinositol, cross-linking, cyclization, disulfide bond formation, demethylation, formation of covalent crosslinks, formation of cystine, formation of pyroglutamate, formylation, gamma carboxylation, glycosylation, GPI anchor formation, hydroxylation, iodination, methylation, myristylation, oxidation, proteolytic processing, phosphorylation, prenylation, racemization, selenoylation, sulfation, transfer-RNA mediated addition of amino acids to proteins such as arginylation, and ubiquitination.
(115) Modifications are made anywhere in the conditionally active binding proteins described herein, including the peptide backbone, the amino acid side-chains, and the amino or carboxyl termini. Certain common peptide modifications that are useful for modification of the conditionally active binding proteins include glycosylation, lipid attachment, sulfation, gamma-carboxylation of glutamic acid residues, hydroxylation, blockage of the amino or carboxyl group in a polypeptide, or both, by a covalent modification, and ADP-ribosylation.
(116) In some embodiments, the conditionally active binding proteins of the disclosure are conjugated with drugs to form antibody-drug conjugates (ADCs). In general, ADCs are used in oncology applications, where the use of antibody-drug conjugates for the local delivery of cytotoxic or cytostatic agents allows for the targeted delivery of the drug moiety to tumors, which can allow higher efficacy, lower toxicity, etc.
(117) Polynucleotides Encoding the Binding Moieties or the Conditionally Active Binding Proteins
(118) Also provided, in some embodiments, are polynucleotide molecules encoding the binding moieties as described herein. In some embodiments, the polynucleotide molecules are provided as a DNA construct. In other embodiments, the polynucleotide molecules are provided as a messenger RNA transcript.
(119) Also provided, in some embodiments, are polynucleotide molecules encoding the conditionally active binding proteins as described herein. In some embodiments, the polynucleotide molecules are provided as a DNA construct. In other embodiments, the polynucleotide molecules are provided as a messenger RNA transcript.
(120) The polynucleotide molecules are constructed by known methods such as by combining the genes encoding the various domains (e.g. binding moiety, target antigen binding domain, etc.) either separated by peptide linkers or, in other embodiments, directly linked by a peptide bond, into a single genetic construct operably linked to a suitable promoter, and optionally a suitable transcription terminator, and expressing it in bacteria or other appropriate expression system such as, for example CHO cells. Depending on the vector system and host utilized, any number of suitable transcription and translation elements, including constitutive and conditionally active promoters, may be used. The promoter is selected such that it drives the expression of the polynucleotide in the respective host cell.
(121) In some embodiments, the polynucleotides described herein are inserted into vectors, such as expression vectors, which represent further embodiments. This recombinant vector can be constructed according to known methods. Vectors of particular interest include plasmids, phagemids, phage derivatives, virii (e.g., retroviruses, adenoviruses, adeno-associated viruses, herpes viruses, lentiviruses, and the like), and cosmids.
(122) A variety of expression vector/host systems may be utilized to contain and express the polynucleotide encoding the polypeptide of the described conditionally active binding protein. Examples of expression vectors for expression in *E. coli* are pSKK (Le Gall et al., J Immunol

Methods. (2004) 285(1):111-27) or pcDNA5 (Invitrogen) for expression in mammalian cells.
(123) Thus, the binding moieties or the conditionally active binding proteins comprising the binding moieties as described herein, in some embodiments, are produced by introducing vectors encoding the binding moieties or the binding proteins as described above into host cells and culturing said host cells under conditions whereby the binding moieties or the binding proteins, or domains thereof are expressed.
(124) Pharmaceutical Compositions
(125) Also provided, in some embodiments, are pharmaceutical compositions comprising a therapeutically effective amount of a conditionally active binding protein of the present disclosure, and at least one pharmaceutically acceptable carrier. The term "pharmaceutically acceptable carrier" includes, but is not limited to, any carrier that does not interfere with the effectiveness of the biological activity of the ingredients and that is not toxic to the patient to whom it is administered. Examples of suitable pharmaceutical carriers are well known in the art and include phosphate buffered saline solutions, water, emulsions, such as oil/water emulsions, various types of wetting agents, sterile solutions etc. Such carriers can be formulated by conventional methods and can be administered to the subject at a suitable dose. Preferably, the compositions are sterile. These compositions may also contain adjuvants such as preservative, emulsifying agents and dispersing agents. Prevention of the action of microorganisms is, in some cases, ensured by the inclusion of various antibacterial and antifungal agents.
(126) The conditionally active binding proteins described herein are contemplated for use as medicaments. Administration is effected by different ways, e.g., by intravenous, intraperitoneal, subcutaneous, intramuscular, topical or intradermal administration. In some embodiments, the route of administration depends on the kind of therapy and the kind of compound contained in the pharmaceutical composition. The dosage regimen will be determined by the attending physician and other clinical factors. Dosages for any one patient depends on many factors, including the patient's size, body surface area, age, sex, the particular compound to be administered, time and route of administration, the kind of therapy, general health and other drugs being administered concurrently. An "effective dose" refers to amounts of the active ingredient that are sufficient to affect the course and the severity of the disease, leading to the reduction or remission of such pathology and may be determined using known methods.
(127) Methods of Treatment
(128) Also provided herein, in some embodiments, are methods and uses for stimulating the immune system of an individual in need thereof comprising administration of a conditionally active binding protein as described herein. In some instances, administration induces and/or sustains cytotoxicity towards a cell expressing a target antigen. In some instances, the cell expressing a target antigen is a cancer or tumor cell, a virally infected cell, a bacterially infected cell, an autoreactive T or B cell, damaged red blood cells, arterial plaques, or fibrotic tissue. In some embodiments, the target antigen is an immune checkpoint protein.
(129) Also provided herein are methods and uses for a treatment of a disease, disorder or condition associated with a target antigen comprising administering to an individual in need thereof a conditionally active binding protein as described herein. Diseases, disorders or conditions associated with a target antigen include, but are not limited to, viral infection, bacterial infection, auto-immune disease, transplant rejection, atherosclerosis, or fibrosis. In other embodiments, the disease, disorder or condition associated with a target antigen is a proliferative disease, a tumorous disease, an inflammatory disease, an immunological disorder, an autoimmune disease, an infectious disease, a viral disease, an allergic reaction, a parasitic reaction, a graft-versus-host disease or a host-versus-graft disease. In one embodiment, the disease, disorder or condition associated with a target antigen is cancer. In one instance, the cancer is a hematological cancer. In another instance, the cancer is a melanoma. In a further instance, the cancer is non-small cell lung cancer. In yet further instance, the cancer is breast cancer.

(130) As used herein, in some embodiments, "treatment" or "treating" or "treated" refers to therapeutic treatment wherein the object is to slow (lessen) an undesired physiological condition, disorder or disease, or to obtain beneficial or desired clinical results. For the purposes described herein, beneficial or desired clinical results include, but are not limited to, alleviation of symptoms; diminishment of the extent of the condition, disorder or disease; stabilization (i.e., not worsening) of the state of the condition, disorder or disease; delay in onset or slowing of the progression of the condition, disorder or disease; amelioration of the condition, disorder or disease state; and remission (whether partial or total), whether detectable or undetectable, or enhancement or improvement of the condition, disorder or disease. Treatment includes eliciting a clinically significant response without excessive levels of side effects. Treatment also includes prolonging survival as compared to expected survival if not receiving treatment. In other embodiments, "treatment" or "treating" or "treated" refers to prophylactic measures, wherein the object is to delay onset of or reduce severity of an undesired physiological condition, disorder or disease, such as, for example is a person who is predisposed to a disease (e.g., an individual who carries a genetic marker for a disease such as breast cancer).
(131) In some embodiments of the methods described herein, the conditionally active binding proteins described herein are administered in combination with an agent for treatment of the particular disease, disorder or condition. Agents include but are not limited to, therapies involving antibodies, small molecules (e.g., chemotherapeutics), hormones (steroidal, peptide, and the like), radiotherapies (y-rays, X-rays, and/or the directed delivery of radioisotopes, microwaves, UV radiation and the like), gene therapies (e.g., antisense, retroviral therapy and the like) and other immunotherapies. In some embodiments, the conditionally active binding proteins described herein are administered in combination with anti-diarrheal agents, anti-emetic agents, analgesics, opioids and/or non-steroidal anti-inflammatory agents. In some embodiments, the conditionally active binding proteins described herein is administered before, during, or after surgery.
EXAMPLES
(132) The examples below further illustrate the described embodiments without limiting the scope of the disclosure.
Example 1: Construction of an Exemplary Binding Moiety which Binds to Albumin and a Target Antigen Binding Domain Whose Target is EGFR
(133) The sequence of an engineered protein scaffold comprising CDR loops capable of binding albumin and non-CDR loops is obtained. Overlapping PCR is used to introduce random mutations in the non-CDR loop regions, thereby generating a library. The resultant sequences are cloned into a phage display vector, thereby generating a phage display library. *Escherichia coli* cells are transformed with the library and used to construct a phage display library. ELISA is performed using an immobilized target antigen binding domain with specificity for EGFR. A clone with high specificity for EGFR is selected. Affinity maturation is performed by re-randomizing residues in the non-CDR loop regions as before.
(134) Sequence alignment of non-CDR loop regions of the resultant proteins is performed to determine sequence conservation between proteins with high affinity for the EGFR binding target antigen binding domain. Site directed mutagenesis of one or more amino acids within these regions of sequence conservation is performed to generate additional proteins. Binding of the resultant proteins to an immobilized target antigen binding domain whose target is EGFR is measured in an ELISA. A protein with the highest affinity for the target antigen binding domain is selected.
(135) The sequence of this binding moiety is cloned into a vector comprising a sequence for a cleavable linker, and sequences for a second target antigen binding domain that binds to a second target antigen, e.g., CD3. The resultant vector is expressed in a heterologous expression system to obtain a conditionally active target binding protein that comprises a binding moiety comprising a cleavable linker and non-CDR loops which provide a binding site specific for the target antigen binding domain whose target is EGFR, and CDR loops which are specific for albumin.

Example 2: Construction of an Exemplary Binding Moiety which Binds to Albumin and a Target Antigen Binding Domain Whose Target is CD3

(136) The sequence of an engineered protein scaffold comprising CDR loops capable of binding albumin and non-CDR loops is obtained. Overlapping PCR is used to introduce random mutations in the non-CDR loop regions, thereby generating a library. The resultant sequences are cloned into a phage display vector, thereby generating a phage display library. *Escherichia coli* cells are transformed with the library and used to construct a phage display library. ELISA is performed using an immobilized target antigen binding domain with specificity for CD3. A clone with high specificity for CD3 is selected. Affinity maturation is performed by re-randomizing residues in the non-CDR loop regions as before.

(137) Sequence alignment of non-CDR loop regions of the resultant proteins is performed to determine sequence conservation between proteins with high affinity for the EGFR binding target antigen binding domain. Site directed mutagenesis of one or more amino acids within these regions of sequence conservation is performed to generate additional proteins. Binding of the resultant proteins to an immobilized target antigen binding domain whose target is CD3 is measured in an ELISA. A protein with the highest affinity for the target antigen binding domain is selected.

(138) The sequence of this binding moiety is cloned into a vector comprising a sequence for a cleavable linker, and sequences for a second target antigen binding domain that binds to a second target antigen, e.g., EGFR. The resultant vector is expressed in a heterologous expression system to obtain a conditionally active target binding protein that comprises a binding moiety comprising a cleavable linker and non-CDR loops which provide a binding site specific for the target antigen binding domain whose target is CD3, and CDR loops which are specific for albumin.

Example 3: A Conditionally Active Binding Protein of this Disclosure Exhibit Reduced Specificity Toward Cell Line which Overexpresses EGFR but is Protease Deficient

(139) Cells overexpressing EGFR and exhibiting low expression of a matrix metalloprotease are separately incubated with an exemplary conditionally active binding protein according to the present disclosure and a non-conditionally active control binding protein. Cells expressing normal levels of EGFR and proteases are also incubated with a conditionally active binding protein according to the present disclosure and a non-conditionally active control binding protein. Both proteins comprise a target antigen binding domain with specificity toward PSMA.

(140) Results indicate that in the absence of protease secretion, the conditionally active binding protein of the present disclosure interacts with the protease expressing cells but does not interact with the EGFR expressed on the surface of the protease deficient cells. In contrast, the non-conditionally active control binding protein lacks the ability to selectively bind the protease expressing cells over the protease deficient ones. Thus, the exemplary conditionally active binding protein receptor of the present disclosure is advantageous, for example, in terms of reducing off-tumor toxicity.

Example 4: Treatment with an Exemplary Conditionally Active Binding Protein of the Present Disclosure Inhibits In Vivo Tumor Growth

(141) Murine tumor line CT26 is implanted subcutaneously in Balb/c mice and on day 7 post-implantation the average size of the tumor is measured. Test mice are treated with an exemplary conditionally active binding protein which has a target antigen binding domain specific for CTLA4 and another target antigen binding domain specific for CD3, wherein either the CTLA4 or the CD3 specific domain is bound to a binding moiety via its non-CDR loops, the binding moiety comprises a cleavable linker, and is bound to albumin. Control mice are treated with binding protein that contains CD3/CTLA4 specific domains but do not contain the binding moiety or the cleavable linker, and are not conditionally active. Results show that treatment with the exemplary conditionally active binding protein of the present disclosure inhibits tumor more efficiently than the comparator binding protein which does not contain the moiety with non-CDR loops.

Example 5: Exemplary Conditionally Active Binding Protein Exhibits Reduced Specificity Toward

Cell Line which Overexpresses Antigen but is Protease Deficient

(142) Cells overexpressing CTLA-4 and exhibiting low expression of a matrix metalloprotease are separately incubated with an exemplary CTLA4 specific conditionally active binding protein of this disclosure, containing a binding moiety that binds to a CTLA4 binding domain via its non-CDR loops and albumin via its CDRs; or a control CTLA-4 binding antibody which does not contain the binding moiety which binds to a CTLA4 binding domain via its non-CDR loops and to albumin via its CDRs. Cells expressing normal levels of antigens and proteases are also incubated with the exemplary CTLA4 specific conditionally active binding protein, or the control CTLA4 binding antibody.

(143) Results indicate that in the absence of protease secretion, the conditionally active binding protein of the present disclosure binds the protease expressing cells but does not bind the protease-deficient antigen expressing cells. In contrast, the control antibody lacks the ability to selectively bind the protease expressing cells over the protease deficient ones. Thus, the exemplary conditionally active binding protein of the present disclosure is advantageous, for example, in terms of reducing off-tumor toxicity.

Example 6: Purification, Pharmacokinetic Analysis, and Potency Assays for Exemplary ProTriTAC Molecules

(144) A) Expression, Purification and Stability of Exemplary ProTriTAC Molecules

(145) Protein Production

(146) Sequences of exemplary ProTriTAC (also referred to as Pro-trispecific) molecules were cloned into mammalian expression vector pcDNA 3.4 (Invitrogen) preceded by a leader sequence and followed by a 6× Histidine Tag (SEQ ID NO: 927). Expi293F cells (Life Technologies A14527) were maintained in suspension in Optimum Growth Flasks (Thomson) between 0.2 to 8×1e6 cells/ml in Expi 293 media. Purified plasmid DNA was transfected into Expi293 cells in accordance with Expi293 Expression System Kit (Life Technologies, A14635) protocols, and maintained for 4-6 days post transfection. Alternatively, sequences of trispecific molecules were cloned into mammalian expression vector pDEF38 (CMC ICOS) transfected into CHO-DG44 dhfr-cells, stable pools generated, and cultured in production media for up to 12 days prior to purification. The amount of the exemplary trispecific proteins in conditioned media was quantitated using an Octet RED 96 instrument with Protein A tips (ForteBio/Pall) using a control trispecific protein for a standard curve. Conditioned media from either host cell was filtered and partially purified by affinity and desalting chromatography. Trispecific proteins were subsequently polished by ion exchange and upon fraction pooling formulated in a neutral buffer containing excipients. Final purity was assessed by SDS-PAGE and analytical SEC using an Acquity BEH SEC 200 1.7u 4.6×150 mm column (Waters Corporation) resolved in an aqueous/organic mobile phase with excipients at neutral pH on a 1290 LC system and peaks integrated with Chemstation CDS software (Agilent). Trispecific proteins purified from CHO host cells were analyzed by running an SDS-PAGE, as shown in FIG. **5**.

(147) Stability Assessment

(148) Purified Protrispecific proteins in two formulations were sub-aliquoted into sterile tubes and stressed by five freeze-thaw cycles each comprising greater than 1 hour at −80° C. and room temperature or by incubation at 37° C. for 1 week. Stressed samples were evaluated for concentration and turbidity by UV spectrometry using UV transparent 96 well plates (Corning 3635) with a SpectraMax M2 and SoftMaxPro Software (Molecular Devices), SDS-PAGE, and analytical SEC and compared to the same analysis of control non-stressed samples. An overlay of chromatograms from analytical SEC of control and stressed samples for a single exemplary trispecific ProTriTAC molecule purified from 293 host cells is depicted in FIG. **6**.

(149) B) ProTriTAC Exhibits Potent, Protease-Dependent, Anti-Tumor Activity in a Rodent Tumor Xenograft Model

(150) An exemplary ProTriTAC molecule (SEQ ID NO: 46) containing an EGFR binding domain

as the target binding domain, a CD3 binding domain and an albumin binding domain comprising a masking moiety (SEQ ID NO: 50) and a cleavable linker (SEQ ID NO: 53) was evaluated for anti-tumor activity in vivo in an HCT116 subcutaneous xenograft tumor admixed with expanded human T cells in immunocompromised NCG mice. A non-cleavable EGFR targeting ProTriTAC molecule (SEQ ID NO: 47) and a GFP targeting ProTriTAC molecule (SEQ ID NO: 49) were also used in the study. Specifically, 5×10.sup.6 HCT116 cells were admixed with 2.5×10.sup.6 expanded T cells per mouse on day 0. Dosing of the test molecules (EGFR targeting ProTriTAC, non-cleavable EGFR targeting Pro-TriTAC, and GFP targeting ProTriTAC) were performed starting on the following day with a q.d.×10 (single daily dose for 10 days) schedule via intraperitoneal injection, at a dose of 0.03 mg/kg. Tumor volumes were determined using caliper measurements and calculated using the formula V=(length×width×width)/2, at the indicated times. Results shown in FIG. **7** indicate that following final dose of test molecules, on day 10, tumor growth was arrested in mice administered the activatable EGFR targeting ProTriTAC molecule. Whereas, administering the GFP targeting ProTriTAC molecule was not effective in inhibiting tumor growth and the administration of the EGFR targeting non-cleavable ProTriTAC molecule was not as potent as the activatable ProTriTAC, in arresting tumor growth.
(151) C) Demonstration of Functional Masking and Stability of ProTriTAC In Vivo in a Three-Week Cynomolgus Monkey Pharmacokinetic Study
(152) Single doses of PSMA targeting ProTriTAC (SEQ ID NO: 43) containing a PSMA binding domain as the target binding domain, a CD3 binding domain, and an albumin binding domain comprising a masking moiety (SEQ ID NO: 50) and a cleavable linker (SEQ ID NO: 53), non-cleavable PSMA targeting ProTriTAC (SEQ ID NO: 44), non-masked/non-cleavable TriTAC (SEQ ID NO: 52), and active drug mimicking protease-activated PSMA targeting ProTriTAC (SEQ ID NO: 45) were dosed into cynomolgus monkeys at 0.1 mg/kg via intravenous injection. Plasma samples were collected at the time points indicated in FIG. **9**. The designs of the above described test molecules are shown in FIG. **8**. Concentrations of the various test molecules, as described above, were determined using ligand binding assays with biotinylated recombinant human PSMA (R&D systems) and sulfo-tagged anti-CD3 idiotype antibody cloned 11D3 in a MSD assay (Meso Scale Diagnostic, LLC). Pharmacokinetic parameters were estimated using Phoenix WinNonlin pharmacokinetic software using a non-compartmental approach consistent with the intravenous bolus route of administration.
(153) To calculate the rate of in vivo conversion of the test molecules (i.e., conversion of PSMA targeting ProTriTAC, non-cleavable PSMA targeting ProTriTAC, non-masked/non-cleavable PSMA targeting ProTriTAC) the concentration of active drug in circulation was estimated by solving the following system of differential equations where P is the concentration of prodrug, A is the concentration of active drug, k.sub.a is the rate of prodrug activation in circulation, k.sub.c,P is the clearance rate of the prodrug, and k.sub.c,A is the clearance rate of the active drug.

(154) $$\frac{dP}{dt} = -k_{c,P}P$$
$$\frac{dA}{dt} = k_a P - k_{c,A}A$$

(155) The clearance rates of the prodrug, active drug, non-masked non-cleavable prodrug control, and a non-cleavable prodrug control (k.sub.c,NCLV) were determined empirically in cynomolgus monkeys. To estimate the rate of prodrug activation in circulation, it was assumed that the difference between the clearance rate of cleavable prodrug and the non-cleavable prodrug arose solely from non-specific activation in circulation. Therefore, the rate of prodrug conversion to active drug in circulation was estimated by subtracting the clearance rate of the cleavable prodrug from the non-cleavable prodrug.
k.sub.a=*k*.sub.c,NCLV−*k*.sub.c,P
(156) The initial concentration of prodrug in circulation was determined empirically and the initial concentration of active drug was assumed to be zero. Further calculations showed that the

ProTriTAC comprising the protease cleavable linker was sufficiently stable in circulation, with 50% non-tumor mediated conversion every 194 hours and the t.sub.1/2 of the molecule was determined, empirically, to be around 211 hours. This indicated that ProTriTAC molecules are sufficiently stable and protected against off-tumor effects. In contrast, the t.sub.1/2 of the active drug fragment mimicking the activated ProTRITAC molecule was determined, empirically, to be 0.97 hours. Thus, active drug was rapidly cleared from circulation. Results are shown in FIG. **10**.
(157) FIG. **11** shows that because the active drug mimicking the activated ProTriTAC molecule was rapidly cleared from circulation and that the ProTriTAC and the control non-masked non-cleavable ProTriTAC had longer half-lives, there is a significant >200-fold differential between the ProTriTAC and the activated ProTriTAC circulating exposure. It was also observed that the masked but non-cleavable ProTriTAC control molecule was present in circulation for a longer time than the non-masked non-cleavable ProTriTAC control, thereby indicating that masking plays a role in increased circulation half-life (t.sub.1/2) and limited peripheral T cell binding. The combination of the functional masking that renders ProTriTAC inert outside the tumor environment and the half-life differential that renders any aberrantly activated ProTriTAC in circulation to be rapidly cleared in vivo works together to ensure on-target, off-tumor activity of ProTriTAC is minimized. Supporting pharmacokinetic parameters are shown in Table 5.
(158) TABLE-US-00006 TABLE 5 Pharmacokinetics of ProTriTAC, Activated ProTriTAC and Control Molecules Terminal t.sub.1/2 C.sub.max AUC, 0-last Clearance Test Article (hr) (nM) (hr*nM) (mL/hr/kg) Control #1 (No mask, 118 48.2 2490 0.735 non-cleavable) Control #2 (Masked, 211 58.0 7000 0.238 non-cleavable) ProTriTAC 101 42.7 2670 0.686 Activated ProTriTAC 0.969 66.6 41.5 58.5
D) Protease Activation of ProTriTAC Molecule Leads to Significantly Enhanced Activity In Vitro
(159) The aim of this study was to assess the relative potency of protease activatable ProTriTAC molecules, non-cleavable ProTriTAC molecules and recombinant active drug fragment mimicking the protease-activated ProTriTAC molecule, in CD3 binding and T cell mediated cell killing. The active drug fragment mimicking the protease activated ProTriTAC molecule contained the CD3 binding domain and the target antigen binding domain but lacked the albumin binding domain. Whereas the protease activatable ProTriTAC molecule contained the albumin binding domain comprising a masking domain and a protease cleavable site, the CD3 binding domain, and the target antigen binding domain. The non-cleavable ProTriTAC molecule lacked the protease cleavable site but was otherwise identical to the protease activatable ProTriTAC molecule.
(160) Purified ProTriTAC (labeled as prodrug in FIGS. **12-14**), non-cleavable ProTriTAC (labeled as prodrug (non-cleavable) in FIGS. **12-14**), and recombinant active drug fragment mimicking the protease-activated ProTriTAC (labeled as active drug in FIGS. **12-14**) were tested for binding to recombinant human CD3 in an ELISA assay (FIG. **12**), binding to purified human primary T cells in a flow cytometry assay (FIG. **13**), and functional potency in a T cell-dependent cellular cytotoxicity assay (FIG. **14**).
(161) For ELISA, soluble test molecules (i.e., active drug, prodrug, and prodrug (non-cleavable) at the indicated concentrations were incubated, in multi-well plates, with immobilized recombinant human CD3ε (R&D Systems) for 1 hour at room temperature in PBS supplemented with 15 mg/mL human serum albumin. Plates were blocked using SuperBlock (Thermo Fisher), washed using PBS with 0.05% Tween-20, and detected using a non-competitive anti-CD3 idiotype monoclonal antibody 11D3 followed by peroxidase-labeled secondary antibody and TMB-ELISA substrate solution (Thermo Fisher). Results shown in FIG. **12** demonstrate that the active drug fragment mimicking the protease-activated ProTriTAC molecule was about 250 times more potent in binding CD3 as compared to prodrug non-cleavable. The EC.sub.50 values are provided in Table 6. The masking ratio is the ratio between the prodrug EC.sub.50 over the active drug EC.sub.50: the higher the number, the higher the fold-shift between prodrug and active drug and thus greater functional masking.

(162) TABLE-US-00007 TABLE 6 CD3 binding potential EC.sub.50 (nM) Masking Ratio Active Drug 0.16 — Prodrug 11.91 74 Prodrug (non-cleavable) 39.44 247

(163) For binding to human primary T cells, determined by flow cytometry, soluble test molecules (i.e., active drug, prodrug, and prodrug (non-cleavable)) at the indicated concentrations (shown in FIG. **13**) were incubated, in multi-well plates, with purified human primary T cells for 1 h at 4° C. in the presence of PBS with 2% fetal bovine serum and 15 mg/ml human serum albumin. Plates were washed with PBS with 2% fetal bovine serum, detected using AlexaFluor 647-labeled non-competitive anti-CD3 idiotype monoclonal antibody 11D3, and data was analyzed using FlowJo 10 (FlowJo, LLC). Results shown in FIG. **13** demonstrate that the active drug fragment mimicking the protease-activated ProTriTAC molecule was greater than 1000 times more potent in binding human primary T cells as compared to prodrug non-cleavable. The EC.sub.50 values are provided in Table 7.

(164) TABLE-US-00008 TABLE 7 Human primary T cell binding potential EC.sub.50 (nM) Masking Ratio Active Drug 1.19 — Prodrug >1000 n/a Prodrug (non-cleavable) >1000 n/a

(165) For functional potency in a T cell-dependent cellular cytotoxicity assays, soluble test molecules (i.e., active drug, prodrug, and prodrug (non-cleavable)) at the indicated concentrations, shown in FIG. **14**, were incubated, in multi-well plates, with purified resting human T cells (effector cell) and HCT116 cancer cell (target cell) at 10:1 effector:target cell ratio for 48 h at 37° C. The HCT116 target cell line had been stably transfected with a luciferase reporter gene to allow specific T cell-mediated cell killing measurement by ONE-Glo (Promega). Results shown in FIG. **14** demonstrate that the active drug fragment mimicking the protease-activated ProTriTAC molecule was about 500 times more potent in T cell mediated killing of cancer cells, as compared to prodrug non-cleavable. The EC.sub.50 values are provided in Table 8.

(166) TABLE-US-00009 TABLE 8 T cell mediated cell killing potential EC.sub.50 (nM) Masking Ratio Active Drug 0.004 — Prodrug 0.485 121 Prodrug (non-cleavable) 2.197 549

Example 7: Anti-Tumor Activity of Exemplary ProTriTAC Molecules Containing Various Exemplary Linkers, in an Admixed Mouse Tumor Model

(167) The aim of this study was to explore the anti-tumor activity of ProTriTAC molecules containing different linkers. NSG female mice, 7 weeks old, were used for this study. At the commencement of the study, on day 0, the NSG female mice were injected with 2.5×10.sup.6 expanded human T cells, and 5×10.sup.6 HCT116 (human colorectal carcinoma) tumor cells. The following day, on day 1, the mice were divided into groups and each group was treated with at least one of the ProTriTAC molecules listed in Table 9 (SEQ ID Nos. 786-790), or with a control GFP TriTAC molecule (SEQ ID No. 792), or with a ProTriTAC molecule that contains a non cleavable linker (NCLV) (SEQ ID No. 791).

(168) The ProTriTAC molecules and the ProTriTAC NCLV molecule used in the following examples were targeted to EGFR and had the following orientation of the individual domains: (anti-albumin binding domain (sdAb): anti-CD3 domain (scFV): anti-EGFR domain (sdAb)). The only differences between the ProTriTAC molecules listed in Table 6 were in the linker sequences. The ProTriTAC molecules, ProTriTAC NCLV molecule, or the GFP TriTAC molecule (the GFP TriTAC molecule had the following orientation of individual domains: anti-GFP sdAb: anti-Alb sdAb: anti-CD3 scFv) were administered daily for a period of 10 days (i.e., final dose was administered on day 10 following injection of tumor cells and expanded cells to the animals) and tumor volumes were measured at regular intervals, beginning a few days prior to the administration of the final dose at day 10.

(169) TABLE-US-00010 TABLE 9 ProTriTAC sequences and linkers Cleavability/ Linker Recognition ProTriTAC in the of the linker molecule ProTriTAC Linker sequence by sequence molecule Sequence enzymes SEQ ID L001 (SEQ KPLGLQARVV MMP9 + matriptase No. 786 ID NO: 58) SEQ ID L040 (SEQ PQASTGRSGG MMP9 + No. 787 ID NO: 59) matriptase + uPA SEQ ID

L041 (SEQ PQGSTGRAAG MMP9 + No. 788 ID NO: 60) matriptase + uPA
SEQ ID L042 (SEQ PPASSGRAGG matriptase + uPA No. 789 ID NO: 61)
SEQ ID L045 (SEQ PIPVQGRAH MMP9 + matriptase No. 790 ID NO: 62)
L043 (SEQ PQGSTARSAG MMP9 + matriptase ID NO: 909)

(170) As shown in FIG. **18**, the ProTriTAC molecules containing linker sequences L001, L045, L040, and L041 demonstrated more potent anti-tumor activity as compared to the GFP TriTAC control of the ProTriTAC NCLV molecule. The statistical significance of the data was determined by repeated measures one-way ANOVA-Dunnett post test. The mean tumor volume of each group of mice were compared to the mean tumor volume of the mice group that received the NCLV molecule.

(171) The pharmacokinetics following administration of the various molecules, as described above, were also assessed and the data is shown in FIG. **19**. The control GFP TriTAC molecule was rapidly cleared from circulation, following administration, whereas the NCLV molecule remained in circulation for the longest time. The pharmacokinetic clearance profile of the test ProTriTAC molecules were in between the GFP TriTAC and NCLV, except for the ProTriTAC containing the linker sequence L001, which was cleared almost as rapidly as the control GFP TriTAC.

Example 8: Individual Tumor Volumes of Admixed Xenograft Tumors, Following Treatment with Exemplary ProTriTAC Molecules Containing Various Exemplary Linkers

(172) The ProTriTAC molecules listed in Table 9, the control GFP TriTAC molecule, and the ProTriTAC NCLV molecule were evaluated in an admixed xenograft model, in order to determine the efficacy of the ProTriTAC molecules containing different linkers, in vivo. As described in previous example (Example 7), the xenograft tumor model was generated by injecting 7 week old NSG mice with 2.5×10.sup.6 expanded human T cells, and 5×10.sup.6 HCT116 (human colorectal carcinoma) tumor cells. The mice were divided into groups and each group was treated with at least one of the ProTriTAC molecules listed in Table 9, with the control GFP TriTAC molecule, or with the ProTriTAC NCLV molecule. Tumor volumes were measured at regular intervals, starting from day 10 post injection of tumor cells and expanded T cells.

(173) It was observed that in animals treated with the exemplary ProTriTAC molecules containing linker L040 there was a statistically significant delay in tumor growth as compared to the mice group which was treated with the control GFP TriTAC molecule, or the mice group that was treated with the ProTriTAC NCLV molecule. Similar observation was also made for the ProTriTAC molecules containing linker sequences L001, L041, and L045. The data is shown in FIG. **20**.

(174) It is also possible to carry out a similar study with xenograft models using other cell lines, such as A549 (non-small cell lung carcinoma) cells, DU-145 (prostate) cells, MCF-7 (breast) cells, Colo 205 (colon) cells, 3T3/]GF-IR (mouse fibroblast) cells, NCI H441 cells, HEP G2 (hepatoma) cells, MDA MB 231 (breast) cells, HT-29 (colon) cells, MDA-MB-435s (breast) cells, U266 cells, SH-SYSY cells, Sk-Mel-2 cells, NCI-H929, RPM18226, and A431 cells.

Example 9: Demonstration of Reduced Cytokine Levels in Cynomolgus Monkeys, Correlated with Masking of TriTAC Molecules

(175) In this study, cynomolgus monkeys were treated with three different concentrations (30 μg/kg; 300 μg/kg; and 1000 μg/kg) of an exemplary EGFR targeting ProTriTAC molecule containing a non-cleavable linker (ProTriTAC (NCLV), or with three different concentrations (10 μg/kg; 30 μg/kg; and 100 μg/kg) of an exemplary EGFR targeting TriTAC molecule (SEQ ID No. 793).

(176) As shown in FIG. **21**, after 4 hours following administering the ProTriTAC (NCLV) molecule, IFN-gamma IL-6 and IL-10 levels were significantly lower in comparison with administering the EGFR targeting TriTAC molecule.

Example 10: Demonstration of Improved Tolerability in Mouse, Conferred by an Exemplary EGFR Targeting ProTriTAC Molecule

(177) In this study, the tolerability of an exemplary EGFR targeting ProTriTAC molecule was

assessed. Seven weeks old NSG female tumor free mice were intraperitoneally injected with 2×10.sup.7 expanded human T cells at the commencement of the study, i.e., at day 0. On day 2, treatment was started by dividing the mice into various groups and administering to them varying concentrations of the exemplary EGFR targeting ProTriTAC molecule, containing the linker sequence L001, an EGFR targeting TriTAC molecule, and an EGFR targeting ProTriTAC molecule containing a non-cleavable linker (ProTriTAC (NCLV). The molecules were administered once daily for 10 days, at the following dosages: 30 μg/kg, 100 μg/kg, 300 μg/kg. Starting from day 2, body weight of the animals were recorded daily.

(178) As shown in FIG. **22**, the EGFR targeting ProTriTAC molecule containing a non-cleavable linker (ProTriTAC (NCLV)) and a GFP TriTAC (used as a negative control) were very well tolerated in mice even at the highest dose of 1000 μg/kg. The EGFR targeted ProTriTAC molecule containing the linker sequence of L001 was well tolerated at the dosage of 100 μg/kg, whereas the EGFR targeted TriTAC was well tolerated at 30 μg/kg. It was thus observed that the ProTriTAC containing the L001 linker sequence conferred about 3 fold increase in tolerability and the ProTriTAC (NCLV) conferred about a 30 fold increase in tolerability, in mouse. The mouse maximum tolerated dose for the ProTriTAC (NCLV) and TriTAC molecules was consistent with what was observed in cynomolgus monkeys.

(179) To further explore the role of the linker in tolerability of the EGFR targeting ProTriTAC molecule in mouse, the linker sequence was changed from L001 to that of L040. In this experiment, seven weeks old NSG female tumor free mice were subcutaneously injected with 5×10.sup.6HCT116 tumor cells, at the commencement of the study, i.e., at day 0. At day 7 following the tumor cell injection, when the tumor volumes were about 180-200 mm.sup.3 (e.g., 183 mm.sup.3), the mice were injected intraperitoneally with 2×10.sup.7 expanded human T cells. Treatment was started on day 9, by dividing the mice into various groups and each group was administered an EGFR targeting TriTAC molecule, an EGFR targeting ProTriTAC molecule with linker sequence L040 (ProTriTAC(L040), and a ProTriTAC molecule containing a non-cleavable linker (ProTriTAC(NCLV). The molecules were administered once daily for 10 days, at the following dosages: 300 μg/kg and 1000 μg/kg. Starting from day 2, body weight of the animals were recorded daily. The results shown in FIG. **23** provide that the EGFR targeting ProTriTAC molecule with the linker sequence L040 conferred better tolerability than when the linker sequence L001 was used. The ProTriTAC (L040) was well-tolerated at 300 μg/kg and at 1000 μg/kg, with body weight percentage change comparable to that with the ProTriTAC (NCLV) molecule. It was thus observed that compared to the TriTAC molecule, the ProTriTAC containing the L040 linker sequence conferred about a 30 fold increase in tolerability, in mouse, similar to the 30 fold increase in tolerability observed with the ProTriTAC (NCLV) molecule.

Example 11: Clinical Pathology Studies in Mice and Cynomolgus Monkeys

(180) In this study, mice were treated with various concentrations of an EFGR targeting TriTAC molecule, an EGFR targeting ProTriTAC molecule containing the linker sequence L001 (ProTriTAC (L001), and an EGFR targeting ProTriTAC molecule containing a non-cleavable linker (ProTriTAC(NCLV)). Tolerability was assessed by measuring serum concentration of ALT (alanine aminotransferase) and AST (aspartate aminotransferase). Results are shown in FIGS. **24**A, **24**B, and **24**C and FIGS. **25**A, **25**B, and **25**C. It was observed that serum concentrations of AST and ALT were not elevated following administering the ProTriTAC (L001) at dosages up to 0.3 mg/kg, and that the serum concentrations of AST and ALT were not elevated following administering the ProTriTAC(NCLV) at dosages up to 1 mg/kg. In contrast, serum concentration of AST and ALT were not elevated following administering the TriTAC molecule at a dosage of 0.3 mg/kg.

(181) In another study, cynomolgus monkeys were treated with various concentrations of an EFGR targeting TriTAC molecule, and an EGFR targeting ProTriTAC molecule containing a non-cleavable linker (ProTriTAC(NCLV)). Tolerability was assessed by measuring serum concentration of ALT (alanine aminotransferase) and AST (aspartate aminotransferase). Results are shown in

FIG. **26**. It was observed that serum concentrations of AST and ALT were not elevated following administering the ProTriTAC (NCLV) at dosages up to 1000 μg/kg. In contrast, serum concentration of AST and ALT were not elevated following administering the TriTAC molecule at a dosage of 10 μg/kg.

Example 12: Demonstration of Therapeutic Window Expansion with an Exemplary ProTriTAC Molecule of this Disclosure in a Tumor-Bearing Mouse Model

(182) The aim of this study was to evaluate the expansion of therapeutic window by measuring anti-tumor activity and observable on-target toxicity in the same tumor-bearing mice. NSG female mice, 7 weeks old, were used for this study.

(183) At the commencement of the study, on day 0, the NSG female mice were injected with 2.5×10.sup.6 expanded human T cells, and 5×10.sup.6 HCT116 (human colorectal carcinoma) tumor cells. The following day, on day 1, the mice were divided into groups and each group was treated with either GFP TriTAC molecule (SEQ ID No. 792), EGFR TriTAC molecule (SEQ ID No. 793), or an EGFR targeting ProTriTAC molecule containing linker L040, (SEQ ID No. 787) at the indicated dose levels in FIGS. **27**A-**297**D (for GFP TriTAC the dosage was 300 μg/kg; for EGFR TriTAC dosages were 10 μg/kg, 30 μg/kg, 100 μg/kg, and 300 μg/kg; for EGFR ProTriTAC dosages were 30 μg/kg, 100 μg/kg, 300 μg/kg, and 1000 μg/kg) and administered daily for a period of 10 days (i.e., final dose was administered on day 10 following injection of tumor cells and expanded cells to the animals) and tumor volumes were measured at regular intervals, beginning a few days prior to the administration of the final dose at day 10. Results shown in FIGS. **27**A-**27**D.

(184) On-target EGFR-related toxicity was determined by measuring the radius of the red scarring skin lesion above the original tumor implantation site with a caliper and applying the equation Area=π*(radius of lesion).sup.2 on day 14. Results provided in FIG. **28** show that an exemplary EGFR ProTriTAC has 30× better tolerability compared to an EGFR TriTAC in the same HCT116 tumor-bearing mice as measured by the onset of red scarring skin lesions above the original tumor implantation site. Therapeutic window is defined as the difference between the minimal dose level required for anti-tumor activity and the highest skin lesion-free dose level.

(185) Results (from FIGS. **27** and **28**) show that the exemplary protease-cleavable EGFR ProTriTAC is 3× less potent but 30× more tolerated (i.e., has a 10× improved therapeutic window) than the EGFR TriTAC when efficacy and toxicity are measured on the same tumor-bearing mice, as summarized in below Table. This would make it possible to dose the ProTriTAC at a dose that is about 3× higher than the TriTAC, to get at least the same efficiency and better tolerability.

(186) TABLE-US-00011 Min Max Therapeutic efficacious dose scar-free dose Window TriTAC 10 μg/kg   10 μg/kg 1 ProTriTAC 30 μg/kg 300 μg/kg 10

Example 13: An Exemplary ProTriTAC Molecule Containing a Binding Moiety with Extended Non-CDR Loop into which a Human CD3ε Epitope is Grafted

(187) The sequence of a binding moiety comprising non-CDR loops (AB, EF, C″D, and CC′) was obtained. A portion of the human CD3ε sequence was grafted into the CC′ loop of the non-CDR loops within the binding moiety, along with glycine residues to further extend the CC′ loop. FIG. **29** illustrates three different variants comprising 10, 12, or 16 amino acid extensions to the CC′ loop. In case of the variant CC10, the portion of human CD38 sequence grafted into the CC′ loop, to replace the wild type sequence of APGKG (SEQ ID NO: 795) was QDGNEE (SEQ ID NO. 801), and in addition 4 glycine residues were inserted to extend the CC′ loop. In case of the variant CC12, the portion of human CD3ε sequence grafted into the CC′ loop, to replace the wild type sequence of APGKG (SEQ ID NO: 795) was QDGNEEMGG (SEQ ID No. 802), and in addition 3 glycine residues were inserted to extend the CC′ loop. In case of the variant CC12, the portion of human CD3ε sequence grafted into the CC′ loop, to replace the wild type sequence of APGKG (SEQ ID NO: 795) was QDGNEEMGG (SEQ ID No. 803), and in addition 7 glycine residues were inserted to extend the CC′ loop. The binding moiety comprising extended non-CDR loops comprising the CD38 sequences, as described above, were cloned into a vector further comprising

coding sequences for a protease cleavable linker, a scFv containing a CD3 binding domain, and an EGFR binding domain, to express a ProTriTAC molecule. The ProTriTAC molecule contained an exemplary binding moiety of this disclosure, a CD3 binding scFv, and an EGFR binding domain. The ProTriTAC molecule was subsequently exposed to a tumor associated protease, matriptase, to assay activation of the molecule upon cleavage of the protease cleavable linker, which separates the binding moiety (depicted as aALB in FIGS. **29** and **30**) comprising the cleavable linker from the rest of the molecule, i.e., the scFv containing the CD3 binding domain (depicted as aCD3 in FIGS. **29** and **30**) and the EGFR binding domain (aEGFR). FIG. **30** shows the activation of the ProTriTAC molecules, containing the CC10, CC12, or CC16 variants of CC′ non-CDR loop, CD3 binding scFv in a VH-VL (left panel) or a VL-VH (right panel) format, upon treatment with matriptase. A ProTriTAC molecule containing a wild-type CC′ loop in the binding moiety was used as a control for the protease activation assay. In addition, a TriTAC molecule that is not in the “pro” form, i.e., a molecule that includes the same domains as the ProTriTAC molecule, except that it has a half-life extension domain, such as albumin, instead of a binding moiety, was also treated with matriptase and used as a control. Results indicated that the ProTriTAC molecules were activated upon cleavage, to generate a free albumin binding domain (depicted as Free aALB in FIG. **30**) whereas the albumin domain did not separate from the TriTAC molecules. Thus, the ProTriTAC molecules containing a binding moiety of this disclosure was able to readily dissociate from half-life extending domain upon cleavage in a tumor microenvironment, unlike the TriTAC versions, and thereby were amenable to rapid clearance from the systemic circulation upon activation.

(188) Further studies were carried out to assay the binding of the binding moiety containing the human CD38 to CD3. It was observed that, in the presence of human serum albumin, the activated forms of ProTriTAC molecules that contained the binding moiety containing the human CD3ε were about 20 times potent in binding CD3 than their activated forms which did not contain the binding moiety. Results are shown in FIG. **31**.

(189) Cell killing potential of a ProTriTAC molecule that contained a binding moiety as described herein was also assayed in a study where CaOV4 cell line was treated with the ProTriTAC molecule or its activated form in the presence of human serum albumin. As shown in FIG. **32**, the ProTriTAC molecule containing the CC16 variant of CC′ non-CDR loop was about 50 times more potent in killing cancer cells compared to its activated form, which was separated from the albumin binding domain. The total observed masking is a combination of steric masking due to binding to human serum albumin and specific masking from the CC16 mask in the CC′ non-CDR loop.

Example 14: Soft Library Mutagenesis Identified CC′ Loop as Most Amenable to Modification

(190) To identify locations within the non-CDR loops (AB, CC′, C″D, and EF) that were most amenable to modification, for creating a masking capability, libraries were assembled and generated using four groups of overlapping DNA oligos containing randomized degenerate “NNK” codons and with different loop lengths, as indicated in the schematic below:

(191) AB loop oligos: WT: LVQPGN (20%) (SEQ ID NO: 903) AB0: XXXXXX (20%) AB1: XXXXXXX (20%) AB2: XXXXXXXX (20%) AB3: XXXXXXXXX (20%)

(192) CC′ loop oligos: WT: APGKG (20%) (SEQ ID NO: 795) CC0; XXXXX (20%) CC1: XXXXXX (20%) CC2: XXXXXXX (20%) CC3: XXXXXXXX (20%)

(193) C″D loop oligos: WT: DSVKGR (20%) (SEQ ID NO: 904) CD0: XXXXXX (20%) CD1: XXXXXXX (20%) CD2: XXXXXXXX (20%) CD3: XXXXXXXXX (20%)

(194) EF loop oligos: WT: SLRPED (20%) (SEQ ID NO: 905) EF0: XXXXXX (20%) EF1: XXXXXXX (20%) EF2: XXXXXXXX (20%) EF3: XXXXXXXXX (20%)

(195) Note: “X” denotes a randomized residue (“NNK” codon) that could be any of the 20 natural amino acids as well as stop codon. The goal was to have approximately 20% of each non-CDR loop be wild-type. These wild-type oligos served as internal benchmarks to gauge the tolerance of each loop to modification (sequence composition and/or length changes). A loop that was less tolerable to change could easily revert to wild-type; in contrast, a loop that was highly amenable to

change would maintain the diverse sequence repertoire. To this end, 24 clones were sequenced from the naive library to verify the randomization of non-CDR loops prior to panning with HAS, as shown in FIG. **33**. After two rounds of phage panning against HSA, 30 clones were sequenced to examine the non-CDR loop composition. The results showed that 3 out of 4 non-CDR loops (AB, C″D, and EF) mostly utilized the 20% wild-type oligo and reverted back to the wild-type, suggesting that they may be less preferred than the wild-type sequence. However, the CC′ loop kept the diverse sequence repertoire (both sequence and length), suggesting that the CC′ loop may be the loop most tolerable to randomization that we could exploit for specific masking of adjacent domains, as shown in FIG. **34**.

Example 15: Screening of Phage Display Library for Identification of EpCAM Binding Domains

(196) Llamas were immunized with purified EpCAM protein expressed in Expi293 cells. A phage display library for expression of heavy variable antibody domains was constructed from circulating B cells. See van der Linden, de Geus, Stok, Bos, van Wassenaar, Verrips, and Frenken. 2000. J Immunol Methods 240:185-195. Phage clones were screened for binding to EpCAM by expressing anti-EpCAM proteins in *E. coli*, preparing periplasmic extracts, and proteins were screened for human and cynomolgus EpCAM binding activity using a colorimetric ELISA. Thirty-eight unique heavy chain only sequences were identified (SEQ ID Nos. 804-841) that produced a signal in the ELISA screening relative to the control with human and/or cynomolgus EpCAM proteins (as shown in Table 10).

(197) TABLE-US-00012 TABLE 10 Binding of Llama anti-Human EpCAM heavy chain only single domain antibodies to Human and Cynomolgus EpCAM, as demonstrated by signal in an ELISA Assay (absorbance readings in a colorimetric ELISA assay), relative to control heavy chain only single domain antibodies ELISA ELISA Human Cynomolgus Sequence Human Cynomolgus ELISA EpCAM/ EpCAM/ name EpCAM EpCAM Control Control Control EPL90 1.6 1.7 0.2 10 10 EPL118 3.4 2.9 0.7 5 4 EPL138 3.1 2.7 0.9 3 3 EPL145 0.6 0.4 0.2 3 2 EPL164 0.6 2.8 0.1 7 34 EPL31 0.6 0.5 0.1 7 6 EPL55 0.4 0.6 0.1 5 7 EPL57 1.7 3.4 0.1 14 27 EPL136 0.9 1.3 0.1 9 13 EPL15 0.7 3.2 0.1 8 35 EPL34 0.9 3.1 0.1 10 36 EPL86 0.8 3.0 0.1 8 31 EPL153 3.1 2.2 0.1 31 21 EPL20 2.9 2.2 0.5 6 4 EPL70 3.2 2.2 0.1 24 16 EPL125 3.5 3.5 0.5 8 8 EPL13 2.9 4.0 0.2 17 23 EPL129 2.0 3.1 0.1 20 29 EPL159 0.4 0.2 0.1 3 2 EPL120 3.2 2.3 0.6 5 4 EPL126 3.4 2.8 0.7 5 4 EPL60 1.1 3.5 0.1 10 33 EPL156 3.6 4.0 0.2 16 17 EPL2 2.2 3.6 0.1 15 24 EPL43 1.8 3.8 0.1 13 27 EPL10 2.7 2.0 0.2 12 9 EPL49 1.1 0.6 0.1 8 4 EPL58 1.1 0.5 0.1 8 4 EPL74 1.4 0.7 0.1 10 5 EPL78 3.1 2.0 0.2 17 11 EPL82 0.9 1.1 0.2 4 5 EPL83 2.0 1.1 0.6 3 2 EPL97 2.5 1.7 0.3 8 5 EPL109 0.4 0.1 0.1 5 2 EPL117 0.9 0.6 0.2 4 3 EPL127 0.4 0.9 0.1 5 11 EPL152 3.4 2.8 0.8 4 4 EPL189 1.3 0.9 0.1 12 8

Example 16: Incorporation of EpCAM Binding Heavy Chain Only Single Domain Antibodies into Fusion Proteins and T Cell Dependent Cellular Cytotoxicity Assays

(198) Selected anti-EpCAM heavy chain only single domain antibodies from Example 15 were cloned into DNA constructs for expression of recombinant proteins. These expression constructs all encoded a signal peptide. One set of anti-EpCAM constructs (SEQ ID Nos. 842 to 868) was designed to express a fusion protein with a humanized anti-CD3 scFv domain on the N-terminus of the mature secreted fusion protein followed by a llama anti-EpCAM domain, with the two domains linked by the sequence GGGGSGGGS (SEQ ID NO: 928), and with a HHHHHH (SEQ ID NO: 927) on the C-terminus. One second of anti-EpCAM constructs (SEQ ID Nos. 869 to 895) was designed to express a fusion protein with a llama anti-EpCAM domain on the N-terminus of the mature secreted fusion protein followed a humanized anti-CD3-scFv domain, with the two domains linked by the sequence GGGGSGGGS (SEQ ID NO: 928), and with a HHHHHH (SEQ ID NO: 927) on the C-terminus.

(199) These anti-EpCAM/anti-CD3 (from N-terminus to C-terminus) or anti-CD3/anti-EpCAM (from N terminus to C terminus) fusion protein constructs were transfected into Expi293 cells. The amount of anti-EpCAM/anti-CD3 fusion protein in the conditioned media from the transfected

Expi293 cells was quantitated using by using an Octet instrument with streptavidin and loaded with biotinylated CD3-Fc fusion protein using an anti-CD3 fusion protein of similar molecular weight to the anti-EPCAM/ant-CD3 proteins as a standard.

(200) The conditioned media were tested in a T-cell dependent cellular cytotoxicity assay. See Nazarian A A, Archibeque I L, Nguyen Y H, Wang P, Sinclair A M, Powers D A. 2015. J Biomol Screen. 20:519-27. In this assay, luciferase labelled NCI-H508 cells, which express EpCAM, were combined with purified human T cells and a titration of the anti-EpCAM/anti-CD3 fusion protein or the anti-CD3/anti-EpCAM. It was hypothesized that if the fusion protein directs T cells to kill the NCI-H508 cells, the signal in a luciferase assay performed at 48 hours after starting the experiment should decrease. FIGS. **36-39** provide the TDCC data in graphical format. EC.sub.50 values from the TDCC assays are listed in Table 11 (lists EC.sub.50 data for SEQ ID Nos. 842 to 868) and Table 12 (lists EC.sub.50 data for SEQ ID Nos. 869-895). The most potent molecule (EPL13) had an EC.sub.5 value of about 1.6 pM. Some of the anti-EpCAM binding proteins were only active when present in an anti-CD3/anti-EpCAM configuration. One anti-EpCAM sequence, EPL34, was only active in the anti-EpCAM/anti-CD3 configuration. A negative control for the TDCC assays was anti-GFP/anti-CD3 protein, and this protein did not direct the T cells to kill the NCI-H508 cells (data not shown).

(201) TABLE-US-00013 TABLE 11 EC.sub.50 Values for Redirected T Cell Killing of NCI-H508 Cells by Anti-CD3/Anti-EPCAM Proteins Containing Llama Anti-EPCAM Sequences (n/a = insufficient activity to calculate an EC.sub.50 using the protein concentrations tested) Anti-EpCAM Sequence NCI-H508 Cell Killing EC.sub.50 (M) EPL10 n/a EPL109 2.2E−09 EPL117 n/a EPL120 5.8e−010 EPL125 1.3E−09 EPL127 n/a EPL13 1.6E−12 EPL136 6.1E−12 EPL138 7.9E−12 EPL145 n/a EPL152 n/a EPL153 1.4E−10 EPL156 n/a EPL164 3.6E−10 EPL189 n/a EPL2 n/a EPL20 n/a EPL34 n/a EPL49 n/a EPL58 n/a EPL74 2.6E−09 EPL78 n/a EPL82 n/a EPL83 3.1E−10 EPL86 4.7E−10 EPL90 9.2E−12 EPL97 n/a

(202) TABLE-US-00014 TABLE 12 EC.sub.50 Values for Redirected T Cell Killing of NCI-H508 Cells by Anti-EpCAM/Anti-CD3 Proteins Containing Llama Anti-EPpCAM Sequences (n/a = insufficient activity to calculate an EC.sub.50 using the protein concentrations tested) Anti-EPCAM Sequence NCI-H508 Cell Killing EC50 (M) EPL10 n/a EPL109 n/a EPL117 n/a EPL120 n/a EPL125 n/a EPL127 n/a EPL13 1.6E−11 EPL136 1.3E−10 EPL138 n/a EPL145 n/a EPL152 n/a EPL153 not expressed EPL156 n/a EPL164 n/a EPL189 n/a EPL2 n/a EPL20 n/a EPL34 3.7E−10 EPL49 n/a EPL58 n/a EPL74 n/a EPL78 n/a EPL82 n/a EPL83 2.1E−11 EPL86 2.7E−10 EPL90 n/a EPL97 n/a

(203) Using conditioned media with known concentrations of anti-EpCAM/anti-CD3 or anti-CD3/anti-EpCAM fusion proteins, the binding affinities of the fusion proteins for human and cynomolgus monkey EpCAM proteins were measured. An Octet instrument with streptavidin tips were loaded with biotinylated human or cynomolgus EpCAM protein, and K.sub.D values were calculated by measuring the on rate and off rate of binding of the anti-EPCAM/anti-CD3 fusion or anti-CD3/anti-EpCAM fusion proteins to the biotinylated EpCAM proteins. The K.sub.D measurements were made using a single 50 nM concentration of the anti-EPCAM/anti-CD3 or anti-CD3/anti-EpCAM fusion proteins, which allowed for rank ordering potency. The measured relative affinities are listed in Table 13. All of the fusion proteins bound to cynomolgus EpCAM, with K.sub.D values ranging from 1.6 to 56 nM. Most, but not all of the fusion proteins were measured binding to human EpCAM with K.sub.D values ranging from 0.8 to 74 nM.

(204) TABLE-US-00015 TABLE 13 Binding Affinities to Human and Cyno EpCAM of Anti-EPCAM/Anti-CD3 or Anti-CD3/Anti-EpCAM Fusion Proteins Containing Llama Anti-EpCAM Sequences Anti-CD3/ Anti-EpCAM/ Anti-EpCAM Anti-CD3 Humanized anti- hu KD cy KD hu KD cy KD EpCAM Binder (nM) (nM) (nM) (nM) EPL13 29 13 14 7.6 EPL136 74 56 32 31 EPL138 2.2 2.2 0.8 1.3 EPL153 n/q 3.5 No expression EPL164 3.1 3.6 2.1 2.9 EPL34 n/q 14 n/q 6.4 EPL83 1.1 3 1.6 3 EPL86 n/q 8.3 n/q 7 EPL90 1.9 1.6 0.8 1.2

Example 17: Humanization of EpCAM Binding Heavy Chain Only Single Domain Antibodies and T Cell Dependent Cellular Cytotoxicity Assays

(205) Three of the llama anti-EpCAM antibodies sequences identified in Example 15 were humanized by grafting their CDR sequences onto human germline sequences, while retaining some llama framework sequences to ensure the antibodies did not lose activity (SEQ ID Nos. 896 to 898).

(206) These sequences were cloned into expression constructs for expression of anti-EpCAM/anti-CD3 fusion proteins (SEQ ID Nos. 899 to 901) in Expi293 cells, as described in Example 16.

(207) The amount of anti-EpCAM/anti-CD3 fusion proteins present in the conditioned medium was quantitated as described in Example 16. The affinities of these humanized proteins for human, cynomolgus, and mouse EpCAM were measured as described in Example 16. The relative K.sub.D values calculated from these measurements are listed in Table 14. All three sequences bound to human and cynomolgus EpCAM, with relative K.sub.D values ranging from about 0.3 to about 18 nM. Two of the sequences also bound to mouse EpCAM, with K.sub.D values ranging from about 1.4 to about 1.8 nM.

(208) TABLE-US-00016 TABLE 14 Binding Affinities to Human, Cynomolgus, and Mouse EpCAM of Anti- EpCAM/Anti-CD3 Fusion Proteins Containing Llama Anti-EpCAM Sequences (n/q = not quantifiable under the experimental conditions used) Humanized Anti- Human EpCAM Cynomolgus Mouse EPCAM EpCAM Sequence (nM) EpCAM (nM) (nM) H13 17 18 n/q H90 0.3 1.3 1.4 H138 0.3 1.8 1.8

(209) T cell killing potential of the anti-EpCAM/anti-CD3 fusion proteins present in the conditioned medium was assessed as described in Example 16. Results are provided in Table 15 and in FIG. **40**.

(210) TABLE-US-00017 TABLE 15 EC.sub.50 Values for Redirected T Cell Killing of NCI-H508 Cells by Purified Anti-CD3/Anti-EpCAM Proteins Containing Humanized Anti-EpCAM Sequences Humanized anti-EpCAM Binder Sequence EC.sub.50 (pM) H13 10 H90 10 H138 16

Example 18: Demonstration of Improved Tolerability in Mouse, Conferred by an Exemplary EpCAM Targeting ProTriTAC Molecule

(211) In this study, the tolerability of an exemplary EpCAM targeting ProTriTAC molecule was assessed. Seven weeks old NSG female tumor free mice were intraperitoneally injected with 2×10.sup.7 expanded human T cells at the commencement of the study, i.e., at day 0. On day 2, treatment was started by dividing the mice into various groups and administering to them varying concentrations of the exemplary EpCAM targeting ProTriTAC molecule, containing the linker sequence L040, an EpCAMR targeting TriTAC molecule, an EpCAM targeting ProTriTAC molecule containing a non-cleavable linker (EpCAM ProTriTAC (NCLV), and a GFP TriTAC molecule (SEQ ID No. 792) as a control. The molecules were administered once daily for 10 days, at the following dosages: 0.03 mg/kg, 0.1 mg/kg, 0.3 mg/kg, and 1 mg/kg. Starting from day 2, body weight of the animals were recorded daily.

(212) As shown in FIGS. **41**A-**41**C, the EpCAM targeting ProTriTAC molecule containing a non-cleavable linker (ProTriTAC (NCLV)) (SEQ ID No. 908) and a GFP TriTAC (used as a negative control) were very well tolerated in mice even at the highest dose of 1 mg/kg. The EpCAM targeted ProTriTAC molecule containing the linker sequence of L040 (SEQ ID No. 907) was well tolerated at the highest tested dosage of 1 mg/kg, whereas the EpCAM targeted TriTAC (SEQ ID No. 906) was well tolerated 0.1 mg/kg. It was thus observed that the EpCAM targeting ProTriTAC containing the L040 linker sequence conferred at least about 10 times improved tolerability, in mouse, compared to the EpCAM targeting TriTAC.

Example 19: Xenograft Tumor Model

(213) An EpCAM targeting fusion protein of this disclosure (e.g., a fusion protein which is a trispecific protein comprising an anti-EpCAM heavy chain only single domain antibody, an anti-CD3 scFv, and an anti-Albumin domain) is evaluated in a xenograft model. In order to determine

efficacy of the exemplary EpCAM targeting fusion protein in vivo, multiple xenograft tumor models are used. Examples of common tumor cell lines for use in xenograft tumor studies include A549 (non-small cell lung carcinoma) cells, DU-145 (prostate) cells, MCF-7 (breast) cells, Colo 205 (colon) cells, 3T3 (mouse fibroblast) cells, NCI H441 cells, HEP G2 (hepatoma) cells, MDA MB 231 (breast) cells, HT-29 (colon) cells, MDA-MB-435s (breast) cells, U266 cells, SH-SYSY cells, Sk-Mel-2 cells, NCI-H929, RPM18226, and A431 cells. Immune-deficient NOD/scid mice are sub-lethally irradiated (2 Gy) and subcutaneously inoculated with 1×10.sup.6 tumor cells (e.g., NCI H441 cells) into their right dorsal flank. When tumors reach 100 to 200 mm.sup.3, animals are allocated into 3 treatment groups. Groups 2 and 3 are intraperitoneally injected with 1.5×10.sup.7 activated human T-cells. Three days later, animals from Group 3 are subsequently treated with the exemplary EPCAM targeting trispecific antigen-binding protein of Groups 1 and 2 are only treated with vehicle. Body weight and tumor volume are determined for 30 days, beginning at least 5 days post treatment with the exemplary EPCAM targeting trispecific protein.

(214) It is expected that animals treated with the exemplary EpCAM targeting trispecific protein have a statistically significant delay in tumor growth in comparison to the respective vehicle-treated control group.

Example 20: Proof-of-Concept Clinical Trial Protocol for Administration of the EpCAM Targeting Trispecific Antigen-Binding Protein of Example 19 to Ovarian Cancer Patients

(215) This is a Phase I/II clinical trial for studying an exemplary EpCAM targeting trispecific antigen-binding protein of this disclosure as a treatment for an epithelial ovarian cancer.

(216) 1. Study Outcomes:

(217) 2. Primary: Maximum tolerated dose of the exemplary EpCAM targeting trispecific protein.

(218) 3. Secondary: To determine whether in vitro response of the exemplary EpCAM targeting trispecific protein is associated with clinical response

(219) Phase I

(220) 4. The maximum tolerated dose (MTD) will be determined in the phase I section of the trial.

(221) 1.1 The maximum tolerated dose (MTD) will be determined in the phase I section of the trial.

(222) 1.2 Patients who fulfill eligibility criteria will be entered into the trial to EPCAM targeting trispecific proteins of the previous examples.

(223) 1.3 The goal is to identify the highest dose of EpCAM targeting trispecific proteins of the previous examples that can be administered safely without severe or unmanageable side effects in participants. The dose given will depend on the number of participants who have been enrolled in the study prior and how well the dose was tolerated. Not all participants will receive the same dose.

Phase II

2.1 A subsequent phase II section will be treated at the MTD with a goal of determining if therapy with therapy of the exemplary EpCAM targeting trispecific protein results in at least a 20% response rate.

Primary Outcome for the Phase II—To determine if therapy of EPCAM targeting trispecific protein results in at least 20% of patients achieving a clinical response (blast response, minor response, partial response, or complete response)

Eligibility: Histologically or cytologically confirmed epithelial ovarian cancer. May have Recurrent epithelial ovarian carcinoma or disease progression following failure of first-line, platinum-based chemotherapy with no more than one prior platinum based regimen therapy Adequate laboratory values of bone marrow function, renal function, liver function, and echocardiogram tests

Phase III

(224) 5.3.1A subsequent phase III section will carried out with the exemplary EpCAM targeting trispecific protein, wherein secondary endpoints such as response rate (RR), patient recorded outcomes (PRO), progression-free survival (PFS), duration of progression free survival, time to progression (TIP), overall survival, health-related quality of life assessment, number of participants with overall survival, duration of response, time to response, number of participants with response,

and time to tumor growth etc. will be assessed.
(225) TABLE-US-00018 SEQUENCE TABLE DESCRIPTION SEQUENCE PSMA Prodrug C1872 EVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQGGG (SEQ ID NO: 43) GGLDGNEEPGGLEWVSSISGSGRDTLYADSVKGRFTISRDNAK TTLYLQMNSLRPEDTAVYYCTIGGSLSVSSQGTLVTVSSGGGGK PLGLQARVVGGGGT QTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPNWVQQKP GQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSGVQPEDEA EYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSGGGGSEVQ LVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLE WVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMNNL KTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVTVSSGGGGS GGGSEVQLVESGGGLVQPGGSLTLSCAASRFMISEYHMHWVR QAPGKGLEWVSTINPAGTTDYAESVKGRFTISRDNAKNTLYLQ MNSLKPEDTAVYYCDSYGYRGQGTQVTVSSHHHHHH PSMA Non-cleavable Prodrug EVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQ**GGG** C1873 **GGLDGNEEPGG**LEWVSSISGSGRDTLYADSVKGRFTISRDNAK (SEQ ID NO: 44) TTLYLQMNSLRPEDTAVYYCTIGGSLSVSSQGTLVTVSS**GGGGS GGGGSGGVVGGGGT** QTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPNWVQQKP GQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSGVQPEDEA EYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSGGGGSEVQ LVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLE WVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMNNL KTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVTVSSGGGGS GGGSEVQLVESGGGLVQPGGSLTLSCAASRFMISEYHMHWVR QAPGKGLEWVSTINPAGTTDYAESVKGRFTISRDNAKNTLYLQ MNSLKPEDTAVYYCDSYGYRGQGTQVTVSSHHHHHH PSMA Active Drug C1875 (SEQ **VVGGGG**TQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYP ID NO: 45) NWVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLS GVQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGS GGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWV RQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNT AYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLV TVSSGGGGSGGGSEVQLVESGGGLVQPGGSLTLSCAASRFMIS EYHMHWVRQAPGKGLEWVSTINPAGTTDYAESVKGRFTISRD NAKNTLYLQMNSLKPEDTAVYYCDSYGYRGQGTQVTVSSHHH HHH EGFR (G8) Prodrug C1486 (SEQ EVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQ**GGG** ID NO: 46) **GGLDGNEEPGG**LEWVSSISGSGRDTLYADSVKGRFTISRDNAK TTLYLQMNSLRPEDTAVYYCTIGGSLSVSSQGTLVTVSS**GGGGK PLGLQARVVGGGGT** QTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPNWVQQKP GQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSGVQPEDEA EYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSGGGGSEVQ LVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLE WVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMNNL KTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVTVSSGGGGS GGGSEVQLVESGGGLVQPGGSLTLSCAASGRTFSSYAMGWFR QAPGKEREFVVAINWASGSTYYADSVKGRFTISRDNSKNTLYL QMNSLRAEDTAVYYCAAGYQINSGNYNFKDYEYDYWGQGTL VTVSSHHHHHH EGFR (G8) Non-cleavable EVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQ**GGG** Prodrug C1756 (SEQ ID NO: 47) **GGLDGNEEPGG**LEWVSSISGSGRDTLYADSVKGRFTISRDNAK

TTLYLQMNSLRPEDTAVYYCTIGGSLSVSSQGTLVTVSS**GGGGS GGGGSGGVVGGGGT**
QTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPNWVQQKP
GQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSGVQPEDEA
EYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSGGGGSEVQ
LVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLE
WVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMNNL
KTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVTVSSGGGGS
GGGSEVQLVESGGGLVQPGGSLTLSCAASGRTFSSYAMGWFR
QAPGKEREFVVAINWASGSTYYADSVKGRFTISRDNSKNTLYL
QMNSLRAEDTAVYYCAAGYQINSGNYNFKDYEYDYWGQGTL VTVSSHHHHHH EGFR
(G8) Active Drug C1300
VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYP (SEQ ID NO: 48)
NWVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLS
GVQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGS
GGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWV
RQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNT
AYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLV
TVSSGGGGSGGGSEVQLVESGGGLVQPGGSLTLSCAASGRTFS
SYAMGWFRQAPGKEREFVVAINWASGSTYYADSVKGRFTISR
DNSKNTLYLQMNSLRAEDTAVYYCAAGYQINSGNYNFKDYEY
DYWGQGTLVTVSSHHHHHH GFP TriTAC C646
QVQLVESGGALVQPGGSLRLSCAASGFPVNRYSMRWYRQAP (SEQ ID NO: 49)
GKEREWVAGMSSAGDRSSYEDSVKGRFTISRDDARNTVYLQM
NSLKPEDTAVYYCNVNVGFEYWGQGTQVTVSSGGGGSGGGS
EVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQAPG
KGLEWVSSISGSGRDTLYADSVKGRFTISRDNAKTTLYLQMNSL
RPEDTAVYYCTIGGSLSVSSQGTLVTVSSGGGGSGGGSEVQLVE
SGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLEWV
ARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMNNLKTE
DTAVYYCVRHANFGNSYISYWAYWGQGTLVTVSSGGGGSGG
GGSGGGGSQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYP
NWVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLS
GVQPEDEAEYYCTLWYSNRWVFGGGTKLTVLHHHHHH Exemplary masking sequence
GGGGGLDGNEEPGG (SEQ ID NO: 50) Exemplary non-cleavable linker
GGGGSGGGGSGGVVGGGGT sequence comprising protease cleavage site (SEQ ID
NO: 51) PSMA Non-masked Non-
EVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQAPG cleavable Prodrug C1874
KGLEWVSSISGSGRDTLYADSVKGRFTISRDNAKTTLYLQMNSL (SEQ ID NO: 52)
RPEDTAVYYCTIGGSLSVSSQGTLVTVSSGGGGSGGGGSGGVV
GGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPNW
VQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSGV
QPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSGG
GGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQ
APGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAY
LQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVTV
SSGGGGSGGGSEVQLVESGGGLVQPGGSLTLSCAASRFMISEY
HMHWVRQAPGKGLEWVSTINPAGTTDYAESVKGRFTISRDN
AKNTLYLQMNSLKPEDTAVYYCDSYGYRGQGTQVTVSSHHHH HH Exemplary
cleavable linker GGGGKPLGLQARVVGGGGT (SEQ ID NO: 53) Exemplary anti-
albumin sdAb EVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQ**GGG** (SEQ

ID NO: 54) **GGLDGNEEPGG**LEWVSSISGSGRDTLYADSVKGRFTISRDNAK TTLYLQMNSLRPEDTAVYYCTIGGSLSVSSQGTLVTVSS**GGGGK PLGLQARVVGGGGT** Exemplary anti-target (anti- EVQLVESGGGLVQPGGSLTLSCAASGRTFSSYAMGWFRQAPG EGFR) sdAb KEREFVVAINWASGSTYYADSVKGRFTISRDNSKNTLYLQMNSL (SEQ ID NO: 55) RAEDTAVYYCAAGYQINSGNYNFKDYEYDYWGQGTLVTVSS Exemplary anti-CD3 scFv QTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPNWVQQKP (SEQ ID NO: 56) GQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSGVQPEDEA EYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSGGGGSEVQ LVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLE WVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMNNL KTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVTVSS Exemplary anti-target (anti- EVQLVESGGGLVQPGGSLTLSCAASRFMISEYHMHWVRQAPG PSMA) sdAb KGLEWVSTINPAGTTDYAESVKGRFTISRDNAKNTLYLQMNSL (SEQ ID NO: 57) KPEDTAVYYCDSYGYRGQGTQVTVSS Exemplary anti-target (anti- EVQLVESGGGLVQPGGSLRLSCAASRFMISEYHMHWVRQAP PSMA) sdAb GKGLEWVSDINPAGTTDYAESVKGRFTISRDNAKNTLYLQMNS (SEQ ID NO: 63) LRAEDTAVYYCDSYGYRGQGTLVTVSS Exemplary anti-target (anti- EVQLVESGGGLVQPGGSLRLSCAASRFMISEYHMHWVRQAP PSMA) sdAb GKGLEWVSTINPAGTTDYAESVKGRFTISRDNAKNTLYLQMNS (SEQ ID NO: 64) LRAEDTAVYYCDSYGYRGQGTLVTVSS Exemplary anti-target (anti- EVQLVESGGGLVQPGGSLRLSCAASRFMISEYSMHWVRQAPG PSMA) sdAb KGLEWVSTINPAKTTDYAESVKGRFTISRDNAKNTLYLQMNSLR (SEQ ID NO: 65) AEDTAVYYCDSYGYRGQGTLVTVSS Exemplary anti-target (anti- EVQLVESGGGLVQPGGSLRLSCAASRFMISPYSMHWVRQAPG PSMA) sdAb KGLEWVSTINPAGTTDYAESVKGRFTISRDNAKNTLYLQMNSL (SEQ ID NO: 66) RAEDTAVYYCDGYGYRGQGTLVTVSS Exemplary anti-target (anti- EVQLVESGGGLVQPGGSLRLSCAASRFMISEYSMHWVRQAPG PSMA) sdAb KGLEWVSTINPAGQTDYAESVKGRFTISRDNAKNTLYLQMNSL (SEQ ID NO: 67) RAEDTAVYYCDGYGYRGQGTLVTVSS Exemplary anti-target (anti- EVQLVESGGGLVQPGGSLRLSCAASRFMISEYSMHWVRQAPG PSMA) sdAb KGLEWVSTINPAGTTDYAEYVKGRFTISRDNAKNTLYLQMNSL (SEQ ID NO: 68) RAEDTAVYYCDGYGYRGQGTLVTVSS Exemplary anti-target (anti- EVQLVESGGGLVQPGGSLRLSCAASRFMISEYHMHWVRQAP PSMA) sdAb GKGLEWVSDINPAKTTDYAESVKGRFTISRDNAKNTLYLQMNS (SEQ ID NO: 69) LRAEDTAVYYCDSYGYRGQGTLVTVSS Exemplary anti-target (anti- EVQLVESGGGLVQPGGSLRLSCAASRFMISPYHMHWVRQAP PSMA) sdAb GKGLEWVSDINPAGTTDYAESVKGRFTISRDNAKNTLYLQMNS (SEQ ID NO: 70) LRAEDTAVYYCDSYGYRGQGTLVTVSS Exemplary anti-target (anti- EVQLVESGGGLVQPGGSLRLSCAASRFMISEYHMHWVRQAP PSMA) sdAb GKGLEWVSDINPAGQTDYAESVKGRFTISRDNAKNTLYLQMN (SEQ ID NO: 71) SLRAEDTAVYYCDSYGYRGQGTLVTVSS Exemplary anti-target (anti- EVQLVESGGGLVQPGGSLRLSCAASRFMISEYHMHWVRQAP PSMA) sdAb GKGLEWVSDINPAGTTDYAEYVKGRFTISRDNAKNTLYLQMNS (SEQ ID NO: 72) LRAEDTAVYYCDSYGYRGQGTLVTVSS Exemplary anti-target (anti- EVQLVESGGGLVQPGGSLTLSCAASRFMISEYHMHWVRQAPG PSMA) sdAb KGLEWVSDINPAGTTDYAESVKGRFTISRDNAKNTLYLQMNSL (SEQ ID NO: 73) KPEDTAVYYCDSYGYRGQGTQVTVSS Exemplary anti-CD3 scFv EVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPG (SEQ ID NO: 74) KGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQM

NNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVTVSSGG
GGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCASSTGAV
TSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGG
KAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVL Exemplary anti-CD3 scFv
EVQLVESGGGLVQPGGSLKLSCAASGFEFNKYAMNWVRQAP (SEQ ID NO: 75)
GKGLEWVARIRSKYNKYATYYADSVKDRFTISRDDSKNTAYLQ
MNNLKTEDTAVYYCVRHGNFGNSYISYWAYWGQGTLVTVSS
GGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCGSSFG
AVTSGNYPNWVQQKPGQAPRGLIGGTKFLAPGTPARFSGSLL
GGKAALTLSGVQPEDEAEYYCVLWYDNRWVFGGGTKLTVL Exemplary anti-CD3 scFv
EVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAMNWVRQAP (SEQ ID NO: 76)
GKGLEWVARIRSKYNNYATYYADSVKDRFTISRDDSKNTAYLQ
MNNLKTEDTAVYYCVRHGNFGNSHISYWAYWGQGTLVTVSS
GGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCGSSTG
YVTSGNYPNWVQQKPGQAPRGLIGGTSFLAPGTPARFSGSLLG
GKAALTLSGVQPEDEAEYYCVLWYSNRWIFGGGTKLTVL Exemplary anti-CD3 scFv
EVQLVESGGGLVQPGGSLKLSCAASGFMFNKYAMNWVRQAP (SEQ ID NO: 77)
GKGLEWVARIRSKSNNYATYYADSVKDRFTISRDDSKNTAYLQ
MNNLKTEDTAVYYCVRHGNFGNSYISYWATWGQGTLVTVSS
GGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCGSSFG
AVTSGNYPNWVQQKPGQAPRGLIGGTKLLAPGTPARFSGSLL
GGKAALTLSGVQPEDEAEYYCVLWYSNSWVFGGGTKLTVL Exemplary anti-CD3 scFv
EVQLVESGGGLVQPGGSLKLSCAASGFTFNTYAMNWVRQAP (SEQ ID NO: 78)
GKGLEWVARIRSKYNNYATYYKDSVKDRFTISRDDSKNTAYLQ
MNNLKTEDTAVYYCVRHGNFGNSPISYWAYWGQGTLVTVSS
GGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCGSSTG
AVVSGNYPNWVQQKPGQAPRGLIGGTEFLAPGTPARFSGSLL
GGKAALTLSGVQPEDEAEYYCVLWYSNRWVFGGGTKLTVL Exemplary anti-CD3 scFv
EVQLVESGGGLVQPGGSLKLSCAASGFTYNKYAMNWVRQAP (SEQ ID NO: 79)
GKGLEWVARIRSKYNNYATYYADEVKDRFTISRDDSKNTAYLQ
MNNLKTEDTAVYYCVRHGNFGNSPISYWAYWGQGTLVTVSS
GGGGGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCGSSKG
AVTSGNYPNWVQQKPGQAPRGLIGGTKELAPGTPARFSGSLL
GGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVL Exemplary anti-CD3 scFv
EVQLVESGGGLVQPGGSLKLSCAASGNTFNKYAMNWVRQAP (SEQ ID NO: 80)
GKGLEWVARIRSKYNNYETYYADSVKDRFTISRDDSKNTAYLQ
MNNLKTEDTAVYYCVRHTNFGNSYISYWAYWGQGTLVTVSSG
GGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCGSSTGA
VTSGYYPNWVQQKPGQAPRGLIGGTYFLAPGTPARFSGSLLGG
KAALTLSGVQPEDEAEYYCVLWYSNRWVFGGGTKLTVL Exemplary anti-CD3 scFv
EVQLVESGGGLVQPGGSLKLSCAASGFTFNNYAMNWVRQAP (SEQ ID NO: 81)
GKGLEWVARIRSKYNNYATYYADAVKDRFTISRDDSKNTAYLQ
MNNLKTEDTAVYYCVRHGNFGNSQISYWAYWGQGTLVTVSS
GGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCGSSTG
AVTDGNYPNWVQQKPGQAPRGLIGGIKFLAPGTPARFSGSLL
GGKAALTLSGVQPEDEAEYYCVLWYSNRWVFGGGTKLTVL Exemplary anti-CD3 scFv
EVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAVNWVRQAPG (SEQ ID NO: 82)
KGLEWVARIRSKYNNYATYYADSVKDRFTISRDDSKNTAYLQM
NNLKTEDTAVYYCVRHGNFGNSYISYWAYWGQGTLVTVSSGG
GGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCGESTGAV

TSGNYPNWVQQKPGQAPRGLIGGTKILAPGTPARFSGSLLGGK
AALTLSGVQPEDEAEYYCVLWYSNRWVFGGGTKLTVL Exemplary anti-CD3 scFv
EVQLVESGGGLVQPGGSLKLSCAASGFTFNKYPMNWVRQAP (SEQ ID NO: 83)
GKGLEWVARIRSKYNNYATYYADSVKDRFTISRDDSKNTAYLQ
MNNLKNEDTAVYYCVRHGNFNNSYISYWAYWGQGTLVTVSS
GGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCGSSTG
AVTKGNYPNWVQQKPGQAPRGLIGGTKMLAPGTPARFSGSLL
GGKAALTLSGVQPEDEAEYYCALWYSNRWVFGGGTKLTVL Exemplary anti-CD3 scFv
EVQLVESGGGLVQPGGSLKLSCAASGFTFNGYAMNWVRQAP (SEQ ID NO: 84)
GKGLEWVARIRSKYNNYATYYADEVKDRFTISRDDSKNTAYLQ
MNNLKTEDTAVYYCVRHGNFGNSPISYWAYWGQGTLVTVSS
GGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCGSSTG
AVVSGNYPNWVQQKPGQAPRGLIGGTEFLAPGTPARFSGSLL
GGKAALTLSGVQPEDEAEYYCVLWYSNRWVFGGGTKLTVL Exemplary anti-CD3 scFv
EVQLVESGGGLVQPGGSLKLSCAASGNTFNKYAMNWVRQAP (SEQ ID NO: 85)
GKGLEWVARIRSKYNNYATYYADSVKDRFTISRDDSKNTAYLQ
MNNLKTEDTAVYYCVRHGNFGDSYISYWAYWGQGTLVTVSS
GGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCGSSTG
AVTHGNYPNWVQQKPGQAPRGLIGGTKVLAPGTPARFSGSLL
GGKAALTLSGVQPEDEAEYYCVLWYSNRWVFGGGTKLTVL Exemplary anti-CD3 scFv
EVQLVESGGGLVQPGGSLKLSCAASGFTFNNYAMNWVRQAP (SEQ ID NO: 86)
GKGLEWVARIRSGYNNYATYYADSVKDRFTISRDDSKNTAYLQ
MNNLKTEDTAVYYCVRHGNFGNSYISYWAYWGQGTLVTVSS
GGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCGSYTG
AVTSGNYPNWVQQKPGQAPRGLIGGTKFNAPGTPARFSGSLL
GGKAALTLSGVQPEDEAEYYCVLWYANRWVFGGGTKLTVL Exemplary anti-CD3 scFv
EVQLVESGGGLVQPGGSLKLSCAASGFEFNKYAMNWVRQAP (SEQ ID NO: 87)
GKGLEWVARIRSKYNNYETYYADSVKDRFTISRDDSKNTAYLQ
MNNLKTEDTAVYYCVRHGNFGNSLISYWAYWGQGTLVTVSS
GGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCGSSSG
AVTSGNYPNWVQQKPGQAPRGLIGGTKFGAPGTPARFSGSLL
GGKAALTLSGVQPEDEAEYYCVLWYSNRWVFGGGTKLTVL Exemplary anti-CD3 scFv
EVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAMNWVRQAP (SEQ ID NO: 88)
GKGLEWVARIRSKYNNYATYYADSVKDRFTISRDDSKNTAYLQ
MNNLKTEDTAVYYCVRHGNFGNSYISYWAYWGQGTLVTVSS
GGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCGSSTG
AVTSGNYPNWVQQKPGQAPRGLIGGTKFLAPGTPARFSGSLL
GGKAALTLSGVQPEDEAEYYCVLWYSNRWVFGGGTKLTVL Exemplary anti-CD3 scFv
EVQLVESGGGLVQPGGSLKLSCAASGFTFNKYALNWVRQAPG (SEQ ID NO: 89)
KGLEWVARIRSKYNNYATEYADSVKDRFTISRDDSKNTAYLQM
NNLKTEDTAVYYCVRHGNFGNSPISYWAYWGQGTLVTVSSGG
GGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCGSSTGAV
TSGNYPNWVQQKPGQAPRGLIGGTNFLAPGTPERFSGSLLGG
KAALTLSGVQPEDEAEYYCVLWYSNRWAFGGGTKLTVL Exemplary anti-CD3 scFv
EVQLVESGGGLVQPGGSLKLSCAASGFTFNEYAMNWVRQAP (SEQ ID NO: 90)
GKGLEWVARIRSKYNNYATYYADDVKDRFTISRDDSKNTAYLQ
MNNLKTEDTAVYYCVRHGNFGNSGISYWAYWGQGTLVTVSS
GGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCGSSTG
AVTVGNYPNWVQQKPGQAPRGLIGGTEFLAPGTPARFSGSLL
GGKAALTLSGVQPEDEAEYYCVLWYSNRWVFGGGTKLTVL Exemplary anti-target

(BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSISPMGWYRQAPGK (SEQ ID NO: 91) QRELVAAIHGFSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTDIFSISPMGWYRQAPGK (SEQ ID NO: 92) QRELVAAIHGGSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVAWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNDFSISPMGWYRQAPG (SEQ ID NO: 93) KQRELVAAIHGGSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPRNVAWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSKSPMGWYRQAPGK (SEQ ID NO: 94) QRELVAAIHGKSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVVWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNDFSISPMGWYRQAPG (SEQ ID NO: 95) KQRELVAAIHGKSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPRNVKWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNQFSISPMGWYRQAPG (SEQ ID NO: 96) KQRELVAAIHGKSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPRNVVWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSSSPMGWYRQAPGK (SEQ ID NO: 97) QRELVAAINGFSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVHWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSSSPMGWYRQAPGK (SEQ ID NO: 98) QRELVAAIHGFSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSTSPMGWYRQAPGK (SEQ ID NO: 99) QRELVAAIHGFQTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVVWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSRSPMGWYRQAPGK (SEQ ID NO: 100) QRELVAAIHGFETLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVLWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSESPMGWYRQAPGK (SEQ ID NO: 101) QRELVAAIHGFTTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVTWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSDSPMGWYRQAPG (SEQ ID NO: 102) KQRELVAAIHGFSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPRNVAWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSNSPMGWYRQAPG (SEQ ID NO: 103) KQRELVAAIHGGSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPRNVHWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSNSPMGWYRQAPG (SEQ ID NO: 104) KQRELVAAIHGRSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPRNVMWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSNSPMGWYRQAPG (SEQ ID NO: 105) KQRELVAAIHGPSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPRNVTWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSNSPMGWYRQAPG (SEQ ID NO: 106) KQRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPRNVRWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSRSPMGWYRQAPGK (SEQ ID NO: 107) QRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE

DTALYYCNKVPWGDYHPRNVTWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSKSPMGWYRQAPGK (SEQ ID NO: 108)
QRELVAAIHGQSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPRNVTWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSSSPMGWYRQAPGK (SEQ ID NO: 109)
QRELVAAIHGHSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPRNVTWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSSSPMGWYRQAPGK (SEQ ID NO: 110)
QRELVAAIHGESTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPRKVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSRSPMGWYRQAPGK (SEQ ID NO: 111)
QRELVAAIHGNSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGIYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSESPMGWYRQAPGK (SEQ ID NO: 112)
QRELVAAIHGNSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGTYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSVSPMGWYRQAPG (SEQ ID NO: 113)
KQRELVAAIHGNSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP
EDTALYYCNKVPWGKYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSVSPMGWYRQAPG (SEQ ID NO: 114)
KQRELVAAIHGNSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP
EDTALYYCNKVPWGDYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSKSPMGWYRQAPGK (SEQ ID NO: 115)
QRELVAAIHGNSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPREVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSDSPMGWYRQAPG (SEQ ID NO: 116)
KQRELVAAIHGTSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP
EDTALYYCNKVPWGDYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSRSPMGWYRQAPGK (SEQ ID NO: 117)
QRELVAAIHGTSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGKYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSHSPMGWYRQAPG (SEQ ID NO: 118)
KQRELVAAIHGTSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP
EDTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSTSPMGWYRQAPGK (SEQ ID NO: 119)
QRELVAAIHGTSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSTSPMGWYRQAPGK (SEQ ID NO: 120)
QRELVAAIHGTSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPRNVQWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSLSPMGWYRQAPGK (SEQ ID NO: 121)
QRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSISPGGWYRQAPGK (SEQ ID NO: 122)
QRELVAAIHGSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNHFSISPMGWYRQAPG (SEQ ID NO: 123)
KQRELVAAIHGSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP
EDTALYYCNKVPWGDYHPRVVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSASPMGWYRQAPG (SEQ ID NO: 124)

KQRELVAAIHGSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP
EDTALYYCNKVPWGDYHPRNVNWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSASPMGWYRQAPG (SEQ ID NO: 125)
KQRELVAAIHGSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP
EDTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNISSISPMGWYRQAPGK (SEQ ID NO: 126)
QRELVAAIHGTSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIESISPMGWYRQAPGK (SEQ ID NO: 127)
QRELVAAIHGTSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSISPYGWYRQAPGKQ (SEQ ID NO: 128)
RELVAAIHGTSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPEDT
ALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIASISPMGWYRQAPGK (SEQ ID NO: 129)
QRELVAAIHGTSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIASISPMGWYRQAPGK (SEQ ID NO: 130)
QRELVAAIHGKSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIASISPMGWYRQAPGK (SEQ ID NO: 131)
QRELVAAIHGSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNITSISPMGWYRQAPGK (SEQ ID NO: 132)
QRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIMSISPMGWYRQAPG (SEQ ID NO: 133)
KQRELVAAIHGNSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP
EDTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNITSISPMGWYRQAPGK (SEQ ID NO: 134)
QRELVAAIHGNSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIVSISPMGWYRQAPGK (SEQ ID NO: 135)
QRELVAAIHGHSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIVSISPMGWYRQAPGK (SEQ ID NO: 136)
QRELVAAIHGKSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNVVSISPMGWYRQAPG (SEQ ID NO: 137)
KQRELVAAIHGKSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP
EDTALYYCNKVPWGDYHPNNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIISISPMGWYRQAPGK (SEQ ID NO: 138)
QRELVAAIHGASTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSITPMGWYRQAPGK (SEQ ID NO: 139)
QRELVAAIHGASTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNITSISPMGWYRQAPGK (SEQ ID NO: 140)
QRELVAAIHGFETLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA)

EVQLVESGGGLVQPGRSLTLSCAASTNIQSISPMGWYRQAPGK (SEQ ID NO: 141) QRELVAAIHGFETLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTSDFSISPMGWYRQAPGK (SEQ ID NO: 142) QRELVAAIHGFETLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIDSISPMGWYRQAPGK (SEQ ID NO: 143) QRELVAAIHGFQTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIMSISPMGWYRQAPG (SEQ ID NO: 144) KQRELVAAIHGFSTVYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIESISPMGWYRQAPGK (SEQ ID NO: 145) QRELVAAIHGFSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSSSPMGWYRQAPGK (SEQ ID NO: 146) QRELVAAIHGFKTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTARYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSSSPMGWYRQAPGK (SEQ ID NO: 147) QRELVAAIHGFSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSNSPMGWYRQAPG (SEQ ID NO: 148) KQRELVAAIHGFSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSTSPMGWYRQAPGK (SEQ ID NO: 149) QRELVAAIHGFSTIYADSVKGRFTISRDNAKNSIYLQMNSLRPED TALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSTSPMGWYRQAPGK (SEQ ID NO: 150) QRELVAAIHGFSTIYADSVKGRFTISRDNAKNSIYLQMNSLRPED TALYYCNKVPWGDYHPLNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCVASTNIFSTSPMGWYRQAPGK (SEQ ID NO: 151) QRELVAAIHGFSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSDSPMGWYRQAPG (SEQ ID NO: 152) KQRELVAAIHGFSTFYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSQSPMGWYRQAPG (SEQ ID NO: 153) KQRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPGNVCWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSQSPMGWYRQAPG (SEQ ID NO: 154) KQRELVAAIHGKSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPSNVYWGKGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSRSPMGWYRQAPGK (SEQ ID NO: 155) QRELVAAIHGESTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGRYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSRSPMGWYRQAPGK (SEQ ID NO: 156) QRELVAAIHGISTLYADSVKGRFTISRDNAKNSIYLQMNSLRPED TALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSRSPMGWYRQAPGK (SEQ ID NO: 157) QRELVAAIHGSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE

DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSGSPMGWYRQAPG (SEQ ID NO: 158) KQRELVAAIHGNSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASSNIFSISPMGWYRQAPGK (SEQ ID NO: 159) QRELVAAIHGSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSIYPMGWYRQAPGK (SEQ ID NO: 160) QRELVAAIHGSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPKNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSKSPMGWYRQAPGK (SEQ ID NO: 161) QRELVAAIHGSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSKSPMGWYRQAPGK (SEQ ID NO: 162) QRELVAAIHGSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNEFSISPMGWYRQAPGK (SEQ ID NO: 163) QRELVAAIHGLSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGAYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNEFSISPMGWYRQAPGK (SEQ ID NO: 164) QRELVAAIHGESTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIPSISPMGWYRQAPGK (SEQ ID NO: 165) QRELVAAIHGESTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVAWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIPSISPMGWYRQAPGK (SEQ ID NO: 166) QRELVAAIHGASTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVAWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIPSISPMGWYRQAPGK (SEQ ID NO: 167) QRELVAAIHGESTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIPSISPMGWYRQAPGK (SEQ ID NO: 168) QRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNITSISPMGWYRQAPGK (SEQ ID NO: 169) QRELVAAIHGVSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVQWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIPSISPMGWYRQAPGK (SEQ ID NO: 170) QRELVAAIHGQSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVQWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIVSISPMGWYRQAPGK (SEQ ID NO: 171) QRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVSWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASSNIFSISPMGWYRQAPGK (SEQ ID NO: 172) QRELVAAIHGESTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVTWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIDSISPMGWYRQAPGK (SEQ ID NO: 173) QRELVAAIHGESTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVTWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIDSISPMGWYRQAPGK (SEQ ID NO: 174)

QRELVAAIHGSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVTWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIRSISPMGWYRQAPGK (SEQ ID NO: 175) QRELVAAIHGSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVVWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNITSISPMGWYRQAPGK (SEQ ID NO: 176) QRELVAAISGFSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPED TALYYCNEVPWGDYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNITSISPMGWYRQAPGK (SEQ ID NO: 177) QRELVAAIHGESTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNITSVSPMGWYRQAPG (SEQ ID NO: 178) KQRELVAAIHGPSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPTNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIGSISPMGWYRQAPGK (SEQ ID NO: 179) QRELVAAIHGQSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPQNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIESISPMGWYRQAPGK (SEQ ID NO: 180) QRELVAAIHGKSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRRVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIVSISPMGWYRQAPGK (SEQ ID NO: 189) QRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRRVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIDSISPMGWYRQAPGK (SEQ ID NO: 190) QRELVAAIHGNSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRMVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFMISPMGWYRQAPG (SEQ ID NO: 191) KQRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFRISPMGWYRQAPGK (SEQ ID NO: 192) QRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSISPMGWYRQAPGK (SEQ ID NO: 193) QRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGEYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIFSISPMGWYRQAPGK (SEQ ID NO: 194) QRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGKYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIESISPMGWYRQAPGK (SEQ ID NO: 195) QRELVAAIHGSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNIESISPMGWYRQAPGK (SEQ ID NO: 196) QRELVAAIHGNSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNISSISPMGWYRQAPGK (SEQ ID NO: 197) QRELVAAIHGFSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGYYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA) EVQLVESGGGLVQPGRSLTLSCAASTNISSISPMGWYRQAPGK (SEQ ID NO: 198) QRELVAAIHGHSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA)

EVQLVESGGGLVQPGRSLTLSCAASTNISSISPMGWYRQAPGK (SEQ ID NO: 199)
QRELVAAIHGFSTVYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSIRPMGWYRQAPGK (SEQ ID NO: 200)
QRELVAAIHGFSTVYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSIYPMGWYRQAPGK (SEQ ID NO: 201)
QRELVAAIHGFSTYYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGSYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFNISPMGWYRQAPGK (SEQ ID NO: 202)
QRELVAAIHGFSTYYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSSSPMGWYRQAPGK (SEQ ID NO: 203)
QRELVAAIHGFSTWYADSVKGRFTISRDNAKNSIYLQMNSLRP
EDTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNISSISPMGWYRQAPGK (SEQ ID NO: 204)
QRELVAAIHGFDTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSINPMGWYRQAPGK (SEQ ID NO: 205)
QRELVAAIHGFDTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPRNVSWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSITPMGWYRQAPGK (SEQ ID NO: 206)
QRELVAAIHGRSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGSYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSITPMGWYRQAPGK (SEQ ID NO: 207)
QRELVAAIHGTSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSITPMGWYRQAPGK (SEQ ID NO: 208)
QRELVAAIHGESTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSITPMGWYRQAPGK (SEQ ID NO: 209)
QRELVAAIHGESTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPRDVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSTSPYGWYRQAPGK (SEQ ID NO: 210)
QRELVAAIHGFSTIYADSVKGRFTISRDNAKNSIYLQMNSLRPED
TALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSTSPGGWYRQAPGK (SEQ ID NO: 211)
QRELVAAIHGFSTIYADSVKGRFTISRDNAKNSIYLQMNSLRPED
TALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSITPYGWYRQAPGK (SEQ ID NO: 212)
QRELVAAIHGASTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
EVQLVESGGGLVQPGRSLTLSCAASTNIFSITPGGWYRQAPGK (SEQ ID NO: 213)
QRELVAAIHGASTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE
DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS Exemplary anti-target (BCMA)
QVQLVESGGGLVQPGESLRLSCAASTNIFSISPMGWYRQAPGK (SEQ ID NO: 214)
QRELVAAIHGFSTLYADSVKGRFTISRDNAKNTIYLQMNSLKPE
DTAVYYCNKVPWGDYHPRNVYWGQGTQVTVSS Exemplary anti-target (MSLN)
QVQLVESGGGLVQPGGSLRLSCAASGRTFSVRGMAWYRQAG (SEQ ID NO: 215)
NNRALVATMNPDGFPNYADAVKGRFTISWDIAENTVYLQMN

SLNSEDTTVYYCNSGPYWGQGTQVTVSS Exemplary anti-target (MSLN) QVQLVESGGGLVQAGGSLRLSCAASGSIPSIEQMGWYRQAPG (SEQ ID NO: 216) KQRELVAALTSGGRANYADSVKGRFTISGDNVRNMVYLQMNS LKPEDTAIYYCSAGRFKGDYAQRSGMDYWGKGTLVTVSS Exemplary anti-target (MSLN) QVQLVESGGGLVQAGGSLRLSCAFSGTTYTFDLMSWYRQAPG (SEQ ID NO: 217) KQRTVVASISSDGRTSYADSVRGRFTISGENGKNTVYLQMNSL KLEDTAVYYCLGQRSGVRAFWGQGTQVTVSS Exemplary anti-target (MSLN) QVQLVESGGGLVQAGGSLRLSCVASGSTSNINNMRWYRQAP (SEQ ID NO: 218) GKERELVAVITRGGYAIYLDAVKGRFTISRDNANNAIYLEMNSL KPEDTAVYVCNADRVEGTSGGPQLRDYFGQGTQVTVSS Exemplary anti-target (MSLN) QVQLVESGGGLVQAGGSLRLSCAASGSTFGINAMGWYRQAP (SEQ ID NO: 219) GKQRELVAVISRGGSTNYADSVKGRFTISRDNAENTVSLQMNT LKPEDTAVYFCNARTYTRHDYWGQGTQVTVSS Exemplary anti-target (MSLN) QVRLVESGGGLVQAGGSLRLSCAASISAFRLMSVRWYRQDPSK (SEQ ID NO: 220) QREWVATIDQLGRTNYADSVKGRFAISKDSTRNTVYLQMNML RPEDTAVYYCNAGGGPLGSRWLRGRHWGQGTQVTVSS Exemplary anti-target (MSLN) QVRLVESGGGLVQAGESLRLSCAASGRPFSINTMGWYRQAPG (SEQ ID NO: 221) KQRELVASISSSGDFTYTDSVKGRFTISRDNAKNTVYLQMNSLK PEDTAVYYCNARRTYLPRRFGSWGQGTQVTVSS Exemplary anti-target (MSLN) QVQPVESGGGLVQPGGSLRLSCVVSGSDFTEDAMAWYRQAS (SEQ ID NO: 222) GKERESVAFVSKDGKRILYLDSVRGRFTISRDIDKKTVYLQMDNL KPEDTGVYYCNSAPGAARNYWGQGTQVTVSS Exemplary anti-target (MSLN) QVQPVESGGGLVQPGGSLRLSCVVSGSDFTEDAMAWYRQAS (SEQ ID NO: 223) GKERESVAFVSKDGKRILYLDSVRGRFTISRDIYKKTVYLQMDNL KPEDTGVYYCNSAPGAARNVWGQGTQVTVSS Exemplary anti-target (MSLN) EVQLVESGGGLVQPGNSLRLSCAASGFTFSSFGMSWVRQAPG (SEQ ID NO: 224) KGLEWVSSISGSGSDTLYADSVKGRFTISRDNAKTTLYLQMNSL RPEDTAVYYCTIGGSLSRSSQGTLVTVSS Exemplary anti-target (MSLN) QVQIVESGGGLVQAGGSLRLSCVASGLTYSIVAVGWYRQAPGK (SEQ ID NO: 225) EREMVADISPVGNTNYADSVKGRFTISKENAKNTVYLQMNSLK PEDTAVYYCHIVRGWLDERPGPGPIVYWGQGTQVTVSS Exemplary anti-target (MSLN) QVQLVESGGGLVQTGGSLRLSCAASGLTFGVYGMEWFRQAP (SEQ ID NO: 226) GKQREWVASHTSTGYVVYYRDSVKGRFTISRDNAKSTVYLQMN SLKPEDTAIYYCKANRGSYEYWGQGTQVTVSS Exemplary anti-target (MSLN) QVQLVESGGGLVQAGGSLRLSCAASTTSSINSMSWYRQAQGK (SEQ ID NO: 227) QREPVAVITDRGSTSYADSVKGRFTISRDNAKNTVYLQMNSLK PEDTAIYTCHVIADWRGYWGQGTQVTVSS Exemplary anti-target (MSLN) QVQLVESGGGLVQAGGSLRLSCAASGRTLSRYAMGWFRQAP (SEQ ID NO: 228) GKERQFVAAISRSGGTTRYSDSVKGRFTISRDNAANTFYLQMN NLRPDDTAVYYCNVRRRGWGRTLEYWGQGTQVTVSS Exemplary anti-target (MSLN) QVQLGESGGGLVQAGGSLRLSCAASGSIFSPNAMIWHRQAPG (SEQ ID NO: 229) KQREPVASINSSGSTNYGDSVKGRFTVSRDIVKNTMYLQMNSL KPEDTAVYYCSYSDFRRGTQYWGQGTQVTVSS Exemplary anti-target (MSLN) QVQLVESGGGLVPSGGSLRLSCAASGATSAITNLGWYRRAPGQ (SEQ ID NO: 230) VREMVARISVREDKEDYEDSVKGRFTISRDNTQNLVYLQMNNL QPHDTAIYYCGAQRWGRGPGTTWGQGTQVTVSS Exemplary anti-target (MSLN) QVQLVESGGGLVQAGGSLRLSCAASGSTFRIRVMRWYRQAPG (SEQ ID NO: 231) TERDLVAVISGSSTYYADSVKGRFTISRDNAKNTLYLQMNNLKP EDTAVYYCNADDSGIARDYWGQGTQVTVSS Exemplary anti-target (MSLN) QVQLVESGGGLVQAGESRRLSCAVSGDTSKFKAVGWYRQAPG (SEQ ID NO: 232)

AQRELLAWINNSGVGNTAESVKGRFTISRDNAKNTVYLQMNR
LTPEDTDVYYCRFYRRFGINKNYWGQGTQVTVSS Exemplary anti-target (MSLN)
QVQLVESGGGLVQAGGSLRLSCAASGSTFGNKPMGWYRQAP (SEQ ID NO: 233)
GKQRELVAVISSDGGSTRYAALVKGRFTISRDNAKNTVYLQME
SLVAEDTAVYYCNALRTYYLNDPVVFSWGQGTQVTVSS Exemplary anti-target (MSLN)
QVQLVESGGGLVQAGGSLRLSCAASGSTSSINTMYWYRQAPG (SEQ ID NO: 234)
KERELVAFISSGGSTNVRDSVKGRFSVSRDSAKNIVYLQMNSLT
PEDTAVYYCNTYIPLRGTLHDYWGQGTQVTVSS Exemplary anti-target (MSLN)
QVQLVESGGGLVQAGGSLRLSCVASGRTDRITTMGWYRQAP (SEQ ID NO: 235)
GKQRELVATISNRGTSNYANSVKGRFTISRDNAKNTVYLQMNS
LKPEDTAVYYCNARKWGRNYWGQGTQVTVSS Exemplary anti-target (MSLN)
QVQLVESGGGLVQARGSLRLSCTASGRTIGINDMAWYRQAPG (SEQ ID NO: 236)
NQRELVATITKGGTTDYADSVDGRFTISRDNAKNTVYLQMNSL
KPEDTAVYYCNTKRREWAKDFEYWGQGTQVTVSS Exemplary anti-target (MSLN)
QVQLVESGGGLVQAGGSLRLSCAASAIGSINSMSWYRQAPGK (SEQ ID NO: 237)
QREPVAVITDRGSTSYADSVKGRFTISRDNAKNTVYLQMNSLK
PEDTAIYTCHVIADWRGYWGQGTQVTVSS Exemplary anti-target (MSLN)
QVQLVESGGGLVQAGGSLRLSCAASGSTSSINTMYWFRQAPG (SEQ ID NO: 238)
EERELVATINRGGSTNVRDSVKGRFSVSRDSAKNIVYLQMNRL
KPEDTAVYYCNTYIPYGGTLHDFWGQGTQVTVSS Exemplary anti-target (MSLN)
QVQLVESGGGLVQAGGSLRLSCTTSTTFSINSMSWYRQAPGN (SEQ ID NO: 239)
QREPVAVITNRGTTSYADSVKGRFTISRDNARNTVYLQMDSLK
PEDTAIYTCHVIADWRGYWGQGTQVTVSS Exemplary anti-target (MSLN)
QVQLVESGGGLVQAGGSLTLSCAASGSTFSIRAMRWYRQAPG (SEQ ID NO: 240)
TERDLVAVIYGSSTYYADAVKGRFTISRDNAKNTLYLQMNNLKP
EDTAVYYCNADTIGTARDYWGQGTQVTVSS Exemplary anti-target (MSLN)
QVQLVESGGGLVQAGGSLRLSCVASGRTSTIDTMYWHRQAPG (SEQ ID NO: 241)
NERELVAYVTSRGTSNVADSVKGRFTISRDNAKNTAYLQMNSL
KPEDTAVYYCSVRTTSYPVDFWGQGTQVTVSS Exemplary anti-target (MSLN)
QVQLVESGGGLVQAGGSLRLSCAASGSTSSINTMYWYRQAPG (SEQ ID NO: 242)
KERELVAFISSGGSTNVRDSVKGRFSVSRDSAKNIVYLQMNSLK
PEDTAVYYCNTYIPYGGTLHDFWGQGTQVTVSS Exemplary anti-target (MSLN)
QVQLVESGGGLVQPGGSLRLSCAASGGDWSANFMYWYRQA (SEQ ID NO: 243)
PGKQRELVARISGRGVVDYVESVKGRFTISRDNAKNTVYLQMN
SLKPEDTAVYYCAVASYWGQGTQVTVSS Exemplary anti-target (MSLN)
EVQLVESGGGLVQPGGSLRLSCAASGGDWSANFMYWYRQAP (SEQ ID NO: 244)
GKQRELVARISGRGVVDYVESVKGRFTISRDNSKNTLYLQMNSL
RAEDTAVYYCAVASYWGQGTLVTVSS Exemplary anti-target (MSLN)
EVQLVESGGGLVQPGGSLRLSCAASGGDWSANFMYWVRQAP (SEQ ID NO: 245)
GKGLEWVSRISGRGVVDYVESVKGRFTISRDNSKNTLYLQMNS
LRAEDTAVYYCAVASYWGQGTLVTVSS Exemplary anti-target (MSLN)
EVQLVESGGGLVQAGGSLRLSCAASGSTSSINTMYWYRQAPG (SEQ ID NO: 246)
KERELVAFISSGGSTNVRDSVKGRFTISRDNSKNTLYLQMNSLR
AEDTAVYYCNTYIPYGGTLHDFWGQGTLVTVSS Exemplary anti-target (MSLN)
EVQLVESGGGLVQPGGSLRLSCAASGSTSSINTMYWYRQAPG (SEQ ID NO: 247)
KERELVAFISSGGSTNVRDSVKGRFTISRDNSKNTLYLQMNSLR
AEDTAVYYCNTYIPYGGTLHDFWGQGTLVTVSS Exemplary anti-target (MSLN)
EVQLVESGGGLVQPGGSLRLSCAASGSTSSINTMYWVRQAPG (SEQ ID NO: 248)
KGLEWVSFISSGGSTNVRDSVKGRFTISRDNSKNTLYLQMNSLR
AEDTAVYYCNTYIPYGGTLHDFWGQGTLVTVSS Exemplary anti-target (MSLN)

QVQLVESGGGVVQAGGSLRLSCAASGSTFSIRAMRWYRQAPG (SEQ ID NO: 249) TERDLVAVIYGSSTYYADAVKGRFTISRDNSKNTLYLQMNSLRA EDTAVYYCNADTIGTARDYWGQGTLVTVSS Exemplary anti-target (MSLN) QVQLVESGGGVVQPGGSLRLSCAASGSTFSIRAMRWYRQAPG (SEQ ID NO: 250) KERELVAVIYGSSTYYADAVKGRFTISRDNSKNTLYLQMNSLRA EDTAVYYCNADTIGTARDYWGQGTLVTVSSGG Exemplary anti-target (MSLN) QVQLVESGGGVVQPGGSLRLSCAASGSTFSIRAMRWVRQAPG (SEQ ID NO: 251) KGLEWVSVIYGSSTYYADAVKGRFTISRDNSKNTLYLQMNSLRA EDTAVYYCNADTIGTARDYWGQGTLVTVSSGG Exemplary anti-target (MSLN) EVQLVESGGGLVQAGGSLRLSCVASGRTSTIDTMYWHRQAPG (SEQ ID NO: 252) NERELVAYVTSRGTSNVADSVKGRFTISRDNSKNTLYLQMNSL RAEDTAVYYCSVRTTSYPVDFWGQGTLVTVSGG Exemplary anti-target (MSLN) EVQLVESGGGLVQPGGSLRLSCAASGRTSTIDTMYWHRQAPG (SEQ ID NO: 253) KERELVAYVTSRGTSNVADSVKGRFTISRDNSKNTLYLQMNSLR AEDTAVYYCSVRTTSYPVDFWGQGTLVTVSS Exemplary anti-target (MSLN) EVQLVESGGGLVQPGGSLRLSCAASGRTSTIDTMYWVRQAPG (SEQ ID NO: 254) KGLEWVSYVTSRGTSNVADSVKGRFTISRDNSKNTLYLQMNSL RAEDTAVYYCSVRTTSYPVDFWGQGTLVTVSS Exemplary anti-target (MSLN) QVQLVESGGGVVQAGGSLTLSCAASGSTFSIRAMRWYRQAPG (SEQ ID NO: 255) TERDLVAVIYGSSTYYADAVKGRFTISRDNSKNTLYLQMNSLRA EDTAVYYCNADTIGTARDYWGQGTLVTVSS Exemplary anti-target (MSLN) QVQLVESGGGVVQAGGSLRLSCAASGSTFSIRAMRWYRQAPG (SEQ ID NO: 256) TERDLVAVIYGSSTYYADAVKGRFTISRDNSKNTLYLQMNSLRA EDTAVYYCNADTIGTARDYWGQGTLVTVSS Exemplary anti-target (MSLN) QVQLVESGGGVVQPGGSLRLSCAASGSTFSIRAMRWYRQAPG (SEQ ID NO: 257) KERELVAVIYGSSTYYADAVKGRFTISRDNSKNTLYLQMNSLRA EDTAVYYCNADTIGTARDYWGQGTLVTVSS Exemplary anti-target (MSLN) QVQLVESGGGVVQPGGSLRLSCAASGSTFSIRAMRWVRQAPG (SEQ ID NO: 258) KGLEWVSVIYGSSTYYADAVKGRFTISRDNSKNTLYLQMNSLRA EDTAVYYCNADTIGTARDYWGQGTLVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVQAGGSLRLSCAASGSIFSIASMGWYRQAPG (SEQ ID NO: 306) KQRELVAVITSFSSTNYADSVKGRFTISRDNAKNTVYLQMNSLK PEDTGVYYCNARYFERTDWGQGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVQAGGSLRLSCAAPGSIFSIASMGWYRQAPG (SEQ ID NO: 307) KQRELVAVITSFSSTNYADSVKGRFTISRDNAKNTVYLQMNSLK PEDTGVYYCNARYFERTDWGQGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVQAGGSLRLSCAASGSIFSIASMAWYRQAPG (SEQ ID NO: 308) KQRELVAAITSFSSTNYADSVKGRFTISRDNAKNTVYLQMNSLK PEDTGVYYCNARYFERTDWGQGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVQAGGSLRLSCAASESIFSINVMAWHRQAPG (SEQ ID NO: 309) KQRELVARITSGGSTNYADSVKGRFTISRDNAKNTVYLQMNSL KPEDTGVYYCGAYQGLYAYWGQGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVQAGGSLRLSCVASGSSFSITSMAWYRQAPG (SEQ ID NO: 310) KQRDLVAAITSFGSTNYADSVKDRFTISRDNAKNTVYLQMNSL KPEDTAVYYCNGRVFDHVYWGQGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVQAGGSLKLSCAASSSIFSISSMSWYRQAPGK (SEQ ID NO: 311) QRELVAAITTFDYTNYADSVKGRFTISRDNAKNMMYLQMNSL KPEDTAVYLCNARAFGRDYWGQGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVQAGGSLKLSCAASSSIFSISSMSWYRQAPGK (SEQ ID NO: 312) QRELVAAITSFGSTNYADSVKGRFTISRDNAKNMMYLQMNSL

KPEDTAVYRCNARTMGRDYWGQGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVQPGGSLRLSCAASGSTLNIKIMAWHRQAPG (SEQ ID NO: 313) KQRELVATLTSGGNTNYADSVKGRFTISRDNAKNTVYLQMNSL QPEDTAVYYCGLWDGVGGAYWGRGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVQPGGSLRISCAASGSTLNIKIMAWHRQAPG (SEQ ID NO: 314) KQRELVATLTSGGNTNYADSVKGRFTISRDNAKNTVYLQMNSL QPEDTAVYYCGLWDGVGGAYWGRGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVQAGGSLRLSCAASGSTFNIKTMAWHRQAP (SEQ ID NO: 315) GNQRELVATLTSGGNTNYADSVKGRFTISRDNAKNTVYLQMN SLKPEDTAVYYCGLWNGVGGAYWGRGTQVTVSS Exemplary anti-target (DLL3) QVQLQDGGGLVQPGGSLRLSCAASGSTFNIKLMAWHRQAPG (SEQ ID NO: 316) NQRELVATLTSGGNTNYADSVKGRFTISRDNASNIVYLQMNSL KPEDTAVYYCGLWDGVGGAYWGRGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVQAGGSLRLSCAASGSTFNFKIMAWHRQAP (SEQ ID NO: 317) GKQRELVASLTSEGLTNYRDSVKGRFTISRDNAKNTVYLQMNN LKPEDTAVYYCGLWDGVGGAYWGRGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVQPGGSLRLSCAASGFMFSSYSMSWYRQAP (SEQ ID NO: 318) GKQRELVAAITTWGSTNYADSVKGRFTISRDNAKNTVWLQM NSLEPEDTAVYFCNARSWNNYWGQGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVQVGGSLRLSCAASGFMFSSYSMSWYRQAP (SEQ ID NO: 319) GKQRELVAAITSYGSTNYADSVKGRFTISRDNAKNTVWLQMN SLKPEDTAVYFCNARSWNNYWGQGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVQPGGSLRLSCAASGFTFSSHSMSWYRQAPG (SEQ ID NO: 320) KQRELVAAITTYGSTNYIDSVKGRFTISRDNTKNTVYLQMNSLK PEDTAVYFCNARSWNNYWGQGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVQAGGSLRLSCVASGSSFSHNTMGWYRQAP (SEQ ID NO: 321) GKQRDLVARITTFGTTNYADSVKGRFTISRDNAKNTVYLQMNS LKPEDTAVYYCNGESFGRIWYNWGQGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVQAGASLRLTCTASGGRFSYATMGWSRQAP (SEQ ID NO: 322) GKQREMVARITSSGFSTNYADSVKGRFTISRDNAKNAVYLQM DSLKPEDTAVYYCNAQHFGTDSWGQGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVQAGASLRLTCTASGSRFSYATMGWSRQAP (SEQ ID NO: 323) GKQRELVARITSSGFSTNYADSVKGRFTISRDNAKNAVYLQMD SLKPEDTAVYYCNAQQFGTDSWGQGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVQAGGSLRLSCAASGSTFTSNVMGWHRQAP (SEQ ID NO: 324) GKQRELVANMHSGGSTNYADSVKGRFTISRDNAKNIVYLQMN NLKIEDTAVYYCRWYGIQRAEGYWGQGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVVAGGSLRLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 325) KKRELVAGISVDGSTNYADSVKGRFTVSRDNAKNTVYLQMNSL QPEDTAVYYCYAYRWVGRDTYWGQGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVVAGGSLRLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 326) KKRELVAGISVDGSTNYADSVKGRFTISRDNAKNTVYLQMNSL QPEDTAVYYCYAYRWVGRDTYWGQGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVVSGGSLRLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 327) KKRELVAGISVDGSTNYADSVKGRFTISRDNAKNTVYLQMNSL QPEDTAAYYCYAYRWVGRDTYWGQGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVVAGGSLRLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 328) KKRELVAGISVDGSTNYADSVKGRFTISRDNAENTVYLQMNSL QPEDTAVYYCYAYRWEGRDTYWGQGTQVTVSS Exemplary anti-target (DLL3) QVQLQESGGGLVVAGGSLRLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 329)

KKRELVAGISVDGSTNYADSVKGRFTISRDNAENTVYLQMNSL
QPEDTAVYYCYAYRWEGRNTYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVQPGGSLRLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 330)
KKRELVAGISTDGSTNYVDSVKGRFTISRDNAKNTVYLQMNSL
QPEDTAVYYCYAYRWVGRYTYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVVAGGSLRLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 331)
KKRELVAGISTDGTTNYVDSVKDRFTISRDNAKNTVYLQMNSL
QPEDTAAYYCYAYRWVGRDTYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVVAGGSLRLSCAASGSSVSFLSIAWYRQAPGK (SEQ ID NO: 332)
KRELVAGISTDGTTNYVDSVKDRFTISRDNAKNTVYLQMNSLQ
PEDTAAYYCYAYRWVGRDTYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVQAGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 333)
KKRELVAGISTDGTTNYVDSVKDRFTISRDNAKNTVYLQMNSL
QPEDTAAYYCYAYRWVGRDTYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVQAGGSLRLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 334)
KKRELVAGISTDGSTNYADSVKGRFTISEGNAKNTVDLQMNSL
QPEDTAVYYCYAYRWVDRYTYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVVAGGSLRLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 335)
KKRELVAGISTDGSTNYADSVKGRFTISEDNAKNTVDLQMNSL
QPEDTAVYYCYAYRWIDRYTYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVQPGGSLRLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 336)
KKRELVAGISTDGSTNYADSVKGRFTISEDNAKNTVDLQMNSL
QPEDTAVYYCYAYRWVDRYTYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVVAGGSLRLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 337)
KKRELVAGISSDGSTNYVDSVKGRFTISRDNAKNIVFLQMNSLQ
PQDTAVYYCYAYRWVGRDTYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVVAGGSLRLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 338)
KKRELVAGISADGSTDYIDSVKGRFTISRDSANNTMYLQMNSL
QPEDTAVYYCYAYRWTTRYTYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVQPGGSLRLSCAASGFTLDYYAIGWYRQAPG (SEQ ID NO: 339)
KKRELVAGISSDGSTHYVDSVKGRFAISRDNAENTVYLQMNDL
QPDDTAVYYCYAYRWVGGYTYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVQAGGSLRLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 340)
KQRELVAGISSDGSKNYADSVKGRFTISRDNAKNTVYLQMNSL
KPEDTAVYYCYYFRTVAASSMQYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVQAGGSLRLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 341)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNTVYLQMNSL
KPEDTAVFYCYYFRTVSGSSMRYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVQAGGSLRLSCAASGITSSVYSMGWYRQAPG (SEQ ID NO: 342)
KQRELVAGSSSDGSTHYVDSVRGRFTISRDNAKNTVYLQMSSL
KPEDTAVYYCYANRGFAGAPSYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVQAGGSLRLSCAASGRTSMFNSMGWHRQA (SEQ ID NO: 343)
PGKQRELVAIIRSGGSSNYADTVKGRFTISRDNTKNTVYLQMN
DLKPEDTAVYYCFYYFQSSYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVQAGGSLRLSCAASGRTSMVNSMGWHRQA (SEQ ID NO: 344)
PGKQRELVALITSGGSSNYADTVKGRFTISRDNTKNTVYLQMN
DLKPEDTAVYYCFYYFQSSYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVQAGGSLRLSCAASGSVSMFNSMGWHRQP (SEQ ID NO: 910)
PGKQRELVAIITSGGSSNYADTVKGRFTISRDNTKNTVYLQMN
DLKPEDTAVYYCFYYFQSSYWGQGTQVTVSS Exemplary anti-target (DLL3)

QVQLQESGGGLVQAGGSLRLSCTASGSIFSIAVMGWYRQVPG (SEQ ID NO: 345)
KRREWVATIFDGSYTNYADSVKGRFTISRDNARNKVYLQMNN
LKPEDTAVYYCQTHWTQGSVPKESWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVQAGGSLRLSCVASSGIFSDMSMVWYRQAP (SEQ ID NO: 346)
GKQRELVASITTFGSTNYADPVKGRFTISRDNAKNTVYLQMNSL
KPEDTAVYYCSGRSYSSDYWGRGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVQAGGSLRLSCVASGSISSIIVMGWSRQAPGK (SEQ ID NO: 347)
QRESVATITRDGTRNYADSLKGRFTISRDNAKNTSYLQINSLKPE
DTAVYSCYARYGDINYWGKGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVQAGGSLRLSCVASGSISSIIVMGWSRQAPGK (SEQ ID NO: 348)
QRESLATISRGGTRTYADSVKGRFTISRDNAKNTSYLQMNSLKP
EDTAVYSCYARYGDINYWGKGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVQAGGSLRLSCVASGSIFTTNSMGWHRQGP (SEQ ID NO: 349)
GKQRELVALIGSAGSTKYADSVKGRFTISRDNAKNTVSLQMDSL
KPEDTAVYYCFYYDSRSYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGMVQPGGSLRLSCAASGSREISTMGWHRQAPG (SEQ ID NO: 350)
KQRELAARITSGGITKYADSVKGRFTISRDNAKKTVYLQMNSLK
SEDTAVYYCFAYDNINAYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGWVQAGGSLRLSCAASGSREISTMGWHRQAP (SEQ ID NO: 351)
GKQRELAARITSGGITKYADSVKGRFTISRDNAKKTVYLQMNSL
KSEDTAVYYCFAYDNINAYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGWVQAGGSLRLSCTASGSREISTMGWHRQAPG (SEQ ID NO: 352)
KQRELAARITSGGITKYADSVKGRFTISRDNAKKTVYLQMDSLK
SEDTAVYYCFAYDNINAYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGSVQAGRSLGLSCAASGSREISTMGWHRQAPG (SEQ ID NO: 353)
KQRELAARITSGGITKYADSVKGRFTISRDNAKKTVYLQMNSLK
SEDTAVYYCFAYDNINAYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVQAGGSLRLSCTASGSIFRGAAMYWHRQAP (SEQ ID NO: 354)
GKQRELVAAITTSGNTSYADSVKGRFTISRDNAKNTMYLQIISLK
PEDTAVYYCAFWIAGKAYWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVQPGGSLRLSCAASGSISSFNFMSWHRQAPG (SEQ ID NO: 355)
KERELAGVITRGGATNYADSVKGRFTISRDNVKNTVYLQMNGL
KPEDTAVYYCHGRSQLGSTWGQGTQVTVSS Exemplary anti-target (DLL3)
QVQLQESGGGLVQAGGSLRLSCLASGTIFTASTMGWHRQPPG (SEQ ID NO: 356)
KQRELVASIAGDGRTNYAESTEGRFTISRDDAKNTMYLQMNSL
KPEDTAVYYCYAYYLDTYAYWGQGTQVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSIFSIASMGWYRQAPGK (SEQ ID NO: 357)
QRELVAVITSFSSTNYADSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCNARYFERTDWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGRTSMFNSMGWHRQAP (SEQ ID NO: 358)
GKQRELVAIIRSGGSSNYADTVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCFYYFQSSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 359)
KKRELVAGISVDGSTNYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYAYRWVGRDTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCTASGSIFRGAAMYWHRQAPG (SEQ ID NO: 360)
KQRELVAAITTSGNTSYADSVKGRFTISRDNAKNSMYLQMNSL
RAEDTAVYYCAFWIAGKAYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSTFNIKTMAWHRQAPG (SEQ ID NO: 361)
NQRELVATLTSGGNTNYADSVKGRFTISRDNAKNSVYLQMNSL

RAEDTAVYYCGLWNGVGGAYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSTLNIKIMAWHRQAPGK (SEQ ID NO: 362)
QRELVATLTSGGNTNYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCGLWDGVGGAYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASSSIFSISSMSWYRQAPGK (SEQ ID NO: 363)
QRELVAAITTFDYTNYADSVKGRFTISRDNAKNSMYLQMNSLR
AEDTAVYYCNARAFGRDYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 364)
KQRELVAGISSDGSKNYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVAASSMQYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGFMFSSYSMSWYRQAPG (SEQ ID NO: 365)
KQRELVAAITSYGSTNYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCNARSWNNYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSVSMFNSMGWHRQPP (SEQ ID NO: 366)
GKQRELVAIITSGGSSNYADTVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCFYYFQSSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCTASGGRFSYATMGWSRQAPG (SEQ ID NO: 367)
KQREMVARITSSGFSTNYADSVKGRFTISRDNAKNSVYLQMNS
LRAEDTAVYYCNAQHFGTDSWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSISSFNFMSWHRQAPG (SEQ ID NO: 368)
KERELAGVITRGGATNYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCHGRSQLGSTWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASESIFSINVMAWHRQAPGK (SEQ ID NO: 369)
QRELVARITSGGSTNYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCGAYQGLYAYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSREISTMGWHRQAPGK (SEQ ID NO: 370)
QRELAARITSGGITKYADSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCFAYDNINAYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCTASGSIFSIAVMGWYRQVPGK (SEQ ID NO: 371)
RREWVATIFDGSYTNYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCQTHWTQGSVPKESWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 372)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASSGIFSDMSMVWYRQAPG (SEQ ID NO: 373)
KQRELVASITTFGSTNYADPVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCSGRSYSSDYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGFMFSSYSMSWYRQAPG (SEQ ID NO: 374)
KQRELVAAITTWGSTNYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCNARSWNNYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCTASGSRFSYATMGWSRQAPG (SEQ ID NO: 375)
KQRELVARITSSGFSTNYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCNAQQFGTDSWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSTFTSNVMGWHRQAP (SEQ ID NO: 376)
GKQRELVANMHSGGSTNYADSVKGRFTISRDNAKNSVYLQM
NSLRAEDTAVYYCRWYGIQRAEGYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 377)
KKRELVAGISTDGSTNYVDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWVGRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSISSIIVMGWSRQAPGK (SEQ ID NO: 378)

QRESVATITRDGTRNYADSLKGRFTISRDNAKNSSYLQMNSLRA
EDTAVYYCYARYGDINYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 379)
KKRELVAGISADGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCLASGTIFTASTMGWHRQPPGK (SEQ ID NO: 380)
QRELVASIAGDGRTNYAESTEGRFTISRDNAKNSMYLQMNSLR
AEDTAVYYCYAYYLDTYAYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 381)
KKRELVAGISVDGSTNYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYAYRWEGRNTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSSFSHNTMGWYRQAPG (SEQ ID NO: 382)
KQRDLVARITTFGTTNYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCNGESFGRIWYNWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSISSIIVMGWSRQAPGK (SEQ ID NO: 383)
QRESLATISRGGTRTYADSVKGRFTISRDNAKNSSYLQMNSLRA
EDTAVYYCYARYGDINYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSSFSITSMAWYRQAPGK (SEQ ID NO: 384)
QRDLVAAITSFGSTNYADSVKDRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCNGRVFDHVYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGRTSMVNSMGWHRQAP (SEQ ID NO: 385)
GKQRELVALITSGGSSNYADTVKGRFTISRDNAKNSVYLQMNS
LRAEDTAVYYCFYYFQSSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSTFNFKIMAWHRQAPG (SEQ ID NO: 386)
KQRELVASLTSEGLTNYRDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCGLWDGVGGAYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGFTLDYYAIGWYRQAPGK (SEQ ID NO: 387)
KRELVAGISSDGSTHYVDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYRWVGGYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSIFTTNSMGWHRQGPG (SEQ ID NO: 388)
KQRELVALIGSAGSTKYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCFYYDSRSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGITSSVYSMGWYRQAPG (SEQ ID NO: 389)
KQRELVAGSSSDGSTHYVDSVRGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYANRGFAGAPSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASSSIFSISSMSWYRQAPGK (SEQ ID NO: 390)
QRELVAAITSFGSTNYADSVKGRFTISRDNAKNSMYLQMNSLR
AEDTAVYYCNARTMGRDYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGFTSSINAMGWYRRAPG (SEQ ID NO: 391)
KQRELVAGISSDGSFVYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRHVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 392)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVSGSSSRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 393)
KQRELVAGISSDGSKVYEDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSPSSINAMGWYRRAPG (SEQ ID NO: 394)
KQRELSAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVRGSSMSYWGQGTLVTVSS Exemplary anti-target (DLL3)

EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 395)
KQRELAAGISSDGSSVYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVSGSSKRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSISSINAMGWYRRAPGK (SEQ ID NO: 396)
QRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRMVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 397)
KQRELVAGISSDGSKLYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVQGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAYGWYRRAPGK (SEQ ID NO: 398)
QRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVYGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAIGWYRRAPGK (SEQ ID NO: 399)
QRELVAGISSDGSKVYIDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYYFRTVSGSSYRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 400)
KQRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVLGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSIINAMGWYRRAPGK (SEQ ID NO: 401)
QRELAAGISSDGSKVIADSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYYFRRVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 402)
KQRELVAGISSDGSKIYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGKTSSINAMAWYRRAPG (SEQ ID NO: 403)
KQRELVAGISSDGSKVYTDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVSGSSARYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGTTSSINAMGWYRRAPG (SEQ ID NO: 404)
KQRELVAGISSDGSLVYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRIVRGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 405)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYYRTVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSGSSINAMGWYRRAPG (SEQ ID NO: 406)
KQRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRHVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGTTSSINAMGWYRRAPG (SEQ ID NO: 407)
KQRELVAGISSDGSKVYVDSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRFVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 408)
KQRELVAGISSDGSKVYVDSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSRINAMGWYRRAPG (SEQ ID NO: 409)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTKSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSRINAMGWYRRAPG (SEQ ID NO: 410)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVYGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 411)
KQRELVAGISSDGSKVYRDSVKGRFTISRDNAKNSVYLQMNSL

RAEDTAVYYCYYFRTVSGSSMGYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 412)
KQRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVSGSSMRSWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 413)
KQRELVAGISSDNSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVGGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGNTSSINAMAWYRRAPG (SEQ ID NO: 414)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 415)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVSGSHMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSIINAMGWYRRAPGK (SEQ ID NO: 416)
QRELVAGISSDGSKVYEDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRAVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 417)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSRINAMGWYRRAPG (SEQ ID NO: 418)
KQRELPAGISSDGSKVYAVSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVSGSPMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSRINAMGWYRRAPG (SEQ ID NO: 419)
KQRELVAGVSSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVSGSSMSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGTTSSINAMGWYRRAPG (SEQ ID NO: 420)
KQRELVAGISSDGSKVYEDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGITSSINAMGWYRRAPGK (SEQ ID NO: 421)
QRELVAGISSDGSKVYAGSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVRGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSDINAMGWYRRAPG (SEQ ID NO: 422)
KQRELVAGISSDKSKVYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVRGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 423)
KQRELVAGISSNGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRQVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAIGWYRRAPGK (SEQ ID NO: 424)
QRELVAGISSDGSKVLADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRIVSGSSMGYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSKNAMGWYRRAPG (SEQ ID NO: 425)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVSGASMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 426)
KQRELVAGISSDNSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVHGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGLTSSINAMGWYRRAPG (SEQ ID NO: 427)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRMVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 428)

KQRELVAGISSDGSKVYTDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTISGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSNNAMAWYRRAPG (SEQ ID NO: 429) KQRELVAGISSDGSKVYTDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTRSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 430) KQRELVAGISSDNSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFRTVSGHSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSHINAMGWYRRAPG (SEQ ID NO: 431) KQRELVAGISSDGSRVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFRTVSGGSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGQTSSINAMGWYRRAPG (SEQ ID NO: 432) KQRELVAGISSDGSQVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFRTKSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSRINGMGWYRRAPG (SEQ ID NO: 433) KQRELPAGISSDGSKAYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFRTASGTSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSVINAMAWYRRAPG (SEQ ID NO: 434) KQRELAAGISSDGSKVYAKSAKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFNTVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 435) KQRELVAGISSDGSKVYNDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFRTVRGSSQRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGKTSSINAMGWYRRAPG (SEQ ID NO: 436) KQRELVAGISSDGSKVIADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTVLGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 437) KQRELVAGISSDGSKVYTDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTRSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSVSSINAMGWYRRAPG (SEQ ID NO: 438) KQRELVAGISSDGSKVYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTVSGLSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGNTSSINAMGWYRRAPG (SEQ ID NO: 439) KQRELVAGISSDGSKVYYDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTVRGSSQRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSTNAMGWYRRAPG (SEQ ID NO: 440) KQRELVAGISSDGSKVYVDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFRTVSGSSMVYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 441) KQRELVAGISSDGSKVYGDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFRTVSRSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 442) KQRELAAGISSDQSKVYADSAKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFRTVSGSSMSYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGGTSSINAMGWYRRAPG (SEQ ID NO: 443) KQRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTVSGSSARYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTRSINAMGWYRRAPG (SEQ ID NO: 444) KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFHTVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)

EVQLVESGGGLVQPGGSLTLSCVASGTTSSINAMGWYRRAPG (SEQ ID NO: 445) KQRELVAGISSDGSKVIADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTVLGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 446) KQRELVAGISSDGSKVDADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFRTVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 447) KQRELVAGISSDGSKVYKDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRNVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGNTSSINAMGWYRRAPG (SEQ ID NO: 448) KQRELVAGISSNGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFRTVTGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSRINAMGWYRRAPG (SEQ ID NO: 449) KQRELVAGISSDGSKVYKDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAIGWYRRAPGK (SEQ ID NO: 450) QRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTVKGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGLTSSINAMGWYRRAPG (SEQ ID NO: 451) KQRELVAGISSDGSKVYQDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFRTNSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 452) KQRELVAGISSDGSKVYAESVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTVSGASMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSTNAMGWYRRAPG (SEQ ID NO: 453) KQRELVAGISSDGSKVLADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTVNLSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAIGWYRRAPGK (SEQ ID NO: 454) QRELVAGISSDGSKYYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTVTGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAIGWYRRAPGK (SEQ ID NO: 455) QRELVAGISSDGSKVYAVSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRKVSGSSARYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 456) KQRELVAGISSDGSKVVADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFRTYSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 457) KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFRTVSKSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 458) KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFKTVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 459) KQRELAAGISSDNSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFRTRSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSKSSINAMGWYRRAPG (SEQ ID NO: 460) KQRELAAGISSDGSKVYAQSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFRTSSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGTTSSINAMGWYRRAPG (SEQ ID NO: 461) KQRELVAGISSDGSKVYVDSVKGRFTISRDNAKNSVYLQMNSL

RAEDTAVYYCYYFRFLSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAFGWYRRAPGK (SEQ ID NO: 462)
QRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYYFRTVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTFSINAMGWYRRAPG (SEQ ID NO: 463)
KQRELVAGISSDGSKVLADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRLVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTRSINAMGWYRRAPG (SEQ ID NO: 464)
KQRELVAGISSDGSKVYNDSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVSGSSMRFWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAIGWYRRAPGK (SEQ ID NO: 465)
QRELVAGISSDGSKVYNDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTQSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 466)
KQRELVAGISSDGSKVYVDSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVSGSSMPYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 467)
KQRELVAGISSDGSKVVADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTLSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGTTSSINAMGWYRRAPG (SEQ ID NO: 468)
KQRELVAGISSDGSKVYGDSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVSGSAMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 469)
KQRELVAGISSDGSKVYTDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTTSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGRTSSINAMGWYRRAPG (SEQ ID NO: 470)
KQRELVAGISSDGSKVYNDSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVSGTSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSRNAMGWYRRAPG (SEQ ID NO: 471)
KQRELVAGISSDGSKVTADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTRSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTKSINAMGWYRRAPG (SEQ ID NO: 472)
KQRELVAGISSDGSKVYRDSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTSSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSRNAMGWYRRAPG (SEQ ID NO: 473)
KQRELVAGISSNGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVSGSSMSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAIGWYRRAPGK (SEQ ID NO: 474)
QRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYYFRPVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSLINAMGWYRRAPG (SEQ ID NO: 475)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRHVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 476)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTKSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 477)
KQRELVAGISSDGSLVYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFTTVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGTTSSINAMGWYRRAPG (SEQ ID NO: 478)

KQRELVAGISSDGTKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFHTVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAFGWYRRAPGK (SEQ ID NO: 479)
QRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVRGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSRNAMGWYRRAPG (SEQ ID NO: 480)
KQRELVAGISSDGSKLYLDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVLGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGNTSSINAMGWYRRAPG (SEQ ID NO: 481)
KQRELVAGISSDGSRVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVSGSSMRSWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGTTSSINAMGWYRRAPG (SEQ ID NO: 482)
KQRELVAGISSDGSKVYNDSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVRGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTASINAMGWYRRAPG (SEQ ID NO: 483)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRYVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAVGWYRRAPGK (SEQ ID NO: 484)
QRELVAGISSDGSKVYVDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVYGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSRNAMGWYRRAPG (SEQ ID NO: 485)
KQRELVAGISSDGSKLYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVLGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTNSINAMGWYRRAPG (SEQ ID NO: 486)
KQRELVAGISSDGSKVYKDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYYRTVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 487)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRSVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSRINAMGWYRRAPG (SEQ ID NO: 488)
KQRELVAGISSDGSKVYQDSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRRVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVPSGSTSNINAMGWYRRAPG (SEQ ID NO: 489)
KQRELPAGISSDGTKIYADSAKVPFTITRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVSGTSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSKINAMGWYRRAPG (SEQ ID NO: 490)
KQRELVAGISSDRSKVYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVAGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINALGWYRRAPGK (SEQ ID NO: 491)
QRELVAGISSDGSLVYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRIVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGKTSSINAMGWYRRAPG (SEQ ID NO: 911)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVSGVSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAVGWYRRAPGK (SEQ ID NO: 912)
QRELVAGISSDGSKVYRDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVQGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSRINAMGWYRRAPG (SEQ ID NO: 492)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTASGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)

EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 493) KQRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTVSGSSSRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAVGWYRRAPGK (SEQ ID NO: 494) QRELVAGISSDGTKVYRDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTVQGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGTTSSINAMGWYRRAPG (SEQ ID NO: 495) KQRELAAGISSDGSKVYNDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFRTVRGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 496) KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFRTKSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGTTSSINAMGWYRRAPG (SEQ ID NO: 497) KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFRTVWGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGKTSSINAMGWYRRAPG (SEQ ID NO: 498) KQRELVAGISSDGSKVYTDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTRSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPFK (SEQ ID NO: 499) QGELPAGISPDGTKAYADSAKVRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFHTVCGTSMGYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSAINAMGWYRRAPG (SEQ ID NO: 500) KQRELVAGISSDGSKVYVDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFRTVSGSSQRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSPSSINAYGWYRRAPGK (SEQ ID NO: 501) QRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYYFRTVSGSSMSYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 502) KQRELVAGISSDGSKVYASSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTVRGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSRSSINAMGWYRRAPG (SEQ ID NO: 503) KQRELVAGISADGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFRTQSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSVSSINAMGWYRRAPG (SEQ ID NO: 504) KQRELVAGISSDGSKVYASSAKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTLSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 505) KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFHTVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAIGWYRRAPGK (SEQ ID NO: 506) QRELVAGISSDGSSVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRRVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 507) KQRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRLVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAVGWYRRAPGK (SEQ ID NO: 508) QRELVAGISSDGSKVYAGSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTVSGSYMRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 509) KQRELAAGISSDNSKVYADSVKGRFTISRDNAKNSVYLQMNSL

RAEDTAVYYCYYFRTVGGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAYGWYRRAPGK (SEQ ID NO: 510)
QRELVAGISSDGSAVYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTHSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAVGWYRRAPGK (SEQ ID NO: 511)
QRELVAGISSDGSSVYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVSTSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSKSSINAMGWYRRAPG (SEQ ID NO: 512)
KQRELPAGISSNGTKVYADSAKVRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVLGTSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 513)
KQRELVAGISSDGSKLYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSVSSINAMGWYRRAPG (SEQ ID NO: 514)
KQRELVAGISSDGSKVYKDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVSGSSMGYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 515)
KQRELVAGISSDGSLVYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVSGSSMRAWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 516)
KQRELVAGISSDGSLVYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRILSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 517)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVQGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSYINAMGWYRRAPG (SEQ ID NO: 518)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVSGQSMGYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 913)
KQRELVAGVSSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVSGSSARYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 914)
KQRELPAGISRDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRYVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 915)
KQRELAAGISSDGSKLYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSRINAMGWYRRAPG (SEQ ID NO: 916)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRRVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 917)
KQRELAAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFHTVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 918)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRQVSGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 919)
KQRELVAGISSDTSKVYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVSGSYMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSTINAMGWYRRAPG (SEQ ID NO: 920)

KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTASGSSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAVGWYRRAPGK (SEQ ID NO: 519)
QRELVAGISSDGSTVYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVSGHSMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 520)
KQRELAAGISKDGSKVYADSAKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRTVSGSSSRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSPSSINAYGWYRRAPGK (SEQ ID NO: 521)
QRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYYFRTVSGSSYSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSPSSINAYGWYRRAPGK (SEQ ID NO: 522)
QRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYYFRTVSGSSQSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVKFLSMAWYRQAPG (SEQ ID NO: 523)
KKRELVAGISADGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTRRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSLAWYRQAPGK (SEQ ID NO: 524)
KRELVAGISADGSTAYIDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYRWTRRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSRVSFLSMAWYRQAPG (SEQ ID NO: 525)
KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRRTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSQVSFLSMAWYRQAPG (SEQ ID NO: 526)
KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRYIYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 527)
KKRELVAGISEAGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLRCAASGSKVSFLSMAWYRQAPG (SEQ ID NO: 528)
KKRELVAGISADGSTDYVDSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYAYRWTRRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVGFLSMAWYRQAPG (SEQ ID NO: 529)
KKRELVAGISADGSTDYIRSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 530)
KKRELVAGISADGSVDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRYIYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSRVSFLSMAWYRQAPG (SEQ ID NO: 531)
KKRELVAGISADGSTLYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRRTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSLAWYRQAPGK (SEQ ID NO: 532)
KRELVAGISTDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 533)
KKRELVAGISGDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVQFLSMAWYRQAPG (SEQ ID NO: 534)
KKRELVAGISADGSTDYINSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)

EVQLVESGGGLVQPGGSLTLSCAASGSNVSFLSMAWYRQAPG (SEQ ID NO: 535) KKRELVAGISARGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYHWTTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSSVKFLSMAWYRQAPG (SEQ ID NO: 536) KKRELVAGISADGSTTYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTRRTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGKSVSFLSMAWYRQAPG (SEQ ID NO: 537) KKRELVAGISKDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSRVSFLSMAWYRQAPG (SEQ ID NO: 538) KKRELVAGISADGSTTYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSHVSFLSMAWYRQAPG (SEQ ID NO: 539) KKRELVAGISANGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTRYAYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 540) KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWVTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 541) KKRELVAGISADGSADYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWVTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 542) KKRELVAGISAHGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWSTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 543) KKRELVAGISADGSTIYIDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 544) KKRELVAGISRDGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTRGTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSHVSFLSMAWYRQAPG (SEQ ID NO: 545) KKRELVAGISADGPTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWDTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGTSVSFLSMAWYRQAPG (SEQ ID NO: 546) KKRELVAGISADGSTTYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTRRTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGTSVSFLSIAWYRQAPGK (SEQ ID NO: 547) KRELVAGISADGSTDYIASVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVKFLSMAWYRQAPG (SEQ ID NO: 548) KKRELVAGISLDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTGRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 549) KKRELVAGISADGSTIYIDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 550) KKRELVAGISAHGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 551) KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR

AEDTAVYYCYAYRWITRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 552)
KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWITRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVVFLSMAWYRQAPG (SEQ ID NO: 553)
KKRELVAGISADGSMDYIDSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 554)
KKRELVAGISADGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTRRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 555)
KKRELVAGISADGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYSWTTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 556)
KKRELVAGISANGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTNRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSRLSMAWYRQAPG (SEQ ID NO: 557)
KKRELVAGISANGSTTYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRYRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSKSFLSMAWYRQAPG (SEQ ID NO: 558)
KKRELVAGISADGSTSYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSRLSMAWYRQAPG (SEQ ID NO: 559)
KKRELVAGISADGSRDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRYKYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVKFLSMAWYRQAPG (SEQ ID NO: 560)
KKRELVAGISADGSTMYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWHTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSGVRFLSMAWYRQAPG (SEQ ID NO: 561)
KKRELVAGISPDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTRRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 562)
KKRELVAGISGDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWMTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVHFLSMAWYRQAPG (SEQ ID NO: 563)
KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRYRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 564)
KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWSTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSKVSFLSMAWYRQAPG (SEQ ID NO: 565)
KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRYTFWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 566)
KKRELVAGISADGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSKVSFLSMAWYRQAPG (SEQ ID NO: 567)
KKRELVAGISTDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRRTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSRVSFLSMAWYRQAPG (SEQ ID NO: 568)

KKRELVAGISADGSTSYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWATRYTYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCAASGSRVSFLSMAWYRQAPG (SEQ ID NO: 569) KKRELVAGISADGSTLYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWHTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 570) KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWGTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCAASYSSVSRLSMAWYRQAPG (SEQ ID NO: 571) KKRELVAGISADGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTRNTYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 572) KKRELVAGISTDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 573) KKRELVAGISADGSTLYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTRYAYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 574) KKRELVAGISADGRTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGTSVSFLSMAWYRQAPG (SEQ ID NO: 575) KKRELVAGISADGSTIYIDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYRWTTRRTYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCAASGSSVKFLSMAWYRQAPG (SEQ ID NO: 576) KKRELVAGISADGSTLYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTRRTYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 577) KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTSRYTYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCAASGSRVSFLSMAWYRQAPG (SEQ ID NO: 578) KKRELVAGISKDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTRVTYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCAASGSSVSVLSMAWYRQAPG (SEQ ID NO: 579) KKRELVAGISADGSTDYIGSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTRTTYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 580) KKRELVAGISVDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCAASGSRVSFLSMAWYRQAPG (SEQ ID NO: 581) KKRELVAGISADGSTGYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWATRYTYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCVASGSSVKFLSMAWYRQAPG (SEQ ID NO: 582) KKRELVAGISGDGSTTYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTRRTYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 583) KKRELVAGISTDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYALRWTTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCAASGSSVSQLSMAWYRQAPG (SEQ ID NO: 584) KKRELVAGISADGSTDYFDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTRGTYWGQGTLVTVSS Exemplary anti-target (DLL3)

EVQLVESGGGLVQPGGSLTLSCAASGSRVSFLSMAWYRQAPG (SEQ ID NO: 585)
KKRELVAGISADGSTSYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASKSSVSFLSMAWYRQAPGK (SEQ ID NO: 586)
KRELVAGISADGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYRWTTRATYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 587)
KKRELVAGISADGSTAYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRRTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVKFLSMAWYRQAPG (SEQ ID NO: 588)
KKRELVAGISADGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWPTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 589)
KKRELVAGISQDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 590)
KKRELVAGISNDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWKTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSRVSFLSMAWYRQAPG (SEQ ID NO: 591)
KKRELVAGISARGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWSTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSLAWYRQAPGK (SEQ ID NO: 592)
KRELVAGISADGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYRWKTRRTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 593)
KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRRTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSKVSFLSMAWYRQAPG (SEQ ID NO: 594)
KKRELVAGISADGSTLYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRYRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 595)
KKRELVAGISADGSTNYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 596)
KKRELVAGISADGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRYKYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSKVSFLSMAWYRQAPG (SEQ ID NO: 597)
KKRELVAGISADGSTTYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWKTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVKFLSMAWYRQAPG (SEQ ID NO: 598)
KKRELVAGISADGSTDYIGSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRVTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVKFLSMAWYRQAPG (SEQ ID NO: 599)
KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRFTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSRVSFLSMAWYRQAPG (SEQ ID NO: 600)
KKRELVAGISADGSTTYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRFTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVLFLSMAWYRQAPG (SEQ ID NO: 601)
KKRELVAGVSSDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR

AEDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSRVSFLSMAWYRQAPG (SEQ ID NO: 602)
KKRELVAGISADGHTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRYTHWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSRVSFLSMAWYRQAPG (SEQ ID NO: 603)
KKRELVAGISADGSTDYFDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTRRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVGFLSMAWYRQAPG (SEQ ID NO: 604)
KKRELVAGISADGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFMSMAWYRQAP (SEQ ID NO: 605)
GKKRELVAGISADGSTDYIASVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYAYRWTTRSTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 606)
KKRELVAGISADGSTDYISSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYSWTTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVTFLSMAWYRQAPG (SEQ ID NO: 607)
KKRELVAGISADGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRGTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 608)
KKRELVAGISADGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWKTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSKVSFLSMAWYRQAPG (SEQ ID NO: 609)
KKRELVAGISADGSTTYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRFTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSKVSFMSMAWYRQAP (SEQ ID NO: 610)
GKKRELVAGISVDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSNLSMAWYRQAPG (SEQ ID NO: 611)
KKRELVAGISADGSTAYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRRTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASNSSVSKLSMAWYRQAPG (SEQ ID NO: 612)
KKRELVAGISADGSTAYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWSTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSKVSFLSMAWYRQAPG (SEQ ID NO: 613)
KKRELVAGISADGSKDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRLTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSQVSFLSMAWYRQAPG (SEQ ID NO: 614)
KKRELVAGISADGSTDYFDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTRRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSKVSFMSMAWYRQAP (SEQ ID NO: 615)
GKKRELVAGISADGSTDYIDSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYAYRWTTRLTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSRVSFLSMAWYRQAPG (SEQ ID NO: 616)
KKRELVAGISADGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTRRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 617)
KKRELVAGISADGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRRTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSKVSFLSMAWYRQAPG (SEQ ID NO: 618)

KKRELVAGISARGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYQWTTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 619)
KKRELVAGISATGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTRRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSIAWYRQAPGK (SEQ ID NO: 620)
KRELVAGISKDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYRWTTRMTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSSSFLSMAWYRQAPGK (SEQ ID NO: 621)
KRELVAGISADGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYRWTTRRTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVKFLSMAWYRQAPG (SEQ ID NO: 622)
KKRELVAGISPDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRYRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVNFLSMAWYRQAPG (SEQ ID NO: 623)
KKRELVAGISADGSTHYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWLTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVKFLSMAWYRQAPG (SEQ ID NO: 624)
KKRELVAGISADGSTDYILSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYEWTTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 625)
KKRELVAGISADGSTDYIHSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 626)
KKRELVAGISVDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSVAWYRQAPGK (SEQ ID NO: 627)
KRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSQVSFLSMAWYRQAPG (SEQ ID NO: 628)
KKRELVAGISADGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWSTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGTSVSFLSMAWYRQAPG (SEQ ID NO: 629)
KKRELVAGISADGSTDYIRSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRLTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSRVSFLSMAWYRQAPG (SEQ ID NO: 630)
KKRELVAGISADGSTMYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRLTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVKFLSMAWYRQAPG (SEQ ID NO: 631)
KKRELVAGISTDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYKWTTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSSAWYRQAPGK (SEQ ID NO: 632)
KRELVAGISADGSTLYIDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYRWTTRSTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVKFLSMAWYRQAPG (SEQ ID NO: 633)
KKRELVAGISADGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 634)
KKRELVAGISATGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWSTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)

EVQLVESGGGLVQPGGSLTLSCAASGSTVSFLSMAWYRQAPG (SEQ ID NO: 635)
KKRELVAGISHDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVQFLSMAWYRQAPG (SEQ ID NO: 636)
KKRELVAGISYDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASRSSVSFLSMAWYRQAPG (SEQ ID NO: 637)
KKRELVAGISTDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWLTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSKVSFLSMAWYRQAPG (SEQ ID NO: 638)
KKRELVAGISADGSTAYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSRVSFLSMAWYRQAPG (SEQ ID NO: 639)
KKRELVAGISADGSTDYIESVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 640)
KKRELVAGISIDGSTDYIKSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYRWTTRYRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSKVSFLSMAWYRQAPG (SEQ ID NO: 641)
KKRELVAGISADGSKDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 642)
KKRELVAGISADGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWPTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVKFLSMAWYRQAPG (SEQ ID NO: 643)
KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRHTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVKFLSMAWYRQAPG (SEQ ID NO: 644)
KKRELVAGISADGSTDYIHSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTRRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSILSMAWYRQAPGK (SEQ ID NO: 645)
KRELVAGISADGSTIYIDSVKGRFTISRDNAKNSVYLQMNSLRAE
DTAVYYCYAYRWHTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSVAWYRQAPGK (SEQ ID NO: 646)
KRELVAGISANGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYRWTNRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 647)
KKRELVAGISTDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRYRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSRVSFLSMAWYRQAPG (SEQ ID NO: 648)
KKRELVAGISYDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRRTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGHSVSFLSMAWYRQAPG (SEQ ID NO: 649)
KKRELVAGISADGSTDYIASVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWSTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVRFLSMAWYRQAPG (SEQ ID NO: 650)
KKRELVAGISADGSTDYIGSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWSTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG (SEQ ID NO: 651)
KKRELVAGISANGSTDYYDSVKGRFTISRDNAKNSVYLQMNSLR

AEDTAVYYCYAYRWRTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSRVSFLSMAWYRQAPG (SEQ ID NO: 652)
KKRELVAGISADGSTSYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVKFLSMAWYRQAPG (SEQ ID NO: 653)
KKRELVAGVSADGSTDYIDSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYAYEWTTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSRLSMAWYRQAPG (SEQ ID NO: 654)
KKRELVAGISARGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRSTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGRSVSFLSMAWYRQAPG (SEQ ID NO: 655)
KKRELVAGISADGSTIYIDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYRWTTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGRSVSFLSMAWYRQAPG (SEQ ID NO: 656)
KKRELVAGISANGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWSTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVKFLSMAWYRQAPG (SEQ ID NO: 657)
KKRELVAGISADGSTDYVDSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYAYRWSTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSKLSMAWYRQAPG (SEQ ID NO: 658)
KKRELVAGISADGSTDYRDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTYRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSRLSMAWYRQAPG (SEQ ID NO: 659)
KKRELVAGISVDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTTRLTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVKFLSLAWYRQAPGK (SEQ ID NO: 660)
KRELVAGISADGSTDYILSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYEWTTRYTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSRLSLAWYRQAPGK (SEQ ID NO: 661)
KRELVAGISVDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYRWTTRLTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGTSSSINAMGWYRRAPG (SEQ ID NO: 662)
KQRELVAGISSDGSKVFNESVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRPAAGSPMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGTTSSINAIGWYRRAPGK (SEQ ID NO: 663)
QRELVAGISSDGSEVYTDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYYFRTVDGSPLRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 664)
KQRELVAGISSDDSNVYYESVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVSGSSKRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGQTYRVNAFGWYRRAPG (SEQ ID NO: 665)
KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFSAGSGTEMSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG (SEQ ID NO: 666)
KQRELVAGISSDESTLYVDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFGSLSGSSTTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSASLTNATGWYRRAPGK (SEQ ID NO: 667)
QRELVAGISSDDSKVYSDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYYFGSVSGSWTRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGYPSLNNAMGWYRRAPG (SEQ ID NO: 668)

KQRELVAGISSDGSQVYGASVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRLVSGSSMSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSSSTINAIGWYRRAPGK (SEQ ID NO: 669)
QRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTGSGTSKSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSYINAMGWYRRAPG (SEQ ID NO: 670)
KQRELVAGISSDGSNMYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFSNMSGTTRRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSVNALGWYRRAPGK (SEQ ID NO: 671)
QRELVAGISSDGSKVYTDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVPGSAMGYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSLSNAVGWYRRAPGK (SEQ ID NO: 672)
QRELVAGISSDGSKVSAESVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYYFRAESGSSMGYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSTNAIGWYRRAPGK (SEQ ID NO: 673)
QRELVAGISSDGSKVYDDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTLYGSSRSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGLTSTINAMGWYRRAPG (SEQ ID NO: 674)
KQRELVAGISSDGSKVYDDSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFSPFSGSDTGYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGVSPSKNAIGWYRRAPGK (SEQ ID NO: 675)
QRELVAGISSDGSAVYVGSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFSTFSGSSISYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAVGWYRRAPGK (SEQ ID NO: 676)
QRELVAGISSDGSYVVYSESVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYYFRTLAGSEMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTTMNNAMAWYRRAP (SEQ ID NO: 677)
GKQRELVAGISSDSSHVYADSVKGRFTISRDNAKNSVYLQMNS
LRAEDTAVYYCYYFRTVSGSGVRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSKINAIGWYRRAPGK (SEQ ID NO: 678)
QRELVAGISSDSSIVYTDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYYFRPGAGHSNSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGQTTALNAMGWYRRAP (SEQ ID NO: 679)
GKQRELVAGISSDGSEVNTDSVKGRFTISRDNAKNSVYLQMNS
LRAEDTAVYYCYYFRRASGTAMSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGATSSINAIGWYRRAPGK (SEQ ID NO: 680)
QRELVAGISSDGSKLSSDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYYFTSASGTDLSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSTINAMGWYRRAPG (SEQ ID NO: 681)
KQRELVAGISSDNSKVYADSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYYFRSANGSSKRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG (SEQ ID NO: 682)
KQRELVAGISSDGSRVYFDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFKTIAGAGMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSLVNAMGWYRRAPG (SEQ ID NO: 683)
KQRELVAGISSDGSLVYAESVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRYGSGSSLSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSLNNAIGWYRRAPGK (SEQ ID NO: 684)
QRELVAGISSDGSVVYVDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYYFRTVPGASMKYWGQGTLVTVSS Exemplary anti-target (DLL3)

EVQLVESGGGLVQPGGSLTLSCVASGSTSPVNAMAWYRRAPG (SEQ ID NO: 685) KQRELVAGISSDGSKVYVDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYYFRTVDGSAISYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGTTSSMNAIGWYRRAPG (SEQ ID NO: 686) KQRELVAGISSDGSKLYDESVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTVKGSGGSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGETSSINAMAWYRRAPG (SEQ ID NO: 687) KQRELVAGISSDYSKLYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTVSGSSRGYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSTINAIGWYRRAPGK (SEQ ID NO: 688) QRELVAGISSDSSKVYTESVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYYFRPGPGSQMAYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTYSMNAMGWYRRAP (SEQ ID NO: 689) GKQRELVAGISSDGSQVYVDSVKGRFTISRDNAKNSVYLQMNS LRAEDTAVYYCYYFRTVAGSASGYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSPSSINAYGWYRRAPGK (SEQ ID NO: 690) QRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYYFRTVSGSSYSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSTINAIGWYRRAPGK (SEQ ID NO: 691) QRELVAGISSDGSKVYVDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFINLKGSSMAYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAIGWYRRAPGK (SEQ ID NO: 692) QRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRMVTGSYGGYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSISSINAMGWYRRAPGK (SEQ ID NO: 693) QRELVAGISSDGSSVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFKSSYGLPMRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTQVNNAMAWYRRAP (SEQ ID NO: 694) GKQRELVAGISSDGSQVYYGSVKGRFTISRDNAKNSVYLQMNS LRAEDTAVYYCYYFKTVSGQSLRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTASFNAMAWYRRAPG (SEQ ID NO: 695) KQRELVAGISSDGSKVYTDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTVTGRAARYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSPLSINAIGWYRRAPGK (SEQ ID NO: 696) QRELVAGISSDGSKVSADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFGPAIGASRTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTTFINAIGWYRRAPGK (SEQ ID NO: 697) QRELVAGISSDGSKVYEDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFRTVSGAPKSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAIGWYRRAPGK (SEQ ID NO: 698) QRELVAGISSDRSKVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYYFHTVSGSSMSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGETDTINAVGWYRRAPGK (SEQ ID NO: 699) QRELVAGISSDGSKVYAESVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYYFRRLEGYSNRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSTSPINAIGWYRRAPGK (SEQ ID NO: 700) QRELVAGISSDGSVVTTESVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYYFRTGSGSSMGYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCVASGSITSSNAMGWYRRAPG (SEQ ID NO: 701) KQRELVAGISSDGSHVHQESVKGRFTISRDNAKNSVYLQMNSL

RAEDTAVYYCYYFTTVTGSSMSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASRYSVSNLSMAWYRQAPG (SEQ ID NO: 702)
KKRELVAGISADGSTVYYVESVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYYWTERRPYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGDSLTLSCAASMSTVSVLSMAWYRQAPG (SEQ ID NO: 703)
KKRELVAGISSDGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAIYYCYAYSWDDAHPYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASDSYVSLLSMAWYRQAPG (SEQ ID NO: 704)
KKRELVAGISVDGSTHYVASVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWMTRLTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASDSAVSVLSIAWYRQAPGK (SEQ ID NO: 705)
KRELVAGISTDGSKHYIDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYDWADAQPYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASHSSVTSLSLAWYRQAPGK (SEQ ID NO: 706)
KRELVAGISYDGSKYYAESVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYSWTDRLPYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASDSVVKFLSMAWYRQAPG (SEQ ID NO: 707)
KKRELVAGISANGSRTYMESVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYAYRWATRLPYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASDPSVWNLSMAWYRQAP (SEQ ID NO: 708)
GKKRELVAGISPDGSTDYVDSVKGRFTISRDNAKNSVYLQMNS
LRAEDTAVYYCYAYKWSNRLPYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGTSVMLLSLAWYRQAPG (SEQ ID NO: 709)
KKRELVAGISPNGSAVYTESVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYGWKTRQPYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASSSPVSNLSLAWYRQAPGK (SEQ ID NO: 710)
KRELVAGISPDGSTAYMESVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWPNRRGYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASWRSVLLLSVAWYRQAPG (SEQ ID NO: 711)
KKRELVAGISNDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYDWTTRQRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASSSSVQYLSMAWYRQAPG (SEQ ID NO: 712)
KKRELVAGISTDGSAVYFDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYNWSYAQPYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGTSVSLLSLAWYRQAPGK (SEQ ID NO: 713)
KRELVAGISTGGSTHYIESVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYNWTDSLQYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASLSSVSNLSIAWYRQAPGKK (SEQ ID NO: 714)
RELVAGISTDGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLRAE
DTAVYYCYAYSWTTSLPYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASMYSVSFLSMAWYRQAPG (SEQ ID NO: 715)
KKRELVAGISNEGSTYYMDSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYAYKWRSRSTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASKSSVSHLSLAWYRQAPGK (SEQ ID NO: 716)
KRELVAGISADGSHVYTNSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYSQTTRDPYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASYTSVLDLSIAWYRQAPGK (SEQ ID NO: 717)
KRELVAGISDDGSRYYTDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYRWTARDTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASMSDVSFLSMAWYRQAPG (SEQ ID NO: 718)

KKRELVAGISAEGSTLYMESVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWTSRLSYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASESSVSFLSSAWYRQAPGK (SEQ ID NO: 719)
KRELVAGISTDGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYSWTTRSRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGDSVSLLSMAWYRQAPG (SEQ ID NO: 720)
KKRELVAGISANGSTSYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYNWTSRYRYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSDVWYLSLAWYRQAPG (SEQ ID NO: 721)
KKRELVAGISDDGSRHYIESVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYSWKTRFPYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASKSAVAFLSIAWYRQAPGK (SEQ ID NO: 722)
KRELVAGISPDGSTVYIESVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYSWTTRYPYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASFSAVAYLSMAWYRQAPG (SEQ ID NO: 723)
KKRELVAGISDDGSTVYVDSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYAYEWTNALPYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASVYSVYDLSTAWYRQAPGK (SEQ ID NO: 724)
KRELVAGISDDGSTVYFDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYSWITRSPYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGDSVSFLSMAWYRQAPG (SEQ ID NO: 725)
KKRELVAGISDEGSTVYIGSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYSWTTRRQYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASSSSVSLLSLAWYRQAPGKK (SEQ ID NO: 726)
RELVAGISDDGSIVYMDSVKGRFTISRDNAKNSVYLQMNSLRA
EDTAVYYCYAYSWITRSPYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASADSVSFLSIAWYRQAPGK (SEQ ID NO: 727)
KRELVAGISDDGSKHYFDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWEESRQYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASASSVTLLSIAWYRQAPGKK (SEQ ID NO: 728)
RELVAGISTDGSTDYLHSVKGRFTISRDNAKNSVYLQMNSLRAE
DTAVYYCYAYTWTTRLPYWGQGTLVTVTS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASADSVSFLSIAWYRQAPGK (SEQ ID NO: 729)
KRELVAGISDDGSKHYFDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWEESRQYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGTSVWLLSMAWYRQAP (SEQ ID NO: 730)
GKKRELVAGISYDGSTVYVESVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYAYSWTTRQPYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSSVSILSIAWYRQAPGKK (SEQ ID NO: 731)
RELVAGISDDGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLRAE
DTAVYYCYAYVWGTRLPYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGTAVSNLSIAWYRQAPGK (SEQ ID NO: 732)
KRELVAGISDDGSTVYVDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYEWTNALPYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGSAVSMLSLAWYRQAPG (SEQ ID NO: 733)
KKRELVAGISDDGSQVYIDSVKGRFTISRDNAKNSVYLQMNSLR
AEDTAVYYCYAYRWEDALTYWGQGTLVTVSS Exemplary anti-target (DLL3)
EVQLVESGGGLVQPGGSLTLSCAASGMTVFFLSMAWYRQAP (SEQ ID NO: 734)
GKKRELVAGISVDGSTVYSDSVKGRFTISRDNAKNSVYLQMNSL
RAEDTAVYYCYAYSWTTRYPYWGQGTLVTVSS Exemplary anti-target (DLL3)

EVQLVESGGGLVQPGGSLTLSCAASQYSVTFLSVAWYRQAPGK (SEQ ID NO: 735) KRELVAGISDDGSNVYIDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYSWIDSLRYWGQGTLVTVSS Exemplary anti-target (DLL3) EVQLVESGGGLVQPGGSLTLSCAASGETVSFLSLAWYRQAPGK (SEQ ID NO: 736) KRELVAGISTDGSTVYFVSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYSWTTPRAYWGQGTLVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQAGGSLRLSCAASGSIAYIYTMDWYRQAPG (SEQ ID NO: 737) KQRELVATSTRDGNVDYAESVKGRFTISRDNAKNTVYLQMNSL KPEDTAVYYCNADLRTAVDLIRANYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQAGDSLRLSCVVSGRTDSWYVMGWFRQA (SEQ ID NO: 738) PGKDREFVAGVSWSYGNTYYADSVKGRFTASRDNAKNTAYLQ MNSLNAEDTAVYYCAARVSREVIPTRWDLYNYWGQGTQVTV SS Exemplary anti-target (EGFR) QVQLQESGGGSVQPGGSLRVSCVVSRTIISINAMTWYHQAPG (SEQ ID NO: 739) KRRELVAIITSGGETNYADSVKGRFTISRDNAKNTAYLQMNNLK PEDTGVYYCNVVPPLGSWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGRVQAGGSLRLSCSASARTLRLYAVGWFRQAPG (SEQ ID NO: 740) KEREFVAGIGRSERTYYTDSVKGRFTLSRDNAKNTVFLEMNDLE PEDTAVYFCALTFQTTDMVDVPTTQHEYDYWGRGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQAGGSLRLSCAASGSIAYIYTMDWYRQAPG (SEQ ID NO: 741) KQRELVATSTHDGNTDYADSVKGRFTISRDNVKNTVYLQMNSL KPEDTAVYYCNADLRTAVDLIRANYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQPGGSLRLSCAASGSIAYIYTMDWYRQAPG (SEQ ID NO: 742) KQRELVATSTRDGNTDYADSVKGRFTISRDNAKDTVYLQMNSL KPEDTAVYYCNADLRTAVDLIRANYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQAGGSLTLSCAASGRYQMAWFRQAPEKER (SEQ ID NO: 743) EFVGTISSGDSTWYTNSVKGRFAISRDSARNTVYLQMNDLKPE DTAIYYCAAALYYRDSRRAADYPYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGRVQAGESLRLSCSTSTRTLKLYAVGWFRQAPG (SEQ ID NO: 744) KERDFVAGIGRSERIYYIDSVKGRFTLSRDNAKNTVFLEMNDLEP EDTAVYFCAATFQTSDNVGVPTVQHEYDYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQAGGSLRLSCAASGSIAYIYTMDWYRQAPG (SEQ ID NO: 745) KQRELVATSTHDGNTDYADSVKGRFTISRDNAKNTVTLQMNSL KPEDTAVYYCNADLRTAVDLIRANYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQAGGSLRLSCAASGSIAYIYTMDWYRQAPG (SEQ ID NO: 746) KQRELVATFTRDGNTDYADSVKGRFTISRDNAKNTVYLQMNSL KPEDTAVYYCNTDLRTAVDLIRANYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQAGGSLRLSCAASGSIAYIYTMDWYRQAPG (SEQ ID NO: 747) KQRELVATSTHDGNTDYADSLKGRFTISRDNAKNTVYLQMNSL KPEDTAVYYCNADLRTAVDLIRANYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGSVQAGGSLRLSCAASGRYQMAWFRQAPEKER (SEQ ID NO: 748) EFVGTISSGDSTWYTNSLKGRFAISRDSARDTVYLQMNDLKPE DTAVYYCAAALYYRDSRRAADYPYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQTGGSLRLSCAVSGSIVTINAMTWYRQAPG (SEQ ID NO: 749) KRRELVAIITSGGETNYADSVKGRFTISRDNAKNTAHLQMNSLN PEDTGVYYCNVVPPLGSWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQTGGSLRLSCAVSRSIVSIKSMTWYRQAPGK (SEQ ID NO: 750) RRELVALITSGGETNYSDSVKGRFTISRDNAKNTVYLQMNSLKP EDTGVYYCNVVPPLGSWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVETGGSLRLSCAGSGSTFRHHAMAWFRQTP (SEQ ID NO: 751) GKEREFVSAINDHGDRTKYLDSVRGRFTISRDNTDNMVYLQM

TDLRPEDTANYSCAAGPLVDYLETTPLVYTYWGHGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQAGGSLRLSCAASGSIAYIYTMDWYRQAPG (SEQ ID NO: 752) KQRELVATSTHDGNTDYADSVKGRFTISRDNAKNTVYLQMNSL KPEDTAVYYCNADLRTAVDLIRANYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGSVQAGGSLTLSCAASGSIAYIYTMDWYRQAPG (SEQ ID NO: 753) KQRELVATSTHDGNTDYADSVKGRFTISRDNAKNTVYLQMNSL KPEDTAVYYCNADLRTAVDLIRANYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQPGGSLRLSCTASVSIFSVNAVDWYRQSPG (SEQ ID NO: 754) KERELVAIMTSDGSTNYGDSVKGRFTISRDNAKNTVYLQMNNL KPEDTAVYYCNTVPPRYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQPGGSLRLSCAASGSIAYIYTMDWYRQAPG (SEQ ID NO: 755) KQRELVATSTRDGNIDYADSVKGRFTISRDSAKNTVYLQMSSLK PEDTAVYYCNADLRTAVDLIRANYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQAGGSLTLSCAASGRYQLAWFRQAPEKVRE (SEQ ID NO: 756) FVGTISSGDSTWYTNSVKGRFAISRDSARNTVYLQMNDLKPED TAVYYCAAALYYRDSRRAADYPYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGSVQAGGSLRLSCAASGRYHMAWFRQAPEKER (SEQ ID NO: 757) EFVGTISSGDSTWYTNSVKGRFAISRDSARNTAYLQMNDLKPE DTAVYYCAAALYYGDSRRAADYPYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGSVQAGGSLRLSCAASGSIAYIYTMDWYRQAPG (SEQ ID NO: 758) KQRELVATSTHDGNTDYTDSVKGRFTISRDNAKNTVYLQMNSL KPEDTAVYYCNADLRTAVDLIRANYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQAGGSLRLSCAASGSIAYIYTMDWYRQTPG (SEQ ID NO: 759) KQRELVATSTRDANTDYAGSVKGRFTISRDNAKDTVYLQMNSL KPEDTAVYYCHADLRTAVDLIRANYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQAGGSLRLSCAASGRYNMAWFRQAPEKER (SEQ ID NO: 760) EFVGTITSADSTWYTNSVKGRFAITQDSARNTVYLQMNDLKPE DTAVYYCAAALYYGDSRRAADYPYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQPGGSLRLSCAASGRYQMAWFRQAPEKER (SEQ ID NO: 761) EFVGTISSGDSTWYTNSVKGRFAISRDSARTTVYLQMNDLKPE DTAVYYCAAALYYRDSWRAADYPYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQPGESLRLSCAATGRYHLAWFRQAPEKERE (SEQ ID NO: 762) FVGTITSADSTWYTNSVKGRFAITRDSARNTVYLQMNDLKPED TAVYYCAAALYYGDSRRAADYPYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQAGGSLKLSCADSGRSFSNYIMGWFRQAP (SEQ ID NO: 763) GKEREFVAGLGWSPGNTYYADSVKGRFTISRDNAKNMVYLQ MNSLNPEDTAVYYCAARRGDVIYTTPWNYVYWGQGTQVTVS S Exemplary anti-target (EGFR) QVQLQESGGGLVQAGGSLRLSCAASGSIAYIYTMDWYRQAPG (SEQ ID NO: 764) KQRELVATSTHDGNTDYADSVKGRFTISRDNAKNTVYLQMNSL KPEDTAVYYCNADLRTPVDLIRANYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGSVQAGGSLRLSCAAPGRYQMAWFRQAPEKER (SEQ ID NO: 765) EFVGTISSGDSTWYTNSVKGRFAISRDSARNTVYLQMNDLKSE DTAVYYCAAALYYRDSRRAIDYPYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGSVQAGGSLRLSCAASGLTFSSYAMAWFRQAP (SEQ ID NO: 766) GKQRELVARITSGGTTDYADSVKGRFTISRDNAKNTMYLQMN SLKPEDTAVYYCAADLTYRNLLLKLPHYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQAGGSLRLSCAASGNIAYIYTMDWYRQAPG (SEQ ID NO: 767) KQRELVATSTHDGSTDYADSVKGRFTISRDNAKNTVYLQMNSL KPEDTAVYYCNADLRTPVDLIRANYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQPGGSLRLSCAASGSIAYIYTMDWYRQAPG (SEQ ID NO: 768)

KQRELVATSTWDGNTDYADSVKGRFTISRDNAKNTVYLQMNS LKPEDTAVYYCNADLRTAVDLIRANYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQAGGSLRLSCAASGSIAYIYTMDWYRQAPG (SEQ ID NO: 769) KQRELVATSTHDGNTDYADSVKGRFTISRDNAKNTVYLQMSSL KPDDTAVYYCNADLRTAVDLIRANYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQAGESLSLSCAASGNDFVITDMHWYRQAP (SEQ ID NO: 770) GKQREWVATITRFATTNYADSVKGRFTISRDNAKNTWYLQMN SLKPDDTAVYYCKAIGLRGVPDVNRQFEVWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQAGGSLRLSCAASGAIAYIYGMGWYRQAP (SEQ ID NO: 771) GNQRELVAAISSGGSTDYADSVKGRFTISRDNAKNTVYLQMSS LKPEDTAVYYCNADVRTSRNLVRSDYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQAGGSLRLSCAASGRYHTAWFRQAPEKERE (SEQ ID NO: 772) FVGTISSGDSTWYTNSVKGRFAISRDSARNTVYLQMNDLKPED TAVYYCAAALYYGDSRRAGDYPYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQAGGSLRLSCAASGNIAYIYTMNWYRQAPG (SEQ ID NO: 773) KQRELVATSTHAGNTDYADSVKGRFTISRDNAKNTVYLQMNSL KPEDTAVYYCNVDLRTAVDLIRANYWGQGTQVTVSS Exemplary anti-target (EGFR) QVQLQESGGGLVQPGGSLRLSCAASGNIAYIYTMGWYRQAPG (SEQ ID NO: 774) KQRELVATSTHDGNSDYADSVKGRFTISRDNAKNTVYLQMNTL KPDDTAVYYCNADLRTPVDRIRGNFWGQGTQVTVSS Exemplary anti-target (EGFR) EVQLLESGGGLVQPGGSLTLSCAASGSIAYIYTMDWYRQAPGK (SEQ ID NO: 775) QRELVATSTRDGNVDYADSVKGRFTISRDNSKNTLYLQMNSLR AEDTAVYYCNADLRTAVDLIRANYWGLGTQVTVSS Exemplary anti-target (EGFR) EVQLLESGGGLVQPGGSLTLSCAASGRTDSWYVMGWFRQAP (SEQ ID NO: 776) GKDREFVAGVSWSYGNTYYADSVKGRFTISRDNSKNTLYLQM NSLRAEDTAVYYCAARVSREVIPTRWDLYNYWGLGTQVTVSS Exemplary anti-target (EGFR) EVQLLESGGGLVQPGGSLTLSCAASARTLRLYAVGWFRQAPGK (SEQ ID NO: 777) EREFVAGIGRSERTYYTDSVKGRFTISRDNSKNTLYLQMNSLRA EDTAVYYCALTFQTTDMVDVPTTQHEYDYWGLGTQVTVSS Exemplary anti-target (EGFR) EVQLLESGGGLVQPGGSLTLSCAASGSIVTINAMTWYRQAPGK (SEQ ID NO: 778) RRELVAIITSGGETNYADSVKGRFTISRDNSKNTLYLQMNSLRAE DTAVYYCNVVPPLGSWGLGTQVTVSS Exemplary anti-target (EGFR) EVQLLESGGGLVQPGGSLTLSCAASGRYHMAWFRQAPGKERE (SEQ ID NO: 779) FVGTISSGDSTWYTNSVKGRFTISRDNSKNTLYLQMNSLRAEDT AVYYCAAALYYGDSRRAADYPYWGLGTQVTVSS Exemplary anti-target (EGFR) EVQLLESGGGLVQPGGSLTLSCAASGSTFRHHAMAWFRQTPG (SEQ ID NO: 780) KEREFVSAINDHGDRTKYLDSVKGRFTISRDNSKNTLYLQMNSL RAEDTAVYYCAAGPLVDYLETTPLVYTYWGLGTQVTVSS Exemplary anti-target (EGFR) EVQLLESGGGLVQPGGSLTLSCAASGRSFSNYIMGWFRQAPG (SEQ ID NO: 781) KEREFVAGLGWSPGNTYYADSVKGRFTISRDNSKNTLYLQMNS LRAEDTAVYYCAARRGDVIYTTPWNYVYWGLGTQVTVSS Exemplary anti-target (EGFR) EVQLLESGGGLVQPGGSLTLSCAASGLTFSSYAMAWFRQAPGK (SEQ ID NO: 782) QRELVARITSGGTTDYADSVKGRFTISRDNSKNTLYLQMNSLRA EDTAVYYCAADLTYRNLLLKLPHYWGLGTQVTVSS Exemplary anti-target (EGFR) EVQLLESGGGLVQPGGSLTLSCAASVSIFSVNAVDWYRQSPGK (SEQ ID NO: 783) ERELVAIMTSDGSTNYDDSVKGRFTISRDNSKNTLYLQMNSLR AEDTAVYYCNTVPPRYWGLGTQVTVSS Exemplary anti-target (EGFR) EVQLLESGGGLVQPGGSLTLSCAASGNDFVITDMHWYRQAPG (SEQ ID NO: 784) KQREWVATITRFATTNYADSVKGRFTISRDNSKNTLYLQMNSL RAEDTAVYYCKAIGLRGVPDVNRQFEVWGLGTQVTVSS Exemplary anti-target (EGFR)

EVQLLESGGGLVQPGGSLTLSCAASGAIAYIYGMGWYRQAPGK (SEQ ID NO: 785)
QRELVAAISSGGSTDYADSVKGRFTISRDNSKNTLYLQMNSLRA
EDTAVYYCNADVRTSRNLVRSDYWGLGTQVTVSS Exemplary EGFR ProTriTAC
EVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQ***GGG*** containing linker
sequence ***GGLDGNEEPGG***LEWVSSISGSGRDTLYADSVKGRFTISRDNAK L001
(identified by residues in TTLYLQMNSLRPEDTAVYYCTIGGSLSVSSQGTLVTVSS
bold and italics) and masking GGGG**KPLGLQARVV**GGGGT sequence
M027(identified by QTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPNWVQQKP
residues that are bold and
GQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSGVQPEDEA underlined)
EYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSGGGGSEVQ (SEQ ID No. 786)
LVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLE
WVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMNNL
KTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVTVSSGGGGS
GGGSEVQLLESGGGLVQPGGSLTLSCAASGRTDSWYVMGWF
RQAPGKDREFVAGVSWSYGNTYYADSVKGRFTISRDNSKNTLY
LQMNSLRAEDTAVYYCAARVSREVIPTRWDLYNYWGLGTQVT VSSHHHHHH
Exemplary EGFR ProTriTAC
EVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQG***GG*** containing linker
sequence ***GGLDGNEEPGG***LEWVSSISGSGRDTLYADSVKGRFTISRDNAK L040
(identified by residues in
TTLYLQMNSLRPEDTAVYYCTIGGSLSVSSQGTLVTVSSGGGG**P** bold and italics) and
masking **QASTGRSGG**GGGGT sequence M027(identified by
QTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPNWVQQKP residues that are bold
and GQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSGVQPEDEA underlined)
EYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSGGGGSEVQ (SEQ ID No. 787)
LVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLE
WVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMNNL
KTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVTVSSGGGGS
GGGSEVQLLESGGGLVQPGGSLTLSCAASGRTDSWYVMGWF
RQAPGKDREFVAGVSWSYGNTYYADSVKGRFTISRDNSKNTLY
LQMNSLRAEDTAVYYCAARVSREVIPTRWDLYNYWGLGTQVT VSSHHHHHH
Exemplary EGFR ProTriTAC
EVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQ***GGG*** containing linker
sequence ***GGLDGNEEPGG***LEWVSSISGSGRDTLYADSVKGRFTISRDNAK L041
(identified by residues in TTLYLQMNSLRPEDTAVYYCTIGGSLSVSSQGTLVTVSS
bold and italics) and masking GGGG**PQGSTGRAAG**GGGGT sequence
M027(identified by QTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPNWVQQKP
residues that are bold and
GQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSGVQPEDEA underlined)
EYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSGGGGSEVQ (SEQ ID No. 788)
LVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLE
WVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMNNL
KTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVTVSS
GGGGSGGGSEVQLLESGGGLVQPGGSLTLSCAASGRTDSWYV
MGWFRQAPGKDREFVAGVSWSYGNTYYADSVKGRFTISRDN
SKNTLYLQMNSLRAEDTAVYYCAARVSREVIPTRWDLYNYWGL GTQVTVSSHHHHHH
Exemplary EGFR ProTriTAC
EVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQ***GGG*** containing linker
sequence ***GGLDGNEEPGG***LEWVSSISGSGRDTLYADSVKGRFTISRDNAK L042

(identified by residues in TTLYLQMNSLRPEDTAVYYCTIGGSLSVSSQGTLVTVSSGGGG**P** bold and italics) and masking **PASSGRAGG**GGGT sequence M027(identified by QTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPNWVQQKP residues that are bold and GQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSGVQPEDEA underlined) EYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSGGGGSEVQ (SEQ ID No. 789) LVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLE WVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMNNL KTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVTVSS GGGGSGGGSEVQLLESGGGLVQPGGSLTLSCAASGRTDSWYV MGWFRQAPGKDREFVAGVSWSYGNTYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCAARVSREVIPTRWDLYNYWGL GTQVTVSSHHHHHH Exemplary EGFR ProTriTAC EVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQ***GGG*** containing linker sequence ***GGLDGNEEPGG***LEWVSSISGSGRDTLYADSVKGRFTISRDNAK L045 (identified by residues in TTLYLQMNSLRPEDTAVYYCTIGGSLSVSSQGTLVTVSSGGGG**P** bold and italics) and masking **IPVQGRAH**GGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVT sequence M027(identified by SGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGK residues that are bold and AALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGS underlined) GGGGSGGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKY (SEQ ID No. 790) AINWVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISR DDSKNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYW GQGTLVTVSSGGGGSGGGS EVQLLESGGGLVQPGGSLTLSCAASGRTDSWYVMGWFRQAP GKDREFVAGVSWSYGNTYYADSVKGRFTISRDNSKNTLYLQM NSLRAEDTAVYYCAARVSREVIPTRWDLYNYWGLGTQVTVSSH HHHHH Exemplary EGFR ProTriTAC EVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQGGG containing a non cleavable GGLDGNEEPGGLEWVSSISGSGRDTLYADSVKGRFTISRDNAK linker sequence(identified by TTLYLQMNSLRPEDTAVYYCTIGGSLSVSSQGTLVTVSSGGGG**S** residues in bold and italics) **GGGGSGGVV**GGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGA and masking sequence VTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLG M027(identified by residues GKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGG that are bold and underlined) GSGGGGSGGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFN (SEQ ID No. 791) KYAINWVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTIS RDDSKNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAY WGQGTLVTVSS GGGGSGGGSEVQLLESGGGLVQPGGSLTLSCAASGRTDSWYV MGWFRQAPGKDREFVAGVSWSYGNTYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCAARVSREVIPTRWDLYNYWGL GTQVTVSSHHHHHH Exemplary GFP TriTAC QVQLVESGGALVQPGGSLRLSCAASGFPVNRYSMRWYRQAP Sequence GKEREWVAGMSSAGDRSSYEDSVKGRFTISRDDARNTVYLQM (SEQ ID No. 792) NSLKPEDTAVYYCNVNVGFEYWGQGTQVTVSSGGGGSGGGS EVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQAPG KGLEWVSSISGSGRDTLYADSVKGRFTISRDNAKTTLYLQMNSL RPEDTAVYYCTIGGSLSVSSQGTLVTVSSGGGGSGGGSEVQLVE SGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLEWV ARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMNNLKTE

DTAVYYCVRHANFGNSYISYWAYWGQGTLVTVSSGGGGSGG GGSGGGGSQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYP NWVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLS GVQPEDEAEYYCTLWYSNRWVFGGGTKLTVLHHHHHH Exemplary EGFR TriTAC EVQLLESGGGLVQPGGSLTLSCAASGRTDSWYVMGWFRQAP Sequence GKDREFVAGVSWSYGNTYYADSVKGRFTISRDNSKNTLYLQM (SEQ ID No. 793) NSLRAEDTAVYYCAARVSREVIPTRWDLYNYWGLGTQVTVSSG GGGSGGGSEVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGM SWVRQAPGKGLEWVSSISGSGRDTLYADSVKGRFTISRDNAKT TLYLQMNSLRPEDTAVYYCTIGGSLSVSSQGTLVTVSSGGGGSG GGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQ APGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAY LQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVTV SSGGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCASST GAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPARFSGSL LGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVLH HHHHH Exemplary anti-CD3 scFv QTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPNWVQQKP (SEQ ID NO: 794) GQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSGVQPEDEA EYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSGGGGSEVQ LVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLE WVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMNNL KTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQAGGSLRLSCAASGFIFRAASMAWYRQSPG (SEQ ID NO: 804) EPL90 NERELVASISSGAFTNYADSVKARFTISRDNAKNTVYLQMNSLK PEDTAVYFCGATFLRSDGHHTINGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQAGGSLRLSCAASGFIFRAASMGWFRQSPG (SEQ ID NO: 805) EPL118 NERELVATVSSGDFTNYADSVKGRFTISRDNAKNTVYLQMNSL KPEDTAVYFCGATFVRSDGHHTIYGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQAGGSLRLSCAASGFIFRAASMDWYRQFPG (SEQ ID NO: 806) EPL138 NERESIATISSGGFTNYADSVKGRFTISRDNAKNTVYLQMNSLK PEDTAVYFCGATFLRSDGHHTINGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQAGGSLRLSCAASGFIFRAASMGWFRQSPG (SEQ ID NO: 807) EPL145 NERELVATVSSGGFTNYADSVKGRFTISRDNAKNTVYLQMNSL KPEDTAVYFCGATFVRSDGHHTIYGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQAGGSLRLSCAASGFIFRAASMDWYRQSPG (SEQ ID NO: 808) EPL164 TQPELVATISSTGFTNYANSVKGRFTISRDNAKNTVYLQMNSLK PEDTAVYFCGATFLRSDGQHSIYGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVHTGGSLRLSCAASGDTFLRYAMGWFRQAP (SEQ ID NO: 809) EPL31 GKEREFVAAITWNGGNTDYAGSLKGRFTISRDNTKNTVYLQM NSLKPEDTAVYYCAADLTFGLASSHYQYDYWGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQAGGSLRLSCAASGDTFLRYAMGWFRQAP (SEQ ID NO: 810) EPL55 GKEREFVAAITWNGGNTDYAGSLKGRFTISRDNTKNTVYLQM NSLKPEDTAVYYCAADLTFGLASSHYQYDYWGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVHTGGSLRLSCAFSGDTFLRYAMGWFRQAP (SEQ ID NO: 811) EPL57 GKEREFVAAITWNGGNTDYADSLKGRFTISRDNTKNTVYLQM NSLRPEDTAVYYCAADLTFGLASSHYQYDYWGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQPGGSLRLSCAASGDTFLRYAMGWFRQAP (SEQ ID NO: 812) EPL136 GKEREFVAAITWNGGNTDYAGSLKGRFTISRDNTKNTVYLQM NSLKPEDTAVYYCAADLTFGLASSHYQYDYWGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGSVLAGGSLRLSCAASGFTFSSYYMSWVRQAPG (SEQ ID

NO: 813) EPL15 KGLEWVSGIHYTGDWTNYADSVKGRFTISRDNAKNELYLEMN NLKPEDTAVYYCARGSDKGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGSVQAGGSLRLSCAASGFTFSSYYMSWVRQAP (SEQ ID NO: 814) EPL34 GKGLEWVSGIHYTGDWTNYADSVKGRFTISRDNAKNELYLEM NNLKPEDTAVYYCARGSDKGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQPGGSLRLSCAASGFTFSDWAMSWVRQAP (SEQ ID NO: 815) EPL86 GKGLEWVSGIHYGDHTTHYADFVKGRFTISRDDAKNTLYLQM NSLKPEDTAVYYCARGSTKGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQPGGSLRLSCAASGFTFSDWAMSWVRQAP (SEQ ID NO: 816) EPL153 GKGLEWVSSIHYGDHTTHYADFVKGRFTISRDDAKNTLYLQMN SLKPEDTAVYYCEKGTTRGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQAGGSLKLSCAASGNVFRAATMAWYRQAP (SEQ ID NO: 817) EPL20 EKQREMVATIASGGTTNYADFVKGRFTISRDNAKNTVYLQMN TLKPEDTAVYYCNAGYLTSLGPKNYWGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQPGGSLRLSCAASGNVFRAATMAWYRQAP (SEQ ID NO: 818) EPL70 EKXREMVATIASGGTTNYADFVKGRFTISRDNAKNTVYLQMNT LKPEDTAVYYCNAGYLTSLGPKNYWGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQAGGSLRLSCAASGNVFRAATMAWYRQVP (SEQ ID NO: 819) EPL125 EKQREMVATIASGGTTNYADFVKGRFTISRDNAKNTVYLQMN TLKPEDTAVYYCNALYLTSLGPKSYWGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQPGGSLRLSCAASGFAFGNHWMYWYRQA (SEQ ID NO: 820) EPL13 PGRGRELVASISSGGSTNYVDSVKGRFTISRDNARNTVYLQMYS LKPEDTAVYYCGTSDNWGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQAGGSLRLSCAASGFAFGNHWMYWYRQA (SEQ ID NO: 821) EPL129 PGRGRELVASISSGGSTNYVDSVKGRFTISRDNARNTVYLQMYS LRPEDTAVYYCGTSDNWGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQAGGSLRLSCAASGFAFGNHWMYWYRQA (SEQ ID NO: 822) EPL159 PGRGRELVASISSGGSTNYVDSVKGRFTISRDNARNTVYLQMYS LKPEDTAVYYCGTSDNWGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQAGGSLRLSCAASGFIFRAASISWYRQSPGN (SEQ ID NO: 823) EPL120 ERELVATINSGGFTNYADSVLGRFTISRDNAKNTGYLQMNSLKP EDTAVYFCAATFLRSDGQPPIWGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQAGGSLRLSCAASGFIFRAASMGWYRQSPG (SEQ ID NO: 824) EPL126 NERELVATINSGGFTNYADSVKGRFTISRDNAKNTGYLQMNSL KPEDTAVYFCAATFLRSDGQPPIWGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQAGGSLRLSCAASEYILSMYRMAWYRQAP (SEQ ID NO: 825) EPL60 GKVRELVADMSSGGTTNYADFVKGRFTISRDNDRNTVYLQMN RLQPEDTAAYYCNVAGRTGPPSYDAFNNWGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQPGASLRVSCAASEYILSMYRMAWYRQAP (SEQ ID NO: 826) EPL156 GKVRELVADMSSGGTTNYADFVKGRFTISRDNDRNTVYLQMN RLQPEDTAAYYCNVAGRTGPPSYDAFNNWGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQPGGSLRLSCAASESISSFIAVGWYRQAPGK (SEQ ID NO: 827) EPL2 ERELVAGINRSGFTYYTDSVKGRFSISRDNAKNTVLLQMTSLKP EDTAVYYCNAGGLYFSNAYTQGDYWGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQTGGSLRLSCAASESISSFIAVGWYRQAPGK (SEQ ID NO: 828) EPL43 ERELVAGINRSGFTYYTDSVKGRFSISRDNAKNTVLLQMTSLKP EDTAVYYCNAGGLYFSNAYTQGDYWGQGTQVTVSS Exemplary anti-target (EpCAM) QVQLQESGGGLVQAGGSLRLSCAASGSVFRANVMGWYRQAP (SEQ ID NO: 829) EPL10 GKQHELVARIDPGGTTTYADPVKGRFTISRDNAKKTVYLQMNS LKPDDTAVYYCNAIILLSGGPKDYWGQGTQVTVSS Exemplary anti-target (EpCAM)

QVQLQESGGGLVQAGGSLRLSCAPSGRTSSIFGMGWFRQAPG (SEQ ID NO: 830)
EPL49 KEREFVASINWSGGSTSYADSVKGRFTISRDNAKNEMYLQMN
SLKFEDTAVYVCAAAVLTNKPSWNFWGQGTQVTVSS Exemplary anti-target (EpCAM)
QVQLQESGGGLVQAGGSLRLSCAASGPIFSDTIRTMGWYRQA (SEQ ID NO: 831)
EPL58 AGKQRELVATIASFPSRTNYVDSVKGRFTISRDIAKNTVYLQMD
SLKPEDTAVYYCNVDLASIPTKTYWGQGTQVTVSS Exemplary anti-target (EpCAM)
QVQLQESGGGLVQAGGSLRLSCAASGSIFGINAMGWYRQAP (SEQ ID NO: 832)
EPL74 GKQRESVAFITIGGNTNYLDSVKGRFTISRDNAKNTVYLQMNG
LKPEDTAVYYCNTNPPLILTAGGLYWGQGTQVTVSS Exemplary anti-target (EpCAM)
QVQLQESGGGLVQPGGSLRLSCATSANRFNINVMGWYRQAP (SEQ ID NO: 833)
EPL78 GQQRELVATINIGGSTDYADSVKGRFTISRDNAKNTVYLQLSDL
KPEDTAVYYCNVKLRVSGPTGPNVYWGQGTQVTVSS Exemplary anti-target (EpCAM)
QVQLQESGGGLVQAGGSLKLSCTASGTILSTMAWYRQAPGKQ (SEQ ID NO: 834)
EPL82 RELVATISRGGTTNYSDSVKGRFAISRDSTKNTVYLQMNSLKPE
DTAVYYCNTPLTDYGMGYNWGQGTQVTVSS Exemplary anti-target (EpCAM)
QVQLQESGGGLVQAGGSLRLSCAVSGSIFSLNTLAWYRQAPGR (SEQ ID NO: 835)
EPL83 QRDLIARITGGGTTVYADSVKGRFTISRDNAKNTVYLQMNSLKP
EDTAVYYCNLMVRHPSGSTYEYWGQGTQVTVSS Exemplary anti-target (EpCAM)
QVQLQESGGGLVQAGGSLRLSCAASGIIFRGTTMGWFRQAPG (SEQ ID NO: 836)
EPL97 KQRESVASISPLGTTSYSGSVEGRFTVSRDNAKNTLFLQMNSLK
SEDTAVYYCNAIQVTNVGPRVYWGQGTQVTVSS Exemplary anti-target (EpCAM)
QVQLQESGGGLVQPGGSLRLSCASSGFTLDDYTIGWFRQAPG (SEQ ID NO: 837)
EPL109 KEREGVSCISRRDDSTYYADSVKGRFTISRDNAKNTVDLQMISL
RPEDTAVYYCAATPRSYTLRCLGKFDFQGQGTQVTVSS Exemplary anti-target (EpCAM)
QVQLQESGGGLVQAGGSLRLSCAASGNIVRMTNMAWYRQA (SEQ ID NO: 838)
EPL117 PGKQREFVATISAGGSTTYVDSVKDRFTISRDNTKNTVYLQMN
YLKPEDTAVYYCATGSILTNRGAIPGSWGHGTQVTVSS Exemplary anti-target (EpCAM)
QVQLQESGGGLVQAGGSLRLSCAAPGFAFNDHAILWFRQAPG (SEQ ID NO: 839)
EPL127 KEREGVSEICRDGTTYYTDSVKGRFTISSDNAKNTVYLQMNSVK
TDDTAVYYCAVDRRRYYCSGNRAFSSDYYYWGQGTQVTVSS Exemplary anti-target
(EpCAM) QVQLQESGGGLVQAGGSLRLSCVHSGSIFRASTMAWYRQAPG (SEQ ID
NO: 840) EPL152 KQRELVAQIMSGGGTNYAGSVKGRFTISRDNANNTVYLQMNS
LKPEDTAVYYCNAAQITSWGPKVYWGQGTQVTVSS Exemplary anti-target (EpCAM)
QVQLQESGGGLVQPGGSLRLSCAASGRINSINTMGWYRQAPG (SEQ ID NO: 841)
EPL189 NQRELVAEITRGGTTNYADSVQGRYAISRDNAKNLVYLQMNSL
KPEDTDVYYCNAQTFPTFSRPTGLDYWGQGTQVTVSS Exemplary anti-CD3/anti-
VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN EpCAM fusion
protein(SEQ ID WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO:
842) EPL10 VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGSVFR
ANVMGWYRQAPGKQHELVARIDPGGTTTYADPVKGRFTISRD
NAKKTVYLQMNSLKPDDTAVYYCNAILLSGGPKDYWGQGTQ VTVSSHHHHHH*
Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN
EpCAM fusion protein(SEQ ID
WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 843) EPL109
VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR

QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGSQVQLQESGGGLVQPGGSLRLSCASSGFTLD
DYTIGWFRQAPGKEREGVSCISRRDDSTYYADSVKGRFTISRDN
AKNTVDLQMISLRPEDTAVYYCAATPRSYTLRCLGKFDFQGQG TQVTVSSHHHHHH*
Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN
EpCAM fusion protein(SEQ ID
WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 844) EPL117
VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGNIVR
MTNMAWYRQAPGKQREFVATISAGGSTTYVDSVKDRFTISRD
NTKNTVYLQMNYLKPEDTAVYYCATGSILTNRGAIPGSWGHGT QVTVSSHHHHHH*
Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN
EpCAM fusion protein(SEQ ID
WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 845) EPL120
VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGFIFR
AASISWYRQSPGNERELVATINSGGFTNYADSVLGRFTISRDNA
KNTGYLQMNSLKPEDTAVYFCAATFLRSDGQPPIWGQGTQVT VSSHHHHHH*
Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN
EpCAM fusion protein(SEQ ID
WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 846) EPL125
VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGNVFR
AATMAWYRQVPEKQREMVATIASGGTTNYADFVKGRFTISRD
NAKNTVYLQMNTLKPEDTAVYYCNALYLTSLGPKSYWGQGTQ VTVSSHHHHHH*
Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN
EpCAM fusion protein(SEQ ID
WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 847) EPL127
VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAAPGFAFN
DHAILWFRQAPGKEREGVSEICRDGTTYYTDSVKGRFTISSDNA
KNTVYLQMNSVKTDDTAVYYCAVDRRRYYCSGNRAFSSDYYY
WGQGTQVTVSSHHHHHH* Exemplary anti-CD3/anti-
VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN EpCAM fusion
protein(SEQ ID WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO:
848) EPL13 VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR

QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGSQVQLQESGGGLVQPGGSLRLSCAASGFAFG
NHWMYWYRQAPGRGRELVASISSGGSTNYVDSVKGRFTISRD
NARNTVYLQMYSLKPEDTAVYYCGTSDNWGQGTQVTVSSHH HHHH* Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN EpCAM fusion protein(SEQ ID WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 849) EPL136 VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGSQVQLQESGGGLVQPGGSLRLSCAASGDTFL
RYAMGWFRQAPGKEREFVAAITWNGGNTDYAGSLKGRFTISR
DNTKNTVYLQMNSLKPEDTAVYYCAADLTFGLASSHYQYDYW
GQGTQVTVSSHHHHHH* Exemplary anti-CD3/anti-
VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN EpCAM fusion protein(SEQ ID WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 850) EPL138 VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGFIFR
AASMDWYRQFPGNERESIATISSGGFTNYADSVKGRFTISRDN
AKNTVYLQMNSLKPEDTAVYFCGATFLRSDGHHTINGQGTQV TVSSHHHHHH*
Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN EpCAM fusion protein(SEQ ID
WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 851) EPL145
VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGFIFR
AASMGWFRQSPGNERELVATVSSGGFTNYADSVKGRFTISRD
NAKNTVYLQMNSLKPEDTAVYFCGATFVRSDGHHTIYGQGTQ VTVSSHHHHHH*
Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN EpCAM fusion protein(SEQ ID
WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 852) EPL152
VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGSQVQLQESGGGLVQAGGSLRLSCVHSGSIFR
ASTMAWYRQAPGKQRELVAQIMSGGGTNYAGSVKGRFTISR
DNANNTVYLQMNSLKPEDTAVYYCNAAQITSWGPKVYWGQ GTQVTVSSHHHHHH*
Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN EpCAM fusion protein(SEQ ID
WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 853) EPL153
VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA

YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGSQVQLQESGGGLVQPGGSLRLSCAASGFTFS
DWAMSWVRQAPGKGLEWVSSIHYGDHTTHYADFVKGRFTIS
RDDAKNTLYLQMNSLKPEDTAVYYCEKGTTRGQGTQVTVSSH HHHHH* Exemplary
anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN EpCAM
fusion protein(SEQ ID WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG
NO: 854) EPL156 VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGSQVQLQESGGGLVQPGASLRVSCAASEYILS
MYRMAWYRQAPGKVRELVADMSSGGTTNYADFVKGRFTISR
DNDRNTVYLQMNRLQPEDTAAYYCNVAGRTGPPSYDAFNN WGQGTQVTVSSHHHHHH*
Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN
EpCAM fusion protein(SEQ ID
WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 855) EPL164
VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGFIFR
AASMDWYRQSPGTQPELVATISSTGFTNYANSVKGRFTISRDN
AKNTVYLQMNSLKPEDTAVYFCGATFLRSDGQHSIYGQGTQV TVSSHHHHHH*
Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN
EpCAM fusion protein(SEQ ID
WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 856) EPL189
VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGSQVQLQESGGGLVQPGGSLRLSCAASGRINSI
NTMGWYRQAPGNQRELVAEITRGGTTNYADSVQGRYAISRD
NAKNLVYLQMNSLKPEDTDVYYCNAQTFPTFSRPTGLDYWGQ GTQVTVSSHHHHHH*
Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN
EpCAM fusion protein(SEQ ID
WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 857) EPL2
VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGSQVQLQESGGGLVQPGGSLRLSCAASESISSFI
AVGWYRQAPGKERELVAGINRSGFTYYTDSVKGRFSISRDNAK
NTVLLQMTSLKPEDTAVYYCNAGGLYFSNAYTQGDYWGQGT QVTVSSHHHHHH*
Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN
EpCAM fusion protein(SEQ ID
WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 858) EPL20
VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT

VSSGGGGSGGGSQVQLQESGGGLVQAGGSLKLSCAASGNVFR
AATMAWYRQAPEKQREMVATIASGGTTNYADFVKGRFTISRD
NAKNTVYLQMNTLKPEDTAVYYCNAGYLTSLGPKNYWGQGT QVTVSSHHHHHH*
Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN
EpCAM fusion protein(SEQ ID
WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 859) EPL34
VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGSQVQLQESGGGSVQAGGSLRLSCAASGFTFS
SYYMSWVRQAPGKGLEWVSGIHYTGDWTNYADSVKGRFTISR
DNAKNELYLEMNNLKPEDTAVYYCARGSDKGQGTQVTVSSHH HHHH* Exemplary anti-
CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN EpCAM
fusion protein(SEQ ID WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG
NO: 860) EPL49 VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAPSGRTSSI
FGMGWFRQAPGKEREFVASINWSGGSTSYADSVKGRFTISRD
NAKNEMYLQMNSLKFEDTAVYVCAAAVLTNKPSWNFWGQG TQVTVSSHHHHHH*
Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN
EpCAM fusion protein(SEQ ID
WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 861) EPL58
VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGPIFS
DTIRTMGWYRQAAGKQRELVATIASFPSRTNYVDSVKGRFTIS
RDIAKNTVYLQMDSLKPEDTAVYYCNVDLASIPTKTYWGQGTQ VTVSSHHHHHH*
Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN
EpCAM fusion protein(SEQ ID
WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 862) EPL74
VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGSIFGI
NAMGWYRQAPGKQRESVAFITIGGNTNYLDSVKGRFTISRDN
AKNTVYLQMNGLKPEDTAVYYCNTNPPLILTAGGLYWGQGTQ VTVSSHHHHHH*
Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN
EpCAM fusion protein(SEQ ID
WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 863) EPL78
VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGSQVQLQESGGGLVQPGGSLRLSCATSANRFN

INVMGWYRQAPGQQRELVATINIGGSTDYADSVKGRFTISRD NAKNTVYLQLSDLKPEDTAVYYCNVKLRVSGPTGPNVYWGQG TQVTVSSHHHHHH* Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN EpCAM fusion protein(SEQ ID WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 864) EPL82 VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT VSSGGGGSGGGSQVQLQESGGGLVQAGGSLKLSCTASGTILST MAWYRQAPGKQRELVATISRGGTTNYSDSVKGRFAISRDSTKN TVYLQMNSLKPEDTAVYYCNTPLTDYGMGYNWGQGTQVTVS SHHHHHH* Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN EpCAM fusion protein(SEQ ID WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 865) EPL83 VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT VSSGGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAVSGSIFSL NTLAWYRQAPGRQRDLIARITGGGTTVYADSVKGRFTISRDNA KNTVYLQMNSLKPEDTAVYYCNLMVRHPSGSTYEYWGQGTQ VTVSSHHHHHH* Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN EpCAM fusion protein(SEQ ID WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 866) EPL86 VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT VSSGGGGSGGGSQVQLQESGGGLVQPGGSLRLSCAASGFTFS DWAMSWVRQAPGKGLEWVSGIHYGDHTTHYADFVKGRFTIS RDDAKNTLYLQMNSLKPEDTAVYYCARGSTKGQGTQVTVSSH HHHHH* Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN EpCAM fusion protein(SEQ ID WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 867) EPL90 VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT VSSGGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGFIFR AASMAWYRQSPGNERELVASISSGAFTNYADSVKARFTISRDN AKNTVYLQMNSLKPEDTAVYFCGATFLRSDGHHTINGQGTQV TVSSHHHHHH* Exemplary anti-CD3/anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN EpCAM fusion protein(SEQ ID WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG NO: 868) EPL97 VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT VSSGGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGIIFRG TTMGWFRQAPGKQRESVASISPLGTTSYSGSVEGRFTVSRDNA KNTLFLQMNSLKSEDTAVYYCNAIQVTNVGPRVYWGQGTQVT VSSHHHHHH*

Exemplary anti-EpCAM/anti-
GGGSQVQLQESGGGLVQAGGSLRLSCAASGSVFRANVMGWY EpCAM fusion
protein(SEQ ID RQAPGKQHELVARIDPGGTTTYADPVKGRFTISRDNAKKTVYL NO:
869) EPL10 QMNSLKPDDTAVYYCNAIILLSGGPKDYWGQGTQVTVSSGGG
GSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINW
VRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKN
TAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTL
VTVSSGGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTC
ASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPARF
SGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLT VLHHHHHH*
Exemplary anti-EpCAM/anti-
GGGSQVQLQESGGGLVQPGGSLRLSCASSGFTLDDYTIGWFR EpCAM fusion
protein(SEQ ID QAPGKEREGVSCISRRDDSTYYADSVKGRFTISRDNAKNTVDLQ NO:
870) EPL109 MISLRPEDTAVYYCAATPRSYTLRCLGKFDFQGQGTQVTVSSG
GGGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAIN
WVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDS
KNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQG
TLVTVSSGGGGGGGGSGGGGSQTVVTQEPSLTVSPGGTVTL
TCASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPA
RFSGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTK LTVLHHHHHH*
Exemplary anti-EpCAM/anti-
GGGSQVQLQESGGGLVQAGGSLRLSCAASGNIVRMTNMAW EpCAM fusion
protein(SEQ ID YRQAPGKQREFVATISAGGSTTYVDSVKDRFTISRDNTKNTVYL NO:
871) EPL117 QMNYLKPEDTAVYYCATGSILTNRGAIPGSWGHGTQVTVSSG
GGGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAIN
WVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDS
KNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQG
TLVTVSSGGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTL
TCASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPA
RFSGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTK LTVLHHHHHH*
Exemplary anti-EpCAM/anti-
GGGSQVQLQESGGGLVQAGGSLRLSCAASGFIFRAASISWYRQ EpCAM fusion
protein(SEQ ID SPGNERELVATINSGGFTNYADSVLGRFTISRDNAKNTGYLQM NO:
872) EPL120 NSLKPEDTAVYFCAATFLRSDGQPPIWGQGTQVTVSSGGGGS
GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCAS
STGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPARFSG
SLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVL HHHHHH* Exemplary
anti-EpCAM/anti- GGGSQVQLQESGGGLVQAGGSLRLSCAASGNVFRAATMAWY
EpCAM fusion protein(SEQ ID
RQVPEKQREMVATIASGGTTNYADFVKGRFTISRDNAKNTVYL NO: 873) EPL125
QMNTLKPEDTAVYYCNALYLTSLGPKSYWGQGTQVTVSSGGG
GSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINW
VRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKN
TAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTL
VTVSSGGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTC
ASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPARF
SGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLT VLHHHHHH*

Exemplary anti-EpCAM/anti-
GGGSQVQLQESGGGLVQAGGSLRLSCAAPGFAFNDHAILWFR EpCAM fusion
protein(SEQ ID QAPGKEREGVSEICRDGTTYYTDSVKGRFTISSDNAKNTVYLQ NO:
874) EPL127 MNSVKTDDTAVYYCAVDRRRYYCSGNRAFSSDYYYWGQGTQ
VTVSSGGGGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTF
NKYAINWVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFT
ISRDDSKNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAY
WGQGTLVTVSSGGGGSGGGGSGGGGSQTVVTQEPSLTVSPG
GTVTLTCASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLV
PGTPARFSGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVF GGGTKLTVLHHHHHH*
Exemplary anti-EpCAM/anti-
GGGSQVQLQESGGGLVQPGGSLRLSCAASGFAFGNHWMYW EpCAM fusion
protein(SEQ ID YRQAPGRGRELVASISSGGSTNYVDSVKGRFTISRDNARNTVYL NO:
875) EPL13 QMYSLKPEDTAVYYCGTSDNWGQGTQVTVSSGGGGSGGGSE
VQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGK
GLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQM
NNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVTVSSGG
GGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCASSTGAV
TSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGG
KAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVLHHHH HH* Exemplary anti-
EpCAM/anti- GGGSQVQLQESGGGLVQPGGSLRLSCAASGDTFLRYAMGWF EpCAM
fusion protein(SEQ ID RQAPGKEREFVAAITWNGGNTDYAGSLKGRFTISRDNTKNTVY
NO: 876) EPL136 LQMNSLKPEDTAVYYCAADLTFGLASSHYQYDYWGQGTQVTV
SSGGGGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKY
AINWVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISR
DDSKNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYW
GQGTLVTVSSGGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGT
VTLTCASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPG
TPARFSGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGG GTKLTVLHHHHHH*
Exemplary anti-EpCAM/anti-
GGGSQVQLQESGGGLVQAGGSLRLSCAASGFIFRAASMDWY EpCAM fusion
protein(SEQ ID RQFPGNERESIATISSGGFTNYADSVKGRFTISRDNAKNTVYLQ NO:
877) EPL138 MNSLKPEDTAVYFCGATFLRSDGHHTINGQGTQVTVSSGGGG
SGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCAS
STGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPARFSG
SLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVL HHHHHH* Exemplary
anti-EpCAM/anti- GGGSQVQLQESGGGLVQAGGSLRLSCAASGFIFRAASMGWF EpCAM
fusion protein(SEQ ID RQSPGNERELVATVSSGGFTNYADSVKGRFTISRDNAKNTVYL
NO: 878) EPL145 QMNSLKPEDTAVYFCGATFVRSDGHHTIYGQGTQVTVSSGGG
GSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINW
VRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKN
TAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTL
VTVSSGGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTC
ASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPARF
SGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLT VLHHHHHH*
Exemplary anti-EpCAM/anti-
GGGSQVQLQESGGGLVQAGGSLRLSCVHSGSIFRASTMAWYR EpCAM fusion

protein(SEQ ID QAPGKQRELVAQIMSGGGTNYAGSVKGRFTISRDNANNTVYL NO: 879) EPL152 QMNSLKPEDTAVYYCNAAQITSWGPKVYWGQGTQVTVSSGG GGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAIN WVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDS KNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQG TLVTVSSGGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTL TCASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPA RFSGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTK LTVLHHHHHH* Exemplary anti-EpCAM/anti- GGGSQVQLQESGGGLVQPGGSLRLSCAASGFTFSDWAMSWV EpCAM fusion protein(SEQ ID RQAPGKGLEWVSSIHYGDHTTHYADFVKGRFTISRDDAKNTLY NO: 880) EPL153 LQMNSLKPEDTAVYYCEKGTTRGQGTQVTVSSGGGGSGGGSE VQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGK GLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQM NNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVTVSSGG GGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCASSTGAV TSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGG KAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVLHHHH HH* Exemplary anti-EpCAM/anti- GGGSQVQLQESGGGLVQPGASLRVSCAASEYILSMYRMAWY EpCAM fusion protein(SEQ ID RQAPGKVRELVADMSSGGTTNYADFVKGRFTISRDNDRNTVY NO: 881) EPL156 LQMNRLQPEDTAAYYCNVAGRTGPPSYDAFNNWGQGTQVT VSSGGGGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNK YAINWVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISR DDSKNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYW GQGTLVTVSSGGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGT VTLTCASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPG TPARFSGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGG GTKLTVLHHHHHH* Exemplary anti-EpCAM/anti- GGGSQVQLQESGGGLVQAGGSLRLSCAASGFIFRAASMDWY EpCAM fusion protein(SEQ ID RQSPGTQPELVATISSTGFTNYANSVKGRFTISRDNAKNTVYLQ NO: 882) EPL164 MNSLKPEDTAVYFCGATFLRSDGQHSIYGQGTQVTVSSGGGG SGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT VSSGGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCAS STGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPARFSG SLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVL HHHHHH* Exemplary anti-EpCAM/anti- GGGSQVQLQESGGGLVQPGGSLRLSCAASGRINSINTMGWYR EpCAM fusion protein(SEQ ID QAPGNQRELVAEITRGGTTNYADSVQGRYAISRDNAKNLVYLQ NO: 883) EPL189 MNSLKPEDTDVYYCNAQTFPTFSRPTGLDYWGQGTQVTVSSG GGGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAIN WVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDS KNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQG TLVTVSSGGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTL TCASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPA RFSGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTK LTVLHHHHHH* Exemplary anti-EpCAM/anti- GGGSQVQLQESGGGLVQPGGSLRLSCAASESISSFIAVGWYRQ EpCAM fusion protein(SEQ ID APGKERELVAGINRSGFTYYTDSVKGRFSISRDNAKNTVLLQMT NO:

884) EPL2 SLKPEDTAVYYCNAGGLYFSNAYTQGDYWGQGTQVTVSSGG
GGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAIN
WVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDS
KNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQG
TLVTVSSGGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTL
TCASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPA
RFSGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTK LTVLHHHHHH*
Exemplary anti-EpCAM/anti-
GGGSQVQLQESGGGLVQAGGSLKLSCAASGNVFRAATMAWY EpCAM fusion
protein(SEQ ID RQAPEKQREMVATIASGGTTNYADFVKGRFTISRDNAKNTVYL NO:
885) EPL20 QMNTLKPEDTAVYYCNAGYLTSLGPKNYWGQGTQVTVSSGG
GGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAIN
WVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDS
KNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQG
TLVTVSSGGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTL
TCASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPA
RFSGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTK LTVLHHHHHH*
Exemplary anti-EpCAM/anti-
GGGSQVQLQESGGGSVQAGGSLRLSCAASGFTFSSYYMSWVR EpCAM fusion
protein(SEQ ID QAPGKGLEWVSGIHYTGDWTNYADSVKGRFTISRDNAKNELY NO:
886) EPL34 LEMNNLKPEDTAVYYCARGSDKGQGTQVTVSSGGGGSGGGS
EVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPG
KGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQM
NNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVTVSSGG
GGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCASSTGAV
TSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGG
KAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVLHHHH HH* Exemplary anti-
EpCAM/anti- GGGSQVQLQESGGGLVQAGGSLRLSCAPSGRTSSIFGMGWFR EpCAM
fusion protein(SEQ ID QAPGKEREFVASINWSGGSTSYADSVKGRFTISRDNAKNEMYL
NO: 887) EPL49 QMNSLKFEDTAVYVCAAAVLTNKPSWNFWGQGTQVTVSSG
GGGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAIN
WVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDS
KNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQG
TLVTVSSGGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTL
TCASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPA
RFSGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTK LTVLHHHHHH*
Exemplary anti-EpCAM/anti-
GGGSQVQLQESGGGLVQAGGSLRLSCAASGPIFSDTIRTMGW EpCAM fusion
protein(SEQ ID YRQAAGKQRELVATIASFPSRTNYVDSVKGRFTISRDIAKNTVYL NO:
888) EPL58 QMDSLKPEDTAVYYCNVDLASIPTKTYWGQGTQVTVSSGGGG
SGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA
YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT
VSSGGGGGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCAS
STGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPARFSG
SLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVL HHHHHH* Exemplary
anti-EpCAM/anti- GGGSQVQLQESGGGLVQAGGSLRLSCAASGSIFGINAMGWYR
EpCAM fusion protein(SEQ ID
QAPGKQRESVAFITIGGNTNYLDSVKGRFTISRDNAKNTVYLQ NO: 889) EPL74
MNGLKPEDTAVYYCNTNPPLILTAGGLYWGQGTQVTVSSGGG

GSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINW
VRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKN
TAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTL
VTVSSGGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTC
ASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPARF
SGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLT VLHHHHHH*
Exemplary anti-EpCAM/anti-
GGGSQVQLQESGGGLVQPGGSLRLSCATSANRFNINVMGWY EpCAM fusion
protein(SEQ ID RQAPGQQRELVATINIGGSTDYADSVKGRFTISRDNAKNTVYL NO:
890) EPL78 QLSDLKPEDTAVYYCNVKLRVSGPTGPNVYWGQGTQVTVSSG
GGGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAIN
WVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDS
KNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQG
TLVTVSSGGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTL
TCASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPA
RFSGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTK LTVLHHHHHH*
Exemplary anti-EpCAM/anti-
GGGSQVQLQESGGGLVQAGGSLKLSCTASGTILSTMAWYRQA EpCAM fusion
protein(SEQ ID PGKQRELVATISRGGTTNYSDSVKGRFAISRDSTKNTVYLQMNS NO:
891) EPL82 LKPEDTAVYYCNTPLTDYGMGYNWGQGTQVTVSSGGGGSGG
GSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAP
GKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQ
MNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVTVSS
GGGGGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCASSTG
AVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLL
GGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVLHH HHHH* Exemplary anti-
EpCAM/anti- GGGSQVQLQESGGGLVQAGGSLRLSCAVSGSIFSLNTLAWYR EpCAM
fusion protein(SEQ ID QAPGRQRDLIARITGGGTTVYADSVKGRFTISRDNAKNTVYLQ
NO: 892) EPL83 MNSLKPEDTAVYYCNLMVRHPSGSTYEYWGQGTQVTVSSGG
GGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAIN
WVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDS
KNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQG
TLVTVSSGGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTL
TCASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPA
RFSGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTK LTVLHHHHHH*
Exemplary anti-EpCAM/anti-
GGGSQVQLQESGGGLVQPGGSLRLSCAASGFTFSDWAMSWV EpCAM fusion
protein(SEQ ID RQAPGKGLEWVSGIHYGDHTTHYADFVKGRFTISRDDAKNTLY NO:
893) EPL86 LQMNSLKPEDTAVYYCARGSTKGQGTQVTVSSGGGGSGGGSE
VQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGK
GLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQM
NNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVTVSSGG
GGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCASSTGAV
TSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGG
KAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVLHHHH HH* Exemplary anti-
EpCAM/anti- GGGSQVQLQESGGGLVQAGGSLRLSCAASGFIFRAASMAWYR EpCAM
fusion protein(SEQ ID QSPGNERELVASISSGAFTNYADSVKARFTISRDNAKNTVYLQ
NO: 894) EPL90 MNSLKPEDTAVYFCGATFLRSDGHHTINGQGTQVTVSSGGGG
SGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR
QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA

YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT VSSGGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTCAS STGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPARFSG SLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVL HHHHHH* Exemplary anti-EpCAM/anti- GGGSQVQLQESGGGLVQAGGSLRLSCAASGIIFRGTTMGWFR EpCAM fusion protein(SEQ ID QAPGKQRESVASISPLGTTSYSGSVEGRFTVSRDNAKNTLFLQ NO: 895) EPL97 MNSLKSEDTAVYYCNAIQVTNVGPRVYWGQGTQVTVSSGGG GSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINW VRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKN TAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTL VTVSSGGGGSGGGGSGGGGSQTVVTQEPSLTVSPGGTVTLTC ASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTPARF SGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLT VLHHHHHH* Exemplary humanized anti- EVQLVESGGGLVQPGGSLTLSCAASGFAFGNHWMYWYRQAP EpCAM sequence (SEQ ID No. GRGRELVASISSGGSTNYVDSVKGRFTISRDNAKNTLYLQMNSL 896) RAEDTAVYYCGTSDNWGQGTLVTVSS Exemplary humanized anti- EVQLLESGGGLVQPGGSLTLSCAASGFIFRAASMDWYRQFPG EpCAM sequence (SEQ ID No. NERESIATISSGGFTNYADSVKGRFTISRDNSKNTLYLQMNSLRA 897) EDTAVYYCGATFLRSDGHHTINGQGTLVTVSS Exemplary humanized anti- EVQLLESGGGLVQPGGSLTLSCAASGFIFRAASMAWYRQSPG EpCAM sequence (SEQ ID No. NERELVASISSGAFTNYADSVKARFTISRDNSKNTLYLQMNSLR 898) AEDTAVYYCGATFLRSDGHHTINGQGTLVTVSS Exemplary Humanized Anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN CD3/Anti-EpCAM sequence WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG name H13 (SEQ ID No. 899) VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT VSSGGGGSGGGSEVQLVESGGGLVQPGGSLTLSCAASGFAFG NHWMYWYRQAPGRGRELVASISSGGSTNYVDSVKGRFTISRD NAKNTLYLQMNSLRAEDTAVYYCGTSDNWGQGTLVTVSSHH HHHH* Exemplary Humanized Anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN CD3/Anti-EpCAM sequence WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG name H138 (SEQ ID No. 900) VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT VSSGGGGSGGGSEVQLLESGGGLVQPGGSLTLSCAASGFIFRA ASMDWYRQFPGNERESIATISSGGFTNYADSVKGRFTISRDNS KNTLYLQMNSLRAEDTAVYYCGATFLRSDGHHTINGQGTLVTV SSHHHHHH* Exemplary Humanized Anti- VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN CD3/Anti-EpCAM sequence WVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLSG name H90 (SEQ ID No. 901) VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVT VSSGGGGSGGGSEVQLLESGGGLVQPGGSLTLSCAASGFIFRA

ASMAWYRQSPGNERELVASISSGAFTNYADSVKARFTISRDNS KNTLYLQMNSLRAEDTAVYYCGATFLRSDGHHTINGQGTLVTV SSHHHHHH* SEQ ID No. 902 QDGNEEMGGITQ SEQ ID No. 903 LVQPGN AB loop oligo WT SEQ ID No. 904 DSVKGR C″D loop oligo WT SEQ ID No. 905 SLRPED EF loop oligo WT EpCAM H90 TriTAC EVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQ C2854 APGKGLEWVSSISGSGRDTLYADSVKGRFTISRDNAKTT (SEQ ID No. 906) LYLQMNSLRPEDTAVYYCTIGGSLSVSSQGTLVTVSSGG GGSGGGSQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSG NYPNWVQQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGG KAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVLG GGGSGGGGSGGGGSEVQLVESGGGLVQPGGSLKLSCAAS GFTENKYAINWVRQAPGKGLEWVARIRSKYNNYATYYAD QVKDRFTISRDDSKNTAYLQMNNLKTEDTAVYYCVRHAN FGNSYISYWAYWGQGTLVTVSSGGGGSGGGSEVQLLESG GGLVQPGGSLTLSCAASGFIFRAASMAWYRQSPGNEREL VASISSGAFTNYADSVKARFTISRDNSKNTLYLQMNSLR AEDTAVYYCGATFLRSDGHHTINGQGTLVTVSSHHHHHH EpCAM H90 ProTriTAC QTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPNWVQ (L040) C2704 QKPGQAPRGLIGGTKFLVPGTPARFSGSLLGGKAALTLS (SEQ ID No. 907) GVQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGG GSGGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTENKY AINWVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFT ISRDDSKNTAYLQMNNLKTEDTAVYYCVRHANFGNSYIS YWAYWGQGTLVTVSSGGGGSGGGSEVQLLESGGGLVQPG GSLTLSCAASGFIFRAASMAWYRQSPGNERELVASISSG AFTNYADSVKARFTISRDNSKNTLYLQMNSLRAEDTAVY YCGATFLRSDGHHTINGQGTLVTVSSHHHHHH EpCAM H90 ProTriTAC EVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQ (NCLV) C2302 GGGGGLDGNEEPGGLEWVSSISGSGRDTLYADSVKGRFT (SEQ ID No. 908) ISRDNAKTTLYLQMNSLRPEDTAVYYCTIGGSLSVSSQG TLVTVSSGGGGSGGGGSGGVVGGGGTQTVVTQEPSLTVS PGGTVTLTCASSTGAVTSGNYPNWVQQKPGQAPRGLIGG TKFLVPGTPARFSGSLLGGKAALTLSGVQPEDEAEYYCT LWYSNRWVFGGGTKLTVLGGGGSGGGGSGGGGSEVQLVE SGGGLVQPGGSLKLSCAASGFTENKYAINWVRQAPGKGL EWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQ MNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVTV SSGGGGSGGGSEVQLLESGGGLVQPGGSLTLSCAASGFI FRAASMAWYRQSPGNERELVASISSGAFTNYADSVKARF TISRDNSKNTLYLQMNSLRAEDTAVYYCGATFLRSDGHH TINGQGTLVTVSSHHHHHH

Claims

1. A conditionally active binding protein comprising a) a binding moiety (M) which comprises a non-CDR loop, wherein the binding moiety (M) comprises a serum albumin binding domain and the non-CDR loop comprises a binding site specific for a CD3ε domain, and wherein the binding moiety further comprises complementarity determining regions (CDRs), (b) a cleavable linker (L), (c) a first target antigen binding domain (T1) comprising an immunoglobulin molecule that binds to CD3ε, and (d) a second target antigen binding domain (T2), wherein the binding moiety (M) is

capable of masking the binding of the first target antigen binding domain (T1) to CD3ε.
2. The conditionally active binding protein of claim 1, wherein the binding moiety is capable of binding to a half-life extending protein.
3. The conditionally active binding protein of claim 1, wherein the binding moiety is a natural peptide, a synthetic peptide, an engineered scaffold, or an engineered serum bulk protein.
4. The conditionally active binding protein of claim 3, wherein the engineered scaffold comprises an sdAb, an scFv, a Fab, a VHH, a fibronectin type III domain, an immunoglobulin-like scaffold, a DARPin, a cystine knot peptide, a lipocalin, a three-helix bundle scaffold, a protein G-related albumin-binding module, or a DNA or RNA aptamer scaffold.
5. The conditionally active binding protein of claim 1, wherein the non-CDR loop is from a variable domain, a constant domain, a C1-set domain, a C2-set domain, an I-domain, or any combinations thereof.
6. The conditionally active binding protein of claim 1, wherein the binding site is capable of binding to a human serum albumin.
7. The conditionally active binding protein of claim 6, wherein the CDRs provide the binding site specific for the serum albumin.
8. The conditionally active binding protein of claim 1, wherein the non-CDR loop provides the binding site specific for binding of the binding moiety to the first target antigen binding domain.
9. The conditionally active binding protein of claim 1, wherein the the second target antigen binding domain binds to a tumor antigen.
10. The conditionally active binding protein of claim 9, wherein the tumor antigen comprises EpCAM, EGFR, HER-2, HER-3, c-Met, FoIR, PSMA, CD38, BCMA, CEA, 5T4, AFP, B7-H3, CDH-6, CAIX, CD117, CD123, CD138, CD166, CD19, CD20, CD205, CD22, CD30, CD33, CD352, CD37, CD44, CD52, CD56, CD70, CD71, CD74, CD79b, DLL3, EphA2, FAP, FGFR2, FGFR3, GPC3, gpA33, FLT-3, gpNMB, HPV-16 E6, HPV-16 E7, ITGA2, ITGA3, SLC39A6, MAGE, mesothelin, Muc1, Muc16, NaPi2b, Nectin-4, CDH-3, CDH-17, EPHB2, ITGAV, ITGB6, NY-ESO-1, PRLR, PSCA, PTK7, RORI, SLC44A4, SLITRK5, SLITRK6, STEAP1, TIM1, Trop2, or WT1.
11. The conditionally active binding protein of claim 1, wherein the first target antigen binding domain binds to a human CD3ε.
12. The conditionally active binding protein of claim 1, wherein the binding moiety (M), the cleavable linker (L), the first target antigen binding domain (T1), and the second target antigen binding domain (T2) are in one of the following configurations: M:L:T1:T2, and T2:T1:L:M.
13. The conditionally active binding protein of claim 1, wherein the cleavable linker comprises a cleavage site.
14. The conditionally active binding protein of claim 13, wherein the cleavage site is recognized by a protease.
15. The conditionally active binding protein of claim 14, wherein the protease cleavage site is recognized by a serine protease, a cysteine protease, an aspartate protease, a threonine protease, a glutamic acid protease, a metalloproteinase, a gelatinase, or an asparagine peptide lyase.
16. The conditionally active binding protein of claim 14, wherein the protease cleavage site is recognized by a Cathepsin B, a Cathepsin C, a Cathepsin D, a Cathepsin E, a Cathepsin K, a Cathepsin L, a kallikrein, a hK1, a hK10, a hK15, a plasmin, a collagenase, a Type IV collagenase, a stromelysin, a Factor Xa, a chymotrypsin-like protease, a trypsin-like protease, a elastase-like protease, a subtilisin-like protease, an actinidain, a bromelain, a calpain, a caspase, a caspase-3, a Mir1-CP, a papain, a HIV-1 protease, a HSV protease, a CMV protease, a chymosin, a renin, a pepsin, a matriptase, a legumain, a plasmepsin, a nepenthesin, a metalloexopeptidase, a metalloendopeptidase, a matrix metalloprotease (MMP), a MMP1, a MMP2, a MMP3, a MMP8, a MMP9, a MMP10, a MMP11, a MMP12, a MMP13, a MMP14, an ADAM10, an ADAM12, an urokinase plasminogen activator (uPA), an enterokinase, a prostate-specific target (PSA, hK3), an

interleukin-1β converting enzyme, a thrombin, a FAP, a dipeptidyl peptidase, or dipeptidyl peptidase IV (DPPIV/CD26), a type II transmembrane serine protease (TTSP), a neutrophil elastase, a cathepsin G, a proteinase 3, a neutrophil serine protease 4, a mast cell chymase, a mast cell tryptase, a dipeptidyl peptidase, or a dipeptidyl peptidase IV (DPPIV/CD26).

17. The conditionally active binding protein of claim 11, wherein the binding moiety comprises a binding site specific the first target antigen binding domain (T1), and wherein the binding site comprises at least one of the following motifs: QDGNE (SEQ ID NO: 921), QDGNEE (SEQ ID NO: 801), DGNE (SEQ ID NO: 922), and DGNEE (SEQ ID NO: 923).

18. The conditionally active binding protein of claim 13, wherein upon cleavage of the cleavable site in a tumor microenvironment, the conditionally active binding protein is activated by separation of the binding moiety.

19. The conditionally active binding protein of claim 1, wherein the binding moiety comprises a sequence selected from the group consisting of SEQ ID Nos. 50 and 259-301.

20. The conditionally active binding protein of claim 1, wherein the binding moiety comprises an sdAb.